



SECRETARIATE GENERAL
THE NATIONAL ENERGY COUNCIL



INDONESIA

ENERGY OUTLOOK

2021



**SECRETARIATE GENERAL
THE NATIONAL ENERGY COUNCIL**

INDONESIA ENERGY OUTLOOK 2021

**Secretariate General of the National Energy Council
2021**

REMARK BY DAILY CHAIRMAN OF NATIONAL ENERGY COUNCIL



I would like to appreciate the effort by the Secretariate General of the National Energy Council for publishing the book of Indonesia Energy Outlook 2021. This book does not only give the overview of the current energy condition, but also show Indonesia energy projection in the future that would be very beneficial as an input in making the policy and planning in Indonesia's energy sector.

The activity restriction during the Covid-19 pandemic has impacted the supply and demand in energy sector as well as the economic growth in 2020. Furthermore, the government has made policies on the utilization of Solar Power Plant, the utilization of biofuel, and the acceleration of electric vehicle. These should be a consideration in evaluating and formulating medium and long-term strategy, including evaluation on National Energy Policy (KEN) and National Energy Plan (RUEN) target.

In this opportunity, allow me to convey my gratitude to Members of the National Energy Council, Secretariate General of the National Energy Council, especially for the Editorial Board for the hard work. I hope the book of Indonesia Energy Outlook 2021 would be beneficial for all the stakeholders and energy policy makers.

Jakarta, December 2021

Daily Chairman of National Energy Council

Arifin Tasrif

FOREWORD BY MEMBER OF NATIONAL ENERGY COUNCIL



Let us praise to God the Almighty for the His blessing that the National Energy Council has finished the book of Indonesia Energy Outlook 2021. This book potrays the energy in Indonesia including the current energy condition and energy projection in the future.

Indonesia Energy Outlook 2021 is jointly formulated by National Energy Council and Editorial Board in Secretariate General of the National Energy Council with the assistance from the experts. The formulation also involves units in the Ministry of Energy and Mineral Resources namely the Data and Information Center, Directorate General of Electricity, Directorate General

of Oil and Gas, Directorate General of New Renewable Energy and Energy Conservation, as well as Directorate General of Mineral and Coal especially on the collection of current data and planning data.

The energy supply and demand projection are carried out for the 2022 projection and 2020-2030 projection. The projection of the next 10 years is done through two scenarios namely Realistic scenario which uses the existing policy and Optimistic scenario that uses the target in National Midterm Development Plan (RPJMN) and discussions on National Energy Grand Strategy (GSEN). Next, the projection from the two scenarios will be compared with the projection that has been set in National Energy Plan (RUEN).

In the future, Indonesia Energy Outlook is expected to be presented based on the regional/provincial approach to give a detailed overview on energy supply and demand projection.

Jakarta, December 2021
Advisor

Herman Darnel Ibrahim

FOREWORD BY SECRETARY GENERAL NATIONAL ENERGY COUNCIL



The formulation of Indonesia Energy Outlook 2021 is an annual publication by the Secretariat General of the National Energy Council and has been published since 2014.

The energy supply and demand projection are conducted based on LEAP (Low Emissions Analysis Platform) modelling using 2019 as the baseline year. The data is taken from the Data and Information Center MEMR and data of planning from related ministries and several industrial associations such as Indonesia Cement Association, Indonesia Fertilizer Producer Association, and Indonesia Pulp and Paper Association.

In the effort to improve the quality of energy supply and demand projection data, Secretariate General of Natioal Energy Council would enhance the relationship with related parties to publish the book of Indonesia Energy Outlook as a reliable and trusted reference.

Jakarta, December 2021
Secretary General of the National Energy Council

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- Members of National Energy Council from stakeholders
- Directorate General of Oil and Gas, Directorate General of Electricity, Directorate General of Mineral and Coal, Directorate General of New Renewable Energy and Energy Conservation, Data and Information Center MEMR and PT PLN (Persero)
- Energy experts for their contribution in formulating Indonesia Energy Outlook 2021

DISCLOSURE

Indonesia Energy Outlook (IEO) 2021 is an analysis on the long-term national energy demand and supply projection (2020-2030) with particular assumptions which are developed for the purpose of future energy scenario planning. The assumption and projection are based on energy technology development including fossil and renewable based on current data and condition. The data in IEO is taken from official publication as well as temporary data or data that are continuously updated by the sources.

EXECUTIVE SUMMARY

Indonesia Energy Outlook (IEO) 2021 presents national energy demand and supply projection in 2020-2030 based on social assumption, economy and technology development in the future by using 2019 as baseline year.

The energy demand and supply analysis is conducted based on LEAP (Low Emissions Analysis Platform) calculation model. LEAP is an energy planning modelling application to take an integrated energy demand and supply analysis.

Population growth and economic growth are the basic assumptions which are developed to obtain an illustration on energy demand up to the year 2030. Furthermore, there are also additional assumptions related to current energy policy such as biofuel mandatory, EV development, green fuel refinery development, and electricity development in RUPTL (Electricity Supply Business Plan).

In 2022, final energy consumption will increase into 129 MTOE compared to the consumption in 2020. The final energy consumption in household sector in 2022 will decline from 16.7% in 2020 into 14.7% in 2022 along with the declining Covid-19 cases. Electricity consumption in 2022 is projected to reach 282 TWh. It increases in all sectors except in household.

Electricity demand in 2030 in Optimistic scenario especially in transportation reaches 16.9 TWh. It is higher compared to the Realistic scenario with 1.08 TWh. Meanwhile, LPG demand in household in 2030 for Optimistic scenario is only 2.2 MTOE. It is lower than in Realistic scenario of 8.4 MTOE. The power plant capacity in 2030 is 109 GW for Optimistic scenario and 90 GW for Realistic scenario.

LIST OF CONTENTS

REMARK BY CHAIRMAN OF NATIONAL ENERGY COUNCIL	i
FOREWORD BY MEMBER OF NATIONAL ENERGY COUNCIL	ii
FOREWORD BY SECRETARY GENERAL	iii
NATIONAL ENERGY COUNCIL	iii
EDITORIAL BOARD	iv
DISCLOSURE	v
EXECUTIVE SUMMARY	vi
LIST OF CONTENTS	vii
LIST OF FIGURES	x
LIST OF TABLES	xiii
CHAPTER I INTRODUCTION	1
1.1 Background	1
1.2 Methodology	1
1.2.1 Modeling Analysis Framework	1
1.2.2 Energy Projection Scenario	3
1.2.3 Energy-related Policy Assumption	4
1.3 Current Energy Condition	7
1.3.1 Oil	7
1.3.2 Gas	9
1.3.3 Coal	11
1.3.4 Renewable Energy	12
1.3.5 New Energy	14
CHAPTER II NATIONAL ENERGY CONSUMPTION UNTIL 2020	17
2.1 Final Energy Consumption per Sector	17
2.2 Electricity Consumption per Sector	17
2.3 Primary Energy Consumption	18
2.4 Fuel and LPG Consumption per Province	21
2.5 Energy Elasticity and Intensity	25
2.6 Energy Price	26
2.6.1 Electricity Price	27
2.6.2 Fuel Price	28
2.6.3 Natural Gas Price	28
2.6.4 LPG Price	29
2.6.5 Coal Price	30
2.7 Emission	31
2.8 Energy Infrastructure	31
2.8.1 Oil Refinery	31
2.8.2 LPG Refinery	32
2.8.3 LNG Refinery	34

2.8.4 City Gas Network	34
2.8.5 Power Plant	35
CHAPTER III 2022 OUTLOOK	39
3.1 Final Energy Consumption	39
3.1.1 Household Sector	41
3.1.2 Transportation Sector	43
3.1.3 Industrial Sector	46
3.1.4 Commercial Sector	47
3.1.5 Other Sectors	48
3.2 Electricity Consumption per Sector	49
3.3 Electricity Consumption per Province	50
3.3.1 Power Plant Capacity by Type 2020-2022	53
3.3.2 Additional Power Plant Capacity in 2021 and 2022	54
3.3.3 Electricity Production by Type in 2020-2022	59
3.4 Primary Energy Supply in 2020-2022	60
3.4.1 Primary Energy Mix	61
3.5 Electricity Consumption per Capita	61
3.5.1 Primary Energy Per Capita	62
3.6 Emission	62
3.7 Energy Balance 2022	64
3.8 Energy Policy	65
3.8.1 Oil and Gas	66
3.8.2 Coal	67
3.8.3 Renewable Energy	68
CHAPTER IV ENERGY OUTLOOK PROJECTION 2021-2030	70
4.1 Comparison of Final Energy Consumption per Sector	70
4.1.1 Industrial Sector	71
4.1.2 Transportation Sector	73
4.1.3 Household Sector	76
4.1.4 Commercial Sector	76
4.1.5 Other Sectors	77
4.2 Electricity Consumption Projection per Sector	78
4.3 Electricity Consumption per Capita	79
4.4 Electricity Projection	80
4.4.1 Electricity Production	80
4.4.2 Power Plant Capacity	81
4.5 Primary Energy Supply Projection per Type	83
4.6 Comparison of Primary Energy Mix in Projection and RUEN	85
4.7 Projection of Carbon Emission per Consumer Sector	86

CHAPTER 5 NATIONAL ENERGY RESILIENCE AND PROGRAMS	
FOR SUSTAINABLE ENERGY DEVELOPMENT	70
5.1 National Energy Resilience Condition	89
5.2 Electricity for Rural Areas and Energy Supply for Outermost, Remote and Disadvantaged Areas	93
5.3 City Gas	96
5.4 Promoting Induction Stove as Fossil Fuel Substitution	97
5.5 The Development of Electric Vehicle	99
5.6 Carbon Tax and Emission Intensity Reduction	100
5.7 Preparing Energy Transition	104

LIST OF FIGURES

Figure 1.1	Modeling Analysis Framework	2
Figure 1.2	Oil Reserves (Proven)	7
Figure 1.3	Crude Oil Production, Export and Import	8
Figure 1.4	Fuel Import and Export	9
Figure 1.5	Gas Reserves	9
Figure 1.6	Gas Production and Export	10
Figure 1.7	LPG Production and Import	11
Figure 1.8	Coal Reserves	11
Figure 1.9	Coal Production, Export and Import	12
Figure 1.10	Biodiesel Production and Export	14
Figure 1.11	Distribution Potential of Uranium Reserves	15
Figure 2.1	Final Energy Consumption by Sector	17
Figure 2.2	Electricity Consumption by Sector	18
Figure 2.3	Primary Energy Consumption by Type of Energy	19
Figure 2.4	Primary Energy Mix	20
Figure 2.5	Comparison of Primary Energy Supply Realization and RUEN	20
Figure 2.6	NRE Primary Energy Mix	21
Figure 2.7	Fuel Consumption 2011-2020	21
Figure 2.8	LPG Consumption in 2011-2020	23
Figure 2.9	Indonesia Energy Elasticity 2011-2020	25
Figure 2.10	Primary Energy Intensity 2011-2020	26
Figure 2.11	Trend of Fuel Price	28
Figure 2.12	Trend of Gas Price	29
Figure 2.13	Trend of LPG Price	30
Figure 2.14	Trend of Coal Price	30
Figure 2.15	CO2 Emission from Fuel Combustion	31
Figure 2.16	Gas Network Infrastructure by Province	35
Figure 2.17	Power Plant Installed Capacity by Type of Energy 2011-2020	36
Figure 2.18	NRE Based-Power Plant Installed Capacity 2011-2020	37
Figure 2.19	Power Plant Capacity by Province 2020	38
Figure 3.1	Final Energy Consumption by Sector	39
Figure 3.2	Comparison of Final Energy Consumption Share by Sector	40
Figure 3.3	Final Energy Consumption by Type of Energy	40
Figure 3.4	Energy Consumption for Cooking	42
Figure 3.5	Energy Consumption for Non-cooking	43
Figure 3.6	Final Energy Consumption by Household	43
Figure 3.7	Energy Consumption by Transportation Mode	44
Figure 3.8	Portion of Gasoline and Diesel Oil Consumption by Transportation Mode	45
Figure 3.9	Final Energy Consumption in Transportation Sector	46
Figure 3.10	Final Energy Consumption in Industrial Sector	47

Figure 3.11	Final Energy Consumption in Commercial Sector	48
Figure 3.12	Final Energy Consumption in Other Sectors	49
Figure 3.13	Electricity Consumption by Sector	50
Figure 3.14	Electricity Consumption by Province in 2020	51
Figure 3.15	Additional Coal Power Plant	55
Figure 3.16	Additional Power Plant Capacity	56
Figure 3.17	Additional Power Plant Capacity in 2022	59
Figure 3.18	Primary Energy Supply in 2020-2022	60
Figure 3.19	Primary Energy Mix in 2022	61
Figure 3.20	Electricity Consumption per Capita	61
Figure 3.21	Primary Energy per Capita	62
Figure 3.22	Emission per Sector in 2020 - 2022	63
Figure 3.23	Emission per Capita in 2020 - 2022	63
Figure 3.24	Energy Flow in 2022	65
Figure 4.1	Comparison of Final Energy Demand in Both Scenarios in 2030	70
Figure 4.2	Share of Final Energy Consumption by Type of Energy in 2020	71
Figure 4.3	Comparison of Final Energy Demand in 2030	71
Figure 4.4	Final Energy Demand in Industrial Sector by Type of Energy	72
Figure 4.5	Share of Energy Demand sub sector Industries in Realistic Scenario in 2030	73
Figure 4.6	Energy Demand in Transportation Sector in Both Scenarios	74
Figure 4.7	Final Energy Consumption by Car in 2030	75
Figure 4.8	Final Energy Consumption by Motorcycle in 2030	75
Figure 4.9	Energy Demand in Household Sector per Type of Energy	76
Pict ure 4.10	Energy Demand in Commercial Sector by Type of Energy	77
Figure 4.11	Energy Demand in Other Sectors by Type of Energy	78
Figure 4.12	Electricity Demand by Sector in 2030	79
Figure 4.13	Electricity Consumption per Capita	79
Figure 4.14	Electricity Production per Type of Energy	80
Figure 4.15	Electricity Production from NRE Power Plant	81
Figure 4.16	Power Plant Capacity per Scenario	82
Figure 4.17	NRE Power Plant Capacity per Scenario	83
Figure 4.18	Primary Energy Mix in both Scenarios in 2025	84

Figure 4.19	Primary Energy Mix in Both Scenarios in 2030	84
Figure 4.20	Comparison of Primary Energy Mix in Both Scenarios and in RUEN in 2025	85
Figure 4.21	Comparison of NRE Mix in both Scenarios and in RUEN	86
Figure 4.22	Carbon Emission in Both Scenarios	87
Figure 4.23	GHG Emission per Capita	88
Figure 5.1	Energy Resilience Model	90
Figure 5.2	Energy Resilience Level	92
Figure 5.3	Eletrified Village Distribution in 2020	94
Figure 5.4	One price fuel distribution until 2020	95
Figure 5.5	Map of Gas Infrastructure Development	96
Figure 5.6	GSEN Program on LPG Import Reduction	99
Figure 5.7	GSEN Program to Stop Fuel Import	100

LIST OF TABLES

Table 1.1 Scenario Assumption	4
Table 1.2 Renewable Energy Reserves	13
Table 2.1 Fuel Consumption per Province in 2011-2020	22
Table 2.2 LPG Consumption by Province in 2011-2020	24
Table 2.3 Energy Price	27
Table 2.4 Average Electricity Price by Costumer Group	27
Table 2.5 Indonesia Oil Refinery Capacity 2020	31
Table 2.6 Indonesia LPG Refinery Capacity 2020	33
Table 2.7 Indonesia LNG Refinery Capacity 2020	34
Table 3.1 Electricity Consumption per Province in 2021 - 2022	52
Table 3.2 Power Plant Capacity by Type 2020 - 2022	53
Table 3.3 Additional Power Plant Capacity 2021 - 2022	54
Table 3.4 Electricity Production by Type 2020-2022	59
Table 3.5 Energy Indicator	64
Table 4.1 Energy Indicator per Scenario	88
Table 5.1 Level in Energy Resilience Assesment	91
Table 5.2 Details of Electricity Cosnumner in Java-Bali	98
Table 5.3 Cap in carbon trade test in 2021	103

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INTRODUCTION

1

INTRODUCTION

1.1 BACKGROUND

Indonesia Energy Outlook (IEO), which is published annually, is a result of an analysis to give an overview on national energy condition especially on energy supply and demand projection until 2030. IEO 2021 presents the energy supply demand projection 2020-2022 and 2020-2030. The 2020-2030 projection uses two scenarios namely Realistic scenario and Optimistic scenario. Realistic scenario uses the assumption based on the current condition while Optimistic scenario uses the assumption based on National Midterm Development Plan (RPJMN) and discussions on National Energy Grand Strategy (GSEN). In addition, calculation assumption also refers to energy related policy that has been determined by the government such as National Energy Policy (KEN), National Energy Plan (RUEN), Electricity Supply Business Plan (RUPTL), Nationally Determined Contributions (NDC), Strategic Planning (Renstra) MEMR and Ministry of Industry, as well as biofuel mandatory.

The main data source in IEO 2021 is Handbook of Energy and Economic Statistics of Indonesia (HEESI) 2019 and 2020, RUPTL PLN 2021-2030, Indonesia Statistics 2020, BPS and energy consumption data from industrial associations such as Indonesian Fertilizer Producers Association (APPI), Indonesian Cement Association (ASI), and Indonesian Ceramic Industry Association (ASAKI).

1.2 METHODOLOGY

1.2.1 Modeling Analysis Framework

Modeling analysis is divided into three stages namely final energy demand analysis, energy transformation, and primary energy supply. Final energy demand analysis is done by using GDP growth assumption and population growth. It also takes into account the policy, strategic planning and roadmap on current energy development. The primary energy supply analysis also takes into account the use of various energy sources and energy resources potential including the current policies as current energy technology development. Meanwhile, energy transformation analysis is done by considering RUPTL, energy infrastructure development plan, and emission reduction. The modeling analysis framework is shown in Figure 1.1.

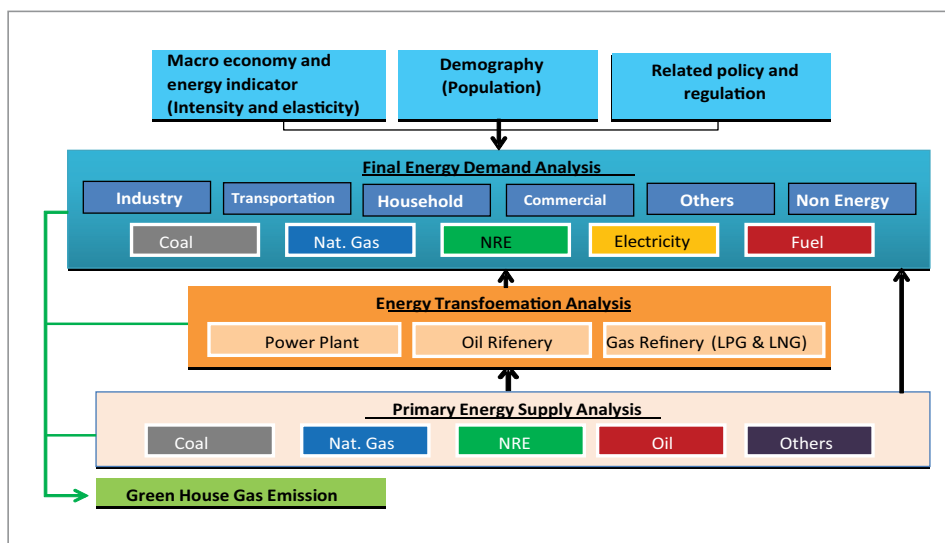


Figure 1.1 Modeling Analysis Framework

The energy supply and demand analysis is conducted based on calculation of LEAP model as a simulation model in energy planning which can conduct an integrated energy supply and demand analysis. In LEAP model, energy demand projection is calculated based on the multiplication of energy consumption activity and energy consumption intensity. Energy activity is described by economic growth, population and production. Meanwhile, energy intensity is the level of energy consumption per GDP or per population or per production in particular period. Energy intensity can be considered as fixed during simulation period or declining to show energy efficiency increase.

Based on the analysis framework in Figure 1.1, the parameters in final energy demand projection are social economic data including population and economic growth, historical data on energy consumption to find out energy intensity and energy consumption pattern as the impact of people's lifestyle that is influenced by the projection of GDP increase or a more efficient technology. The base year data uses 2019 energy data, and 2020 data to find out the trend of energy supply and demand movements during the pandemic.

1.2.2 ENERGY PROJECTION SCENARIO

1.2.2.1 Realistic Scenario (BaU)

Statistics Indonesia shows that Indonesia's economic growth is 5.02% in 2019 and 0.27% in 2020. Based on State Budget 2022, Indonesia's economic growth is targeted to reach 5.2% in 2020.

Realistic scenario refers to current condition and the future projection is assumed based on historical data in the last several years such as additional city gas, electric stove, electric vehicle, and others. The power plant development refers to RUPTL 2021-2030 with construction status only.

1.2.2.2 Optimistic Scenario

Optimistic scenario uses the assumptions in RPJMN and GSEN. The assumption on population and economic growth is similar to the previous scenario. The use of EV, gas fueled vehicle, electric stove, city gas and DME is a quite different assumption compared to Realistic scenario. In addition, the increase of oil and gas production as well as refinery capacity enhancement will become assumptions to be considered in the modeling. Meanwhile, the power plant development also refers to RUPTL 2021-2030 with the status of the plan until construction.

The detail of assumptions in both scenarios above is shown in Table 1.1.

Table 1.1 Scenario Assumption

INDICATOR	Unit	Realistic (BaU)	Optimistic
Demography & Economy			
Population Growth (average)	%	1	1
Economic growth (average)	%	5	5
Demand			
Biodiesel Target 2030	%	30	30
Biogasoline Target 2030	Million KL	1	4.4
Electric Car 2030	Unit	2 Thousand	3 Million
Electric Motorcycle 2030	Unit	12 Thousand	14 Million
City Gas	Household/Year	100 Thousand	1 Million
Induction stove	Household/Year	100 Thousand	1 Million
DME 2030	Million Ton	1.4	3.04
Supply			
Coal Production 2030	Million Ton	600	680
Crude Oil Production 2030	Million BOPD	0.5	1
Gas Production 2030	MMSCFD	4,800	12,000
Electricity			
Power Plant Capacity		RUPTL 2021 (Construction Stage)	RUPTL 2021

1.2.3 Energy-related Policy Assumption

In order to support energy demand projection, several energy-related policies are being considered, including:

1. National Energy Policy

National Energy Policy (KEN) mandates renewable energy mixed target in primary energy mix of at least 23% in 2025 and minimize the use of oil of less than 25% in 2025. Furthermore, energy efficiency is also targeted to decline 1% per year to promote energy saving in all sectors. Several targets in KEN also becomes the consideration in energy demand projection such as optimization of gas for domestic and coal downstreaming.

2. National Energy Plan

National Energy Plan (RUEN) is a mandate of Law No. 30 of 2007 on Energy. Based on Article 17 paragraph (1) of Law No. 30, the government formulates the draft of RUEN based on KEN. The targets in RUEN that are taken into consideration in energy demand projection are LPG to city gas substitution, DME utilization, biofuel utilization, and EV utilization.

3. MEMR Strategic Planning

Several strategic planning in MEMR taken into consideration in calculating the energy demand projection is the development of city gas, gas for public vehicle and biofuel utilization.

4. Strategic Planning of the Ministry of Transportation

There are a number of programs in the Strategic Planning of the Ministry of Transportation which become considerations in the energy demand projection such as the operation plan of BRT (Bus Rapid Transit), Mass rapid Transit (MRT), and Light Rail Transit (LRT). In addition, there is also plan to use biofuel for land transportation, gas fueled vehicle, and electric bus in DKI Jakarta.

5. Strategic Planning of the Ministry of Industry

One of the future planning data in industrial sector is the roadmap of electric vehicle industry, both two wheeled or four wheeled or more wheeled vehicles.

6. RUPTL 2021-2030

The data on capacity of planned power plant refers to RUPTL 2021-2030. In Realistic scenario, the power plant development refers to power plan in construction stage with the consideration on power plant operation based on Commercial Operation Date (COD) planning. For Optimistic scenario, additional power plant uses the data of power plan development plan from planning to construction status.

7. Biofuel Roadmap

Biofuel mandatory is taken into consideration as one of the assumptions in final energy demand projection in transportation, industry, commercial and power plant sector.

8. Special Industry

In energy demand projection analysis, industry specifically is divided into two types based on data availability. They are special industry and other industry. Special industry for this outlook analysis consists of energy consuming industry such as fertilizer industry, cement and ceramic industry. In order to count energy demand projection in special industry, energy consumption intensity data from a number of industrial associations is used. The associations are Indonesia Fertilizer Producers Association, Indonesian Cement Association, and Indonesian Ceramic Association. Beside that, production growth in each industry is also a reference in the energy demand projection analysis.

Meanwhile, other industry includes food and beverage industry, textile, wood, metal, non metal, engine, and other industries which use GDP per industry type approach in calculating the energy intensity.

9. National Energy Grand Strategy 2020-2040

In the end of 2020, National Energy Council and MEMR prepared a National Energy Grand Strategy (GSEN) until 2040 as an energy supply demand projection in the future as well as strategy. In principle, GSEN's mission is to improve Indonesia trade balance by reducing dependency on import especially on kerosene and LPG by increasing oil and gas production, enhancing oil refinery, substituting kerosene with electric vehicle, gas and biofuel as well as substituting LPG with DME, induction stove and city gas.

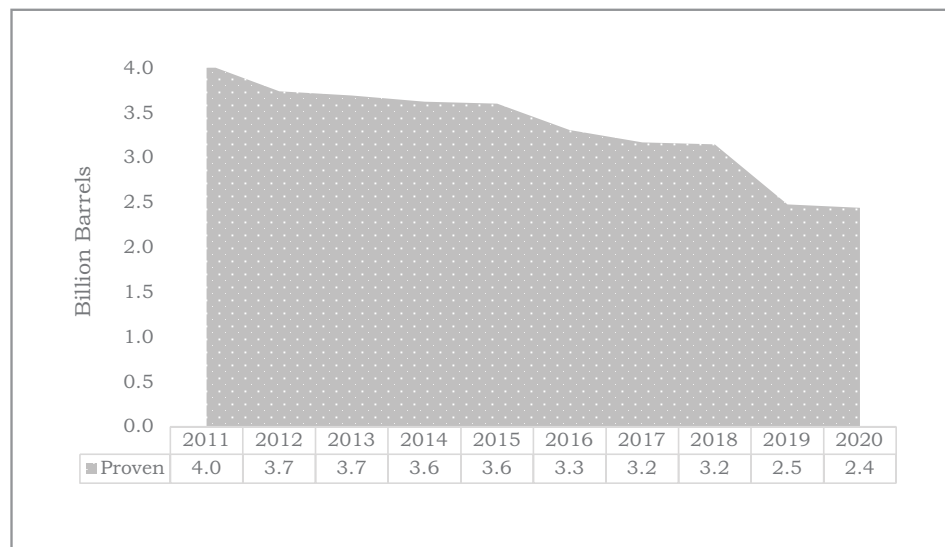
1.3 CURRENT ENERGY CONDITION

In 2020, total primary energy production consisting of oil, gas, coal and renewable energy reaches 450.6 MTOE. Around 57.8% or 260.5 MTOE from the total production is exported, especially coal and LNG. However, Indonesia is also importing energy especially crude oil and fuel products amounting to 40.3 MTOE and small amount of high rank coal to meet the demand of steel industry.

The total final energy consumption in 2020 is around 118.3 MTOE from transportation (43.1%), industry (34.1%), household (16.8%), commercial sector (4.8%) and other sectors (1.2%).

1.3.1 Oil

Based on BP Statistic Review 2020 data, the number of Indonesia oil reserves is only 0.1% from the world reserves. Indonesia oil reserves are declining from 7.3 Billion Barrels in 2015 (3.7 Billion Barrels of potential reserves and 3.6 Billion Barrels of proven reserves) into 4.2 Billion Barrels (1.7 Billion Barrels of potential reserves and 2.4 Billion Barrels of proven reserves) in 2020 (Figure 1.2).



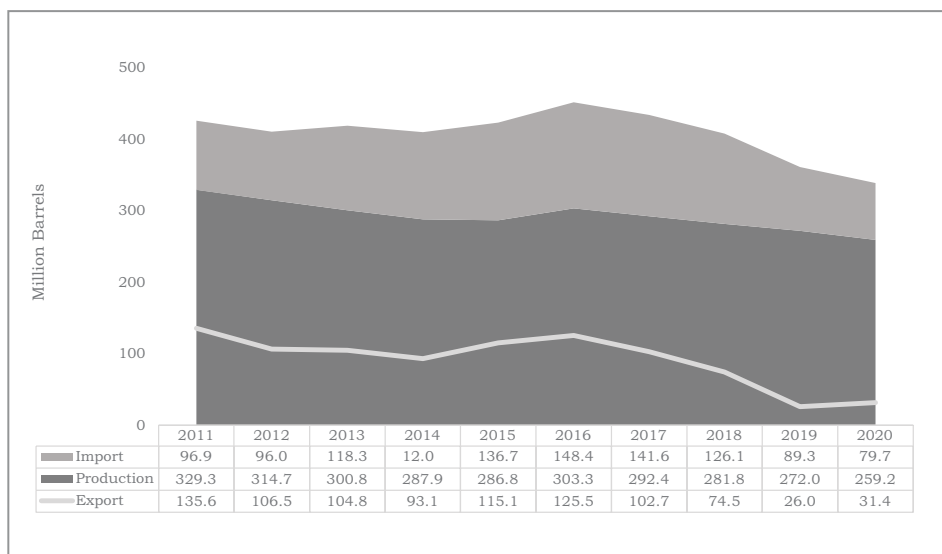
Source: HEESI, 2020

Figure 1.2 Oil Reserves (Proven)

Oil production in the last 10 years shows the declining trend, from 329.3 Million Barrels (902 thousand bph) in 2011 into 259.2 Million Barrels (710.3 thousand bph) in 2020. The declining production occurs since the oil producing wells are mainly mature wells, while new producing wells are relatively limited.

The declining production and the increasing refinery input demand cause Indonesia to import oil especially from Middle East. The import tends to decline from 96.9 Million Barrel in 2011 into 79.9 Million Barrels in 2020 that is influenced by the fluctuating crude oil demand for refinery input. The crude oil import for refinery input in 2011-2018 is around 30-40%, but it declines into 15-20% in 2019-2020 (Figure 1.3).

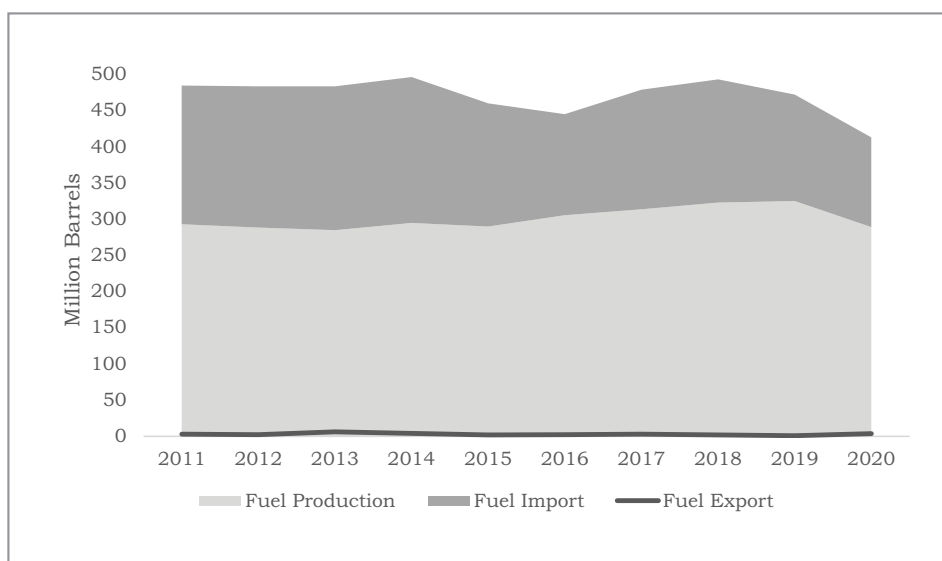
On the other hand, Indonesia oil export continues to decline from 135.6 Million Barrels in 2011 into 31.4 Million Barrels in 2020. This is based on the Minister of Energy and Mineral Resources Regulation No. 42 of 2018 on the Priority of Crude Oil Utilization for Domestic Need. It means that domestic oil production is used maximally for the domestic need. The trend of oil production, export and import since 2011 to 2020 can be seen in Figure 1.3.



Source: HEESI, 2020

Figure 1.3 Crude Oil Production, Export and Import

In demand side, fuel demand including domestic biodiesel in 2020 reaches 413 Million Barrels/year that is fulfilled from the domestic refinery production and import. Fuel production from domestic refinery in a year is approximately around 300.7 Million Barrels and the import is around 170.3 Million Barrels. The trend of fuel production, export, and import in the last 10 years can be seen in Figure 1.4.

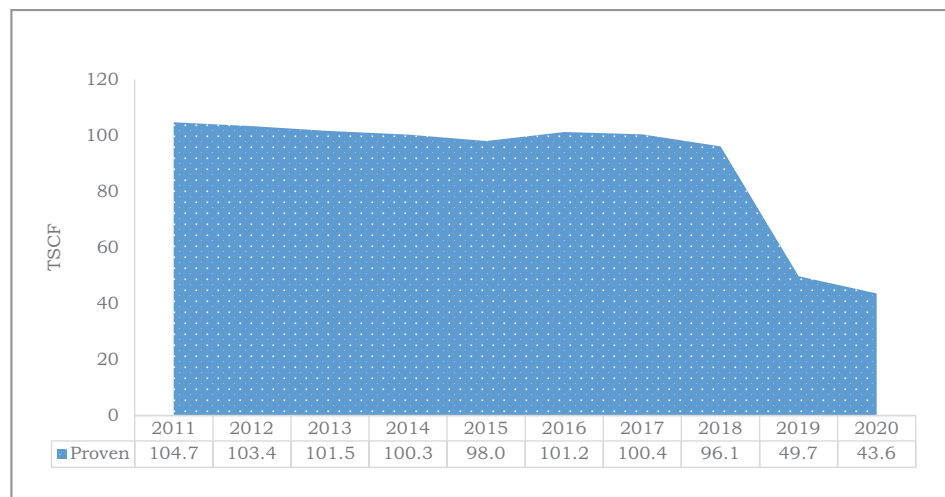


Source: HEESI, 2020

Figure 1.4 Fuel Import and Export

1.3.2 Gas

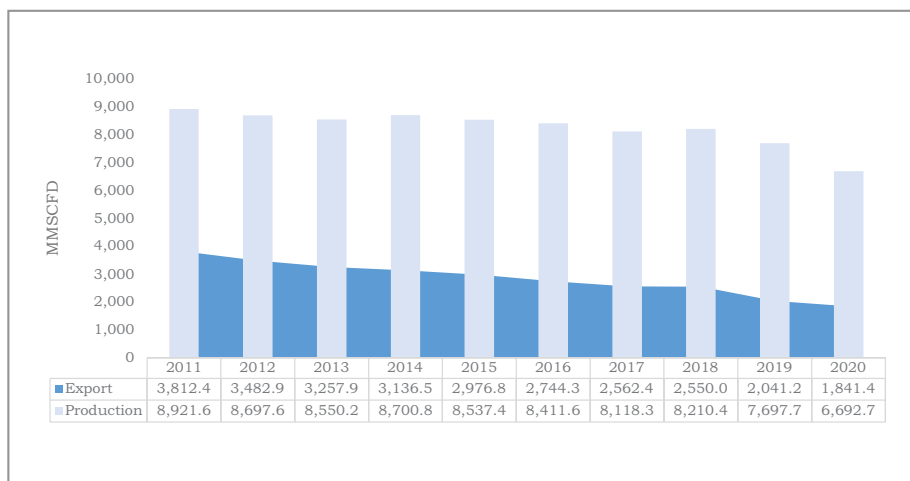
Based on HEESI, Indonesia's total proven gas reserves in 2020 is 43.6 TSCF. It declines from 104.7 TSCF in 2011 as shown in Figure 1.5.



Source: HEESI, 2020

Figure 1.5 Gas Reserves

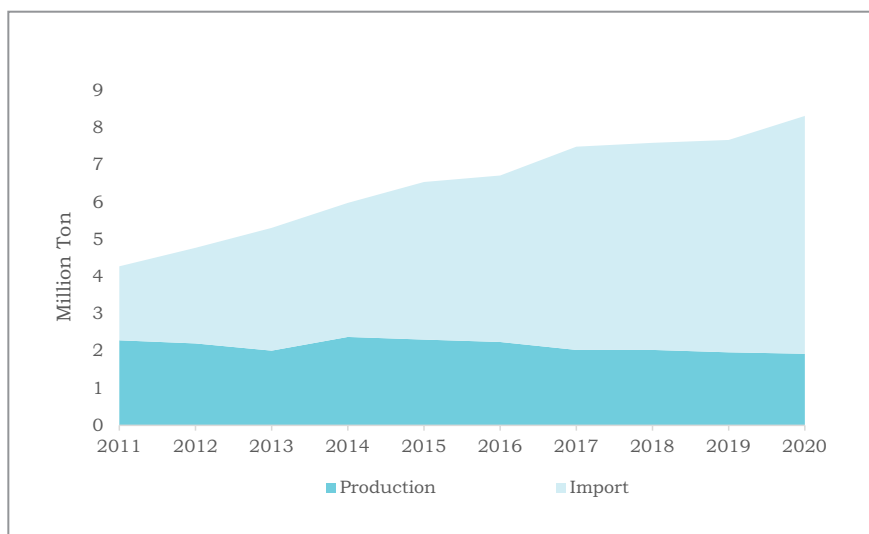
In 2020, Indonesia's total gas production is around 6,692.7 MMSCFD that is used mainly to meet the domestic consumption in industrial sector such as feed stock in petrochemical industry or energy, power plant and city gas (household and commercial), also gas for transportation. Gas production is declining from around 8,921.6 MMSCFD in 2011. One of them is influenced by the limited reserves in Arun. Furthermore, the activity of gas production in Tangguh field (Train 3) is not yet started. In addition, gas has become the export commodity in the form of LNG and piped gas of around 1,841.4 MMSCFD in 2020 but the gas export is declining due to the program in prioritizing gas for domestic need. Gas for domestic is increasing from 57.3% in 2011 into 72.5%. The export percentage through pipeline and LNG toward the total gas production is declining from 42.7% in 2011 into 27.5% in 2020 (Figure 1.6).



Source: HEESI, 2020

Figure 1.6 Gas Production and Export

Gas primary energy including LPG is derived from LPG refinery production and LPG import. In 2020, LPG consumption is 8.3 Million Ton consisting of 1.9 Million Ton (23.1%) from domestic LPG production and 6.4 Million Ton (76.9%) from import. LPG consumption is increasing up to 8 Million Ton in 2020. The success of kerosene to LPG conversion program leads to the increasing LPG consumption, while LPG supply from domestic (LPG refinery and oil refinery) is limited. LPG import increases four times since 2011 due to the increasing LPG. The increasing LPG consumption especially the subsidized 3 kg LPG needs to be anticipated by the government since the 3 kg LPG has not met the target. The trend of LPG production and import is shown in Figure 1.7.

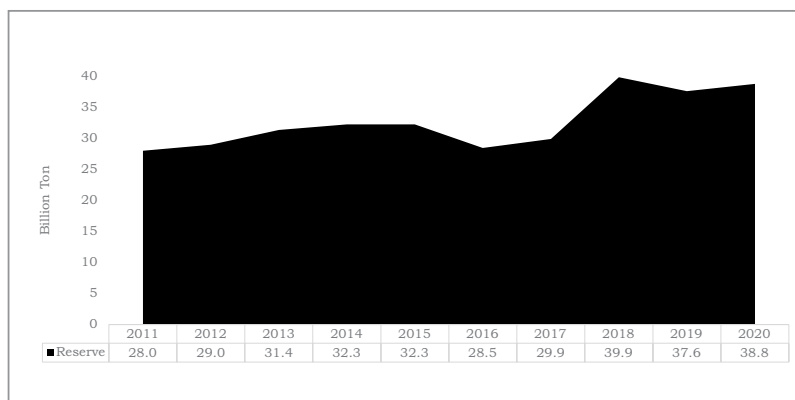


Source: HEESI, 2020

Figure 1.7 LPG Production and Import

1.3.3 Coal

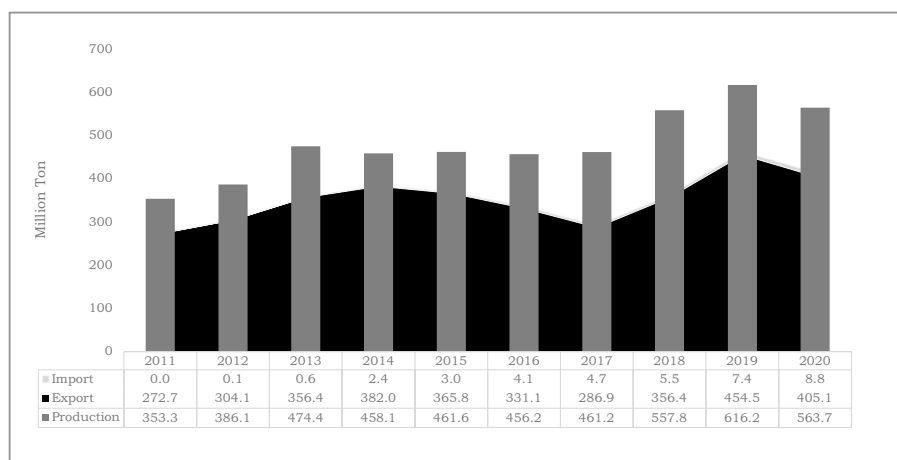
Indonesia's total coal reserves in 2020 are 38.8 Billion Ton. It increased since 2011 despite of the increasing production (Figure 1.8). It is due to the increasing exploration in the last 10 years.



Source: HEESI, 2020

Figure 1.8 Coal Reserves

Coal production in 2011-2020 increased significantly from 353 Million Ton into 563.7 Million Ton in 2020. However, the declining demand in industrial sector due to Covid-19 pandemic has given an impact to the declining coal production of 52.4 Million Tonnes from 616.2 Million Ton in 2019. Coal export is around 72% (405.1 Million Ton) from the total production. It is mostly exported to China and India. The high percentage of Indonesia coal export has made Indonesia one of the biggest coal exporters in the world besides Australia. The domestic coal is used to supply Coal-fired Power Plant as well as cement, textile, pulp and paper, and food industry. Indonesia is also importing high rank coal to supply the steel industry. The coal production, export and import in the last 10 years is shown in Figure 1.9.



Source: HEESI, 2020

Figure 1.9 Coal Production, Export and Import

1.3.4 Renewable Energy

The declining fossil energy production especially oil and gas as well as the global commitment in reducing emission have encouraged the government to continuously enhance the role of new and renewable energy to maintain energy security and independence. Based on Government Regulation No. 79 of 2014 on National Energy Policy, the new and renewable energy mix target is at least 23% by 2025 and 31% by 2050. Indonesia has quite abundant new renewable energy potential to meet the primary energy mix target as seen in Table 1.2 below.

Table 1.2 Renewable Energy Reserves

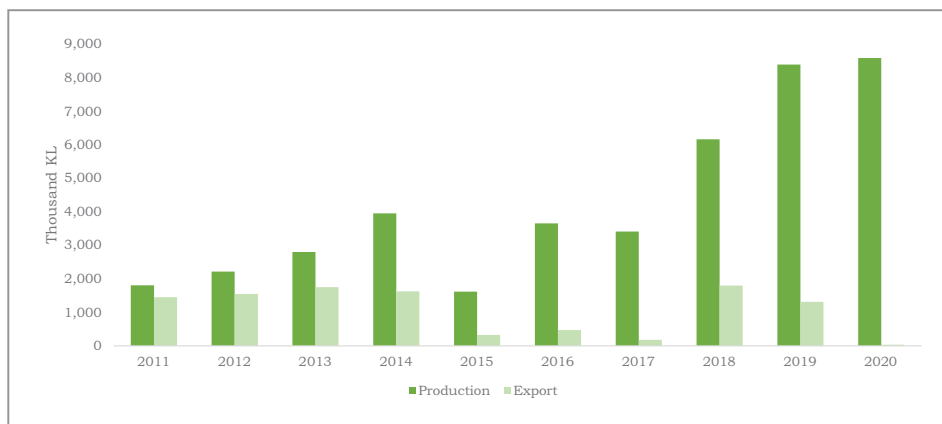
Types of Energy	Potential (GW)	Realization (GW)	Utilization
Geothermal	23.8	2.1	9.0%
Solar	207.8	0.2	0.1%
Wind	60.7	0.2	0.3%
Water	94.6	6.1	6.5%
Bioenergy	32.7	1.9	5.8%
Tidal	17.9	-	-

Source: HEESI, 2020 and DG of NREEC, 2020

The total renewable energy potential for power plant is 437 GW, but only 2.4% is utilized. Meanwhile, the utilization of biodiesel in 2020 reached 8.6 million KL as the biodiesel mixture. The utilization of new renewable energy for power plant in 2020 is 10.5 GW or 14.4% from the total power plant capacity.

The low NRE utilization for electricity generation is due to the relatively high cost of NRE-based power plant. Thus, it is difficult to compete with fossil power plant especially coal. In addition, the lack of support from domestic industry in NRE component and the issue in low interest funding have contributed to the slow development of renewable energy.

The utilization of NRE especially biodiesel in transportation has been increasing significantly along with the mandatory of biofuel which mandates the 30% mixture of biofuel and fuel (B30) in all sectors in 2020. It also leads to the declining biodiesel export along with the optimization of biodiesel for domestic (B30). The trend of biodiesel production, export and utilization is shown in Figure 1.10.



Source: HEESI, 2020

Figure 1.10 Biodiesel Production and Export

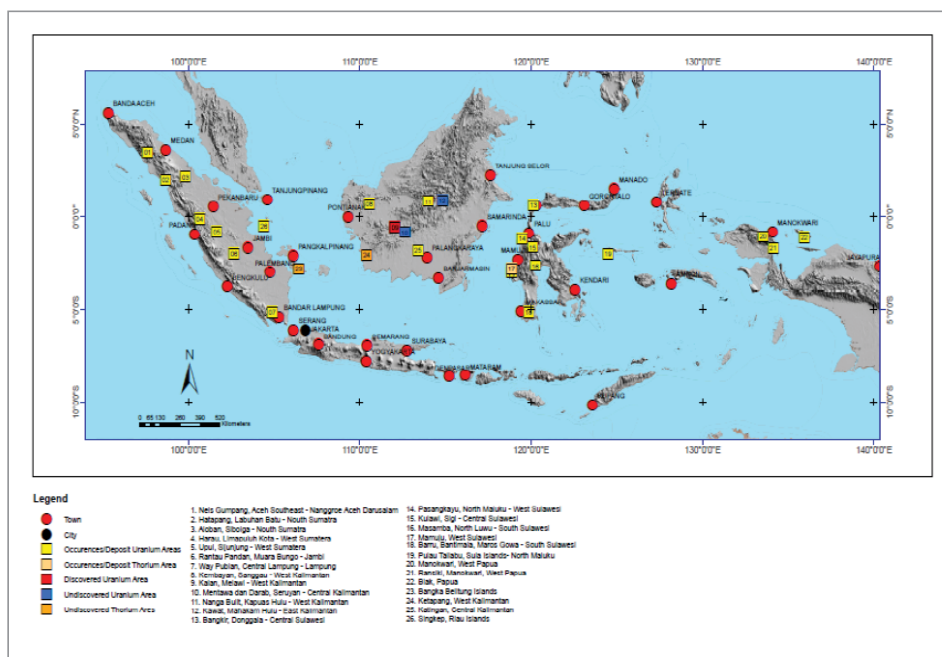
1.3.5 New Energy

The definition of new energy is based on the general provision of Law No. 30 of 2007 that energy from new energy sources may be derived from new technology, both from renewable energy sources or non-renewable energy sources. Several energy types that are classified into new energy are nuclear, hydrogen, coal bed methane, liquified coal, and gasified coal but until now these energy types have not been developed maximally. In RUEN, type of energy that is classified into new energy is only gasified coal. It already has the capacity development target, while other types of energy have not determined the development target. RUEN sets the target to develop gasified coal power plant with the capacity of 44 MW until 2025. The issues on price and technology have become the challenges in developing new energy.

a. Nuclear

Based on data published by BATAN in 2019, Indonesia has uranium and thorium reserves that can be used as fuel for nuclear power plants of 0.08 Million Tons and 0.14 Million Tons, respectively, most of which are in West Kalimantan, Papua, Bangka Belitung and West Sulawesi. Based on the last study done in Mamuju, West Sulawesi, and preliminary detection, uranium content in those locations is between 100 to 1,500 ppm (mg/kg). A study by BATAN also says that energy development potential for generating electricity from 1 gram of uranium is equivalent to energy derived from 112 kg of coal. Until now, Indonesia has three nuclear reactors namely Triga Mark Reactor in Bandung, Kartini Reactor in Yogyakarta, and Serbaguna Reactor in Serpong. These three reactors are mostly used for research and development study by national nuclear scientists. Nuclear energy development in Indonesia has been going for years. It is seen from the feasibility study and a number of scientific studies by the national scientists. However, utilizing nuclear as a source of energy, in this case

the nuclear power plant, needs careful considerations in terms of its characteristic, generation system reliability, environmental to socio politics issue. The distribution of uranium reserves can be seen in Figure 1.11.



Source: Jurnal Pengembangan Energi Nuklir, BATAN

Figure 1.11 Distribution Potential of Uranium Reserves

b. Coal Bed Methane

The abundant coal reserves and resources in Indonesia contain another energy source potential including coal bed methane (CBM). CBM that is formed together with coal seams underground can be used as an alternative gas energy source. Based on Research and Development Agency MEMR data, Indonesia CBM resources are 453 Trillion Cubic Feet (TCF) spread in 11 coal and oil and gas basins. One of them is South Sumatera Basin with 180 TCF resources but until today, the potential is not yet used massively.

c. Coal Gasification

The government through the Ministry of Energy and Mineral Resources planned to develop Coal Gasification in the form of Dimethyl Ether (DME) as an alternative substitution to LPG for household. DME is planned to be developed from the abundant resources of domestic coal. The Presidential Regulation on RUEN mandates the development of 1 Million Ton DME in 2025. Furthermore, National Energy Grand Strategy (GSEN) has targeted DME development from coal to stop LPG import by

2030 from State Owned Enterprise IUP and extension of PKP2B. There are three business entities that have been planning to develop DME. They are PT Bukit Asam with production plan of 1.4 million tonnes per year, PT KPC with production plan of 1.2 million tonnes per year and PT Arutmin with production plan of 2 million tonnes per year. They are going to start the production in 2025.

In addition, three coal business entities located in East Kalimantan and North Kalimantan plan to develop Underground Coal Gasification but it is still in pilot project stage. These business entities are PT Kideco Jaya Agung in East Kalimantan, PT Indominco in East Kalimantan, and PT Medco Energi Mining International (MEMI) and Phoenix Energi Ltd in North Kalimantan.



2

NATIONAL ENERGY
CONSUMPTION
UNTIL 2020

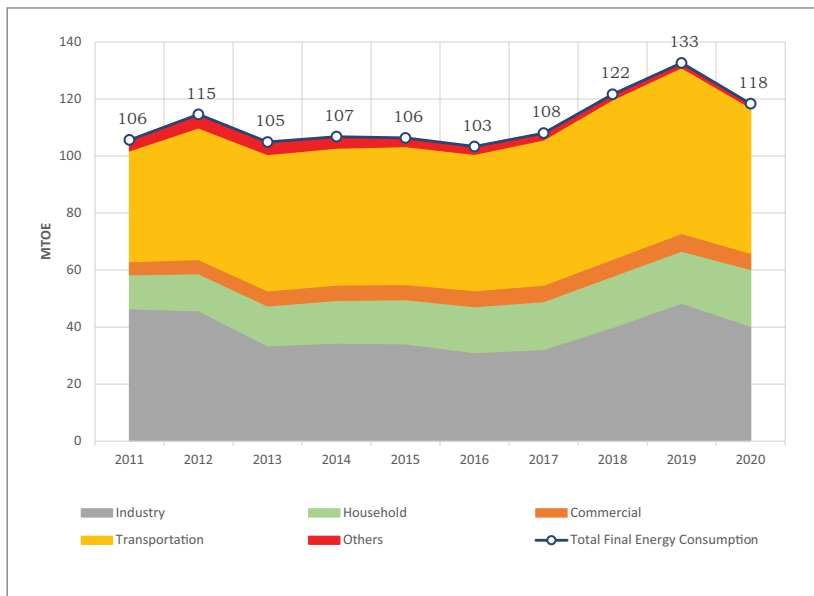
2

NATIONAL ENERGY CONSUMPTION UNTIL 2020

2.1 FINAL ENERGY CONSUMPTION BY SECTOR

The social activity restriction due to Covid19 pandemic has become one of the main factors in the declining total energy consumption. The total final energy consumption in 2020 was 118.3 million TOE from 132.6 million TOE in 2019.

Based on energy consuming sectors, the national final energy demand is still dominated by transportation and industry. Transportation becomes the biggest energy consuming sector with 51 million TOE (43.1%) followed by industry with 40.3 million TOE (34.1%). The rest consuming sectors are household, commercial and others as seen in Figure 2.1.



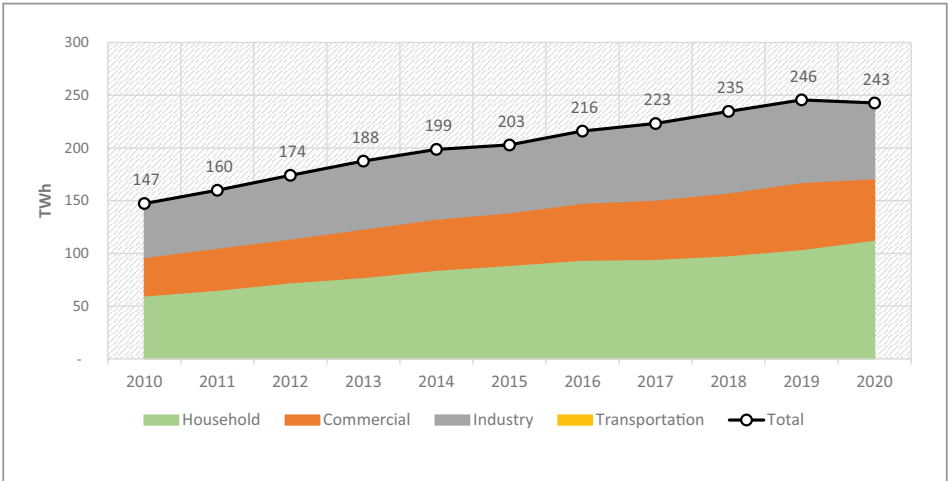
Source: HEESI, 2020

Figure 2.1 Final Energy Consumption by Sector

2.2 ELECTRICITY CONSUMPTION BY SECTOR

The trend of electricity consumption in the last 10 years shows an increase from 159.9 TWh in 2011 into 242.6 TWh in 2020. It means that electricity reaches its highest growth compared to other energy sources. Compared to the amount in 2019, the electricity consumption in 2020 slightly declined due to the decreasing activity in industry and commercial sector during the pandemic.

Household is the biggest electricity consumer in 2020 followed by industry, commercial and transportation. Electricity demand in household is 112.7 TWh or 46.4% from the total national electricity demand. It is influenced by the high number of households in Indonesia of around 69 million households in 2020. The electricity demand in industry is 71.5 TWh (29.5%) and commercial sector is 58.2 TWh (24%). The rest of the demand is from transportation of 0.3 TWh (0.1%). The electricity demand per sector is shown in Figure 2.2.



Source: HEESI, 2020

Figure 2.2 Electricity Consumption by Sector

Electricity demand in transportation sector is going to increase along with the development of electric vehicle and electric motorcycle that is continuously promoted by the government. It is also supported by electric based-transportation projects such as the already established MRT, LRT, and monorail.

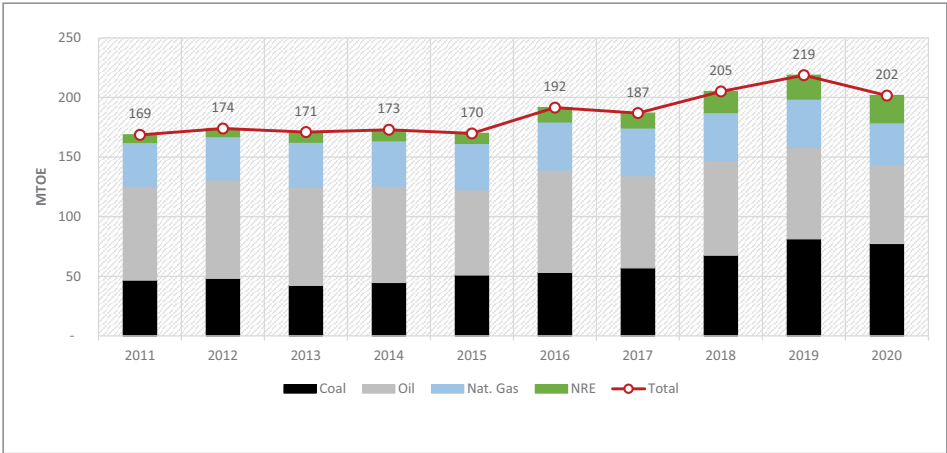
2.3 PRIMARY ENERGY CONSUMPTION

A number of energy related-policies such as energy diversification, energy efficiency and Indonesia’s commitment to the international world would give an impact on a more rational primary energy consumption pattern. The household appliances producers, for instance, are competing to produce energy-saving or more efficient appliances. Energy producers also work hard to increase energy production from a more environmentally friendly energy sources.

The primary energy types that dominate energy supply mix in 2020 are coal of approximately 38.5% from the national total followed by oil of approximately 32.8%, gas of approximately 17.4%, and NRE of approximately 11.3%. This condition is different from the energy condition in 2011 when oil became the energy type that dominated the



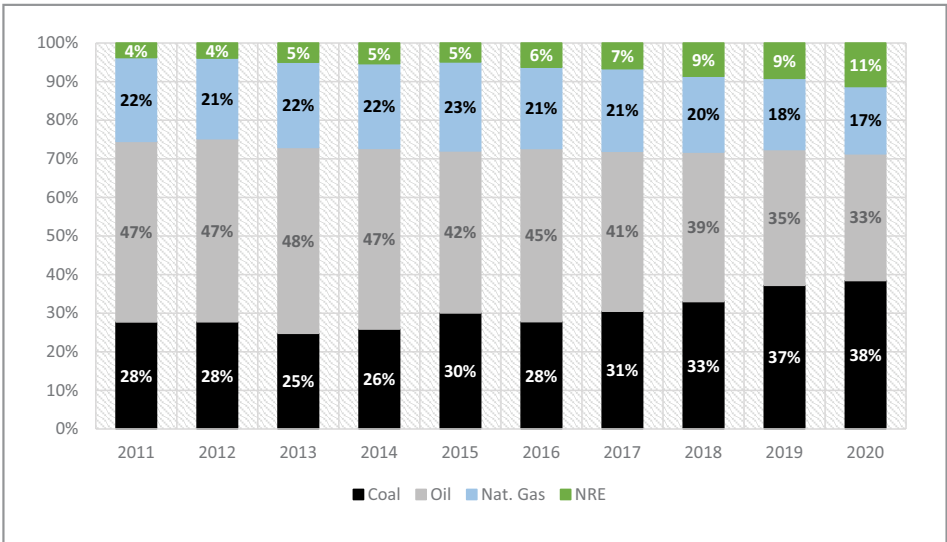
primary energy supply of approximately 46.8%, while NRE contributes of approximately 3.8% from the total national primary energy supply. Figure 2.3 and 2.4 show the change of volume and type in national energy supply.



Source: HEESI, 2020

Figure 2.3 Primary Energy Consumption by Type of Energy

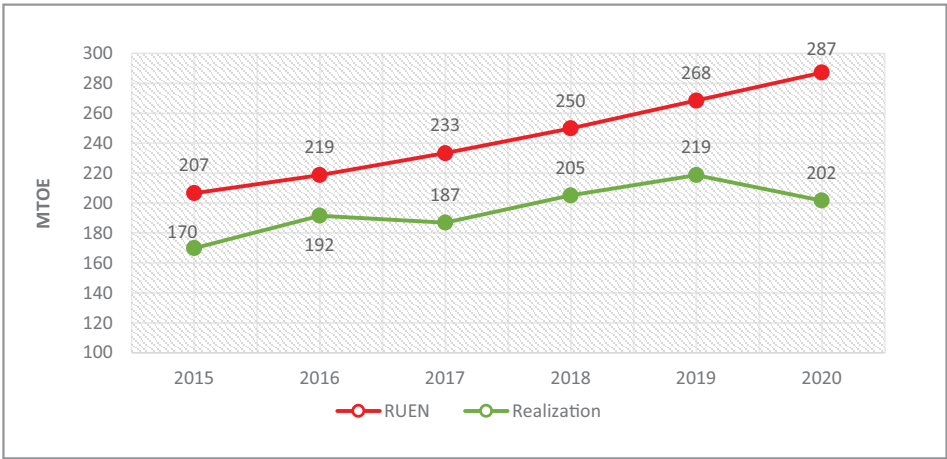
The increase of energy sources from NRE that is continuously promoted by the government is based on the mandates in KEN and RUEN. Thus, NRE in national energy mix is increasing. NRE mix in the last ten years has been increasing from 3.8% in 2011 to 11.3% in 2020.



Source: HEESI, 2020

Figure 2.4 Primary Energy Mix

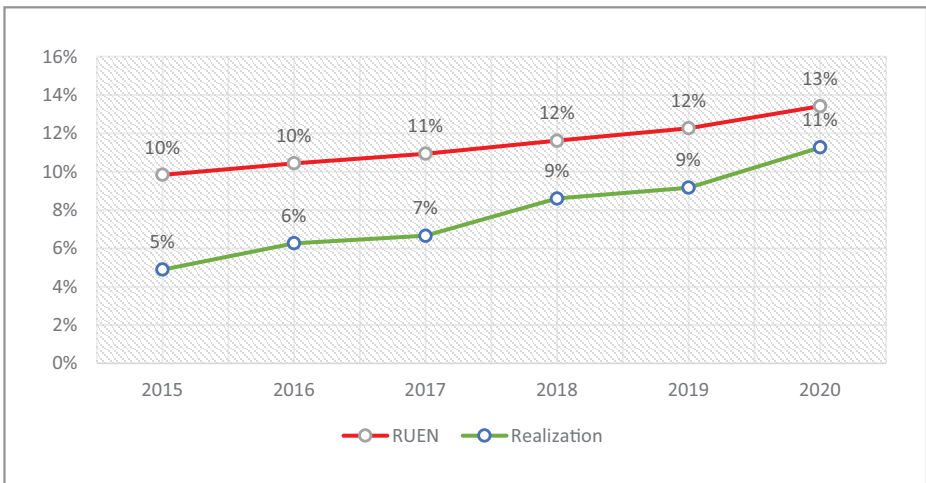
The actual total primary energy supply is still below the projection in RUEN (Figure 2.5). One of the influencing factors is the gap between economic growth in RUEN that is assumed to reach 7% with the real average GDP growth of approximately 5%. In addition, the economic growth in 2020 is -2.7% due to Covid19 pandemic.



Source: RUEN, 2017 and HEESI, 2020

Figure 2.5 Comparison of Primary Energy Supply Realization and RUEN

The trend of increase in NRE mix each year is an evidence that Indonesia is committed to carry out mandates in KEN and RUEN as well as in other international agreements. However, NRE mix that is continuously increasing is still below the target in RUEN. Figure 2.6 shows the gap between the target in RUEN and the declining realization from 4.9% in 2015 into 2.1% in 2020. It takes effort and involvement from all related stakeholders in increasing the use of NRE since NRE resources and reserves in Indonesia are still abundant but not yet used maximally.

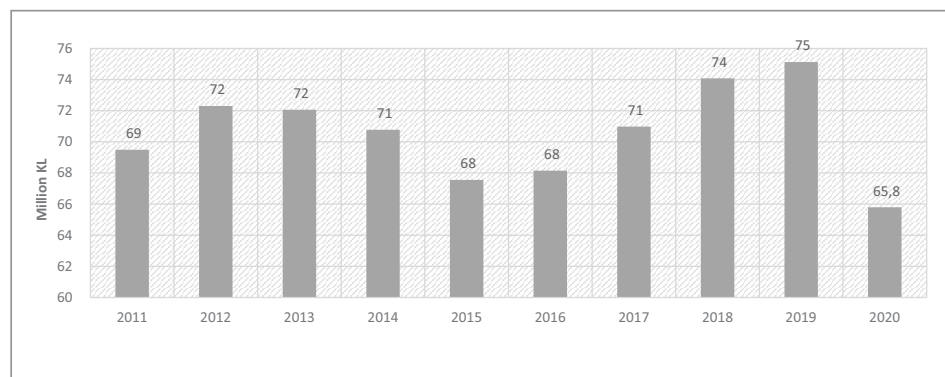


Source: RUEN, 2017 and HEESI, 2020

Figure 2.6 NRE Primary Energy Mix

2.4 FUEL AND LPG CONSUMPTION PER PROVINCE

Fuel consumption is fluctuating. However, accumulatively it delines 5.6% from 2011 to 2020 (Figure 2.7). Until today, fuel diversification is not succeeded yet except for the use of 30% biodiesel in 2020.



Source: HEESI 2011-2020

Figure 2.7 Fuel Consumption 2011-2020

Based on the area, fuel is mostly consumed in Java-Bali region reaching 48%. It is followed by Sumatera with 21% and Kalimantan with 18%. Based on the province, the biggest fuel consumer is East Java, followed by West Java and Central Java (Table 2.1). The high fuel consumption in Java Bali is influenced by the high mobility and number of vehicles in this region.

Table 2.1 Fuel Consumption per Province in 2011-2020

(Million KL)

Provinsi	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Aceh	1.1	1.1	1.1	1.1	1.1	1.1	1.1	1.2	1.2	1.1
North Sumatera	3.3	3.4	3.4	3.4	3.2	3.2	3.4	3.5	3.5	3.0
West Sumatera	1.2	1.3	1.3	1.2	1.2	1.2	1.2	1.3	1.3	1.1
Riau	2.2	2.3	2.3	2.3	2.1	2.2	2.3	2.4	2.4	2.1
Riau Islands	1.1	1.2	1.2	1.1	1.1	1.1	1.1	1.2	1.2	1.0
Jambi	0.8	0.9	0.9	0.8	0.8	0.8	0.9	0.9	0.9	0.8
Bengkulu	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.5	0.4	0.4
South Sumatera	2.1	2.2	2.2	2.2	2.1	2.1	2.2	2.3	2.2	2.0
Bangka Belitung	0.9	0.9	0.9	0.9	0.9	0.9	0.9	1.0	0.8	0.7
Lampung	1.6	1.6	1.6	1.6	1.5	1.5	1.6	1.7	1.6	1.4
DKI Jakarta	5.3	5.5	5.5	5.4	5.1	5.2	5.4	5.6	4.7	4.1
Banten	4.6	4.8	4.8	4.7	4.5	4.5	4.7	4.9	4.9	4.3

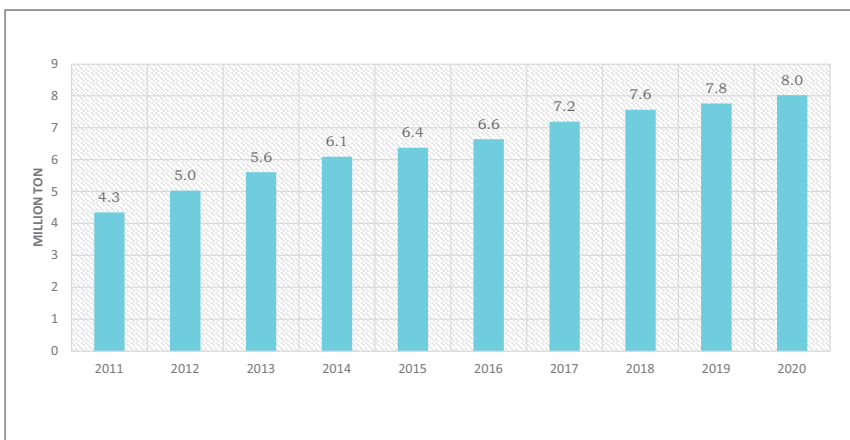
Table 2.1 Fuel Consumption per Province in 2011-2020 (continued)

(Million KL)

Province	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
West Java	7.4	7.7	7.6	7.5	7.2	7.2	7.5	7.8	8.1	7.1
Central Java	5.5	5.8	5.8	5.7	5.4	5.4	5.7	5.9	6.1	5.3
DIY	0.8	0.8	0.8	0,8	0,8	0,8	0,8	0.9	0.9	0.8
East Java	7.7	8.0	8.0	7.8	7.5	7.5	7.8	8.2	8.4	7.3
Bali	2.0	2.1	2.1	2,1	2.0	2.0	2.1	2.2	2.1	1.8
West Nusa Tenggara	1.2	1.2	1.2	1,2	1.1	1.1	1.2	1,2	1.4	1.2
East Nusa Tenggara	0.8	0.8	0.8	0.8	0.7	0.7	0.8	0.8	0.9	0.8
West Kalimantan	1.4	1.5	1.5	1.5	1.4	1.4	1.5	1.5	1.5	1.3
Central Kalimantan	1.2	1.2	1.2	1.2	1.1	1.1	1.2	1.2	1.2	1.1
South Kalimantan	3.5	3.6	3.6	3.6	3.4	3.4	3.6	3.7	3.6	3.2
East Kalimantan	6.1	6.3	5.7	5.6	5.4	5.4	5.7	5.9	6.8	5.9
North Kalimantan	-	-	0.6	0.6	0.5	0.5	0.6	0.6	0.6	0.6
North Sulawesi	0.8	0.8	0.8	0.8	0.7	0.8	0.8	0.8	0.9	0.9
Gorontalo	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Central Sulawesi	0.7	0.8	0.8	0.8	0.7	0.7	0.8	0.8	0.8	0.7
South East Sulawesi	0.9	0.9	0.9	0.9	0.9	0.9	0.9	1.0	1.1	1.0
South Sulawesi	2.0	2.1	2.1	2.1	2.0	2.0	2.1	2.2	2.4	2.1
West Sulawesi	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2	0.2
Maluku	0.5	0.5	0.5	0.5	0.5	0.5	0.5	0.6	0.6	0.5
North Maluku	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.4	0.5	0.4
Papua	1.1	1.1	1.1	1.1	1.0	1.0	1.1	1.1	1.1	1.0
West Papua	0.5	0.5	0.5	0.5	0.4	0.5	0.5	0.5	0.5	0.5

Source: BPH Migas and Pusdatin MEMR, processed by Sec.Gen. DEN

Meanwhile, LPG consumption since 2012 shows an increase each year. The average LPG consumption increase each year is 6% as shown in Figure 2.8.



Source: HEESI 2020 and DG of Oil and Gas

Figure 2.8 LPG Consumption in 2011-2020

Based on the region, the biggest LPG consumption share is Java Bali with 66% followed by Sumatera with 19% and Sulawesi with 7% (Table 2.2). The biggest LPG consumer with 96% consumption is household, while the rest is commercial and industrial sector.

Table 2.2 LPG Consumption by Province in 2011-2020

(Thousand Ton)

Provinsi	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Aceh	63	73	82	89	93	99	110	116	119	123
North Sumatera	225	264	294	319	334	351	381	401	414	439
West Sumatera	47	56	62	67	70	87	101	111	116	124
Riau	89	104	116	126	132	140	157	166	169	181
Riau Islands	36	43	47	51	54	56	59	63	67	70
Bangka Belitung Islands	19	22	24	27	28	31	36	40	42	44
Jambi	43	51	57	61	64	68	75	78	79	85
Bengkulu	26	31	35	37	39	42	45	47	49	52
South Sumatera	135	158	176	192	200	202	216	223	228	231
Lampung	108	126	141	153	160	170	186	194	198	206
Banten	214	250	279	303	317	322	346	366	372	388
DKI Jakarta	316	370	412	448	469	469	487	499	489	482
West Java	750	878	978	1,063	1,112	1,341	1,417	1,473	1,508	1,558
Central Java	604	707	788	856	896	940	1,021	1,081	1,123	1,176
DI Yogyakarta	78	91	102	111	116	120	129	137	142	140
East Java	750	878	978	1,063	1,112	1,138	1,222	1,281	1,310	1,344

Table 2.2 LPG Consumption by Province in 2011-2020

(Thousand Ton)

Province	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020
Bali	118	138	153	167	174	182	201	216	225	212
West Kalimantan	77	90	100	109	114	120	132	139	141	147
Central Kalimantan	21	24	27	29	31	43	58	65	71	74
South Kalimantan	46	54	60	65	68	85	98	105	109	114
North Kalimantan	-	-	5	5	5	6	9	11	11	12
East Kalimantan	84	102	109	118	124	125	134	141	145	150
NTT	0	0	0	0	0	0	1	1	1	1
NTB	47	55	61	67	70	74	84	91	98	112
North Sulawesi	38	44	49	54	56	63	71	76	79	81
Gorontalo	15	18	20	21	22	25	29	31	33	34
West Sulawesi	14	16	18	19	20	24	27	29	30	30
Central Sulawesi	12	14	15	17	18	36	47	51	56	60
South Sulawesi	158	184	206	223	234	245	266	280	286	287
South East Sulawesi	18	21	23	25	27	35	42	46	50	53
Maluku	1	1	1	1	1	1	1	1	1	1
North Maluku	0	0	1	1	1	1	1	1	1	1
West Papua	1	1	1	1	2	1	1	1	2	2
Papua	1	1	1	1	1	2	2	3	4	4

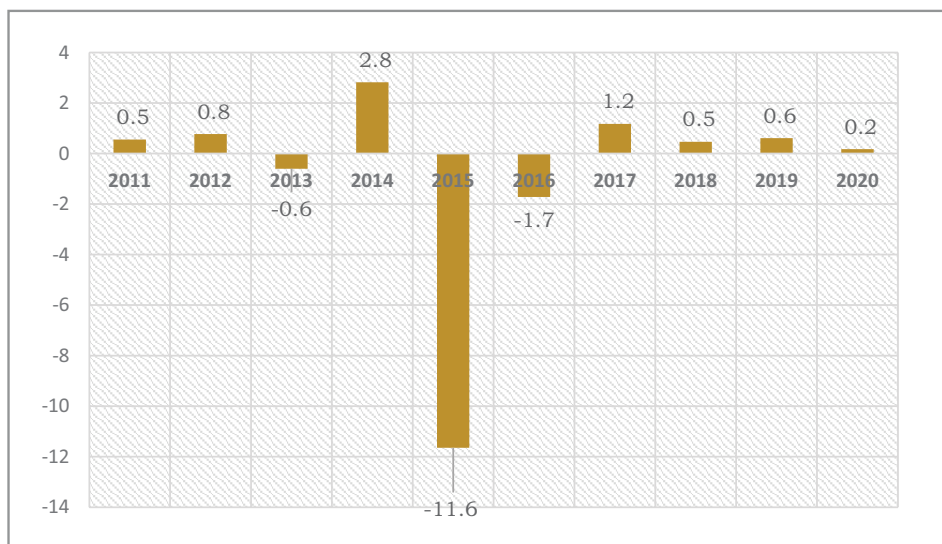
Source: PT. Pertamina and Pusdatin MEMR, processed by Sec.Gen NEC

2.5 ENERGY ELASTICITY AND INTENSITY

Energy elasticity is a ratio of final energy consumption growth and GDP growth in the same period. Energy elasticity that is low or below one shows an efficient energy use since to increase 1% of GDP growth only needs energy demand growth below 1%.

Based on KEN target, energy elasticity must be below one in 2025. Energy elasticity is gradually declining through the years. In 2050, energy elasticity is targeted at 0.46 which shows that national energy use will become more efficient.

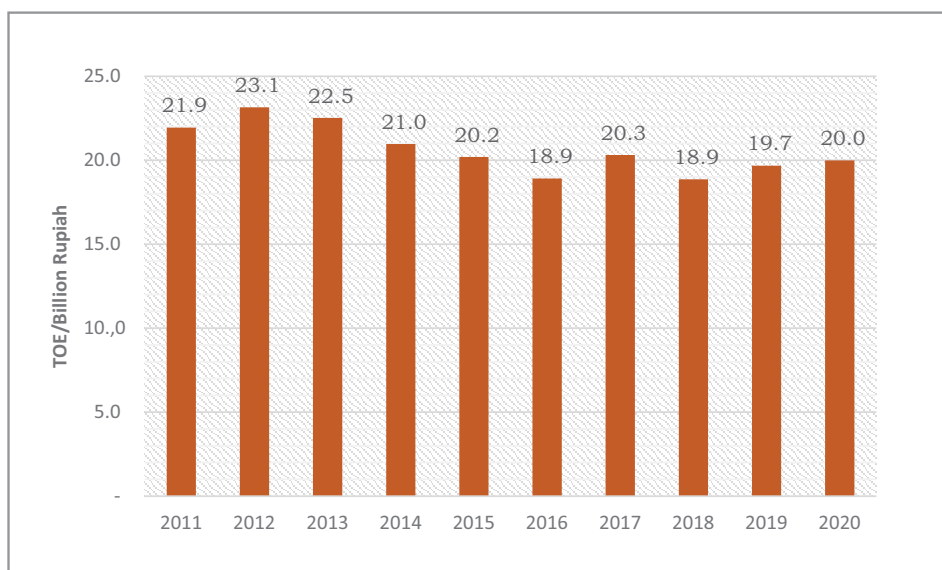
Based on the processed data, Indonesia energy elasticity from 2011 to 2020 is fluctuating. In 2015, final energy consumption elasticity is declining into negative due to the declining final energy consumption (Figure 2.9). Many factors influence final energy elasticity, including energy data availability that is not entirely covered in HEESI such as non-PLN electricity production data and other data.



Source: HEESI, 2020

Figure 2.9 Indonesia Energy Elasticity 2011-2020

Meanwhile, primary energy intensity is a comparison between primary energy supply and Gross Domestic Product (TOE/Billion Rupiah). Based on RUEN, the projection of primary energy intensity should be reduced by 1% each year but the real Indonesia's primary energy intensity from 2012-2020 has been fluctuating as shown in Figure 2.10.



Source: HEESI, 2020

Figure 2.10 Primary Energy Intensity 2011-2020

2.6 ENERGY PRICE

Energy price is one of the influencing factors in energy demand trend. Energy price paid by consumer will influence the use of technology and equipment. From supply side, energy price will influence the decision of production and investment from the chosen technology.

The declining energy price in 2020 occurs almost in all energy types including coal. Coal production is relatively stable despite of its production decline. Meanwhile, coal demand is declining due to restriction in several countries that causes coal reference price to be in its lowest level. Coal Reference Price in 2020 has been the lowest level since 2015. If energy price is compared in the same unit (Thousand Rupiah/MTOE), coal becomes the cheapest energy source with the price of IDR 29,960/TOE as presented in Table 2.3.

Table 2.3 Energy Price

(Thousand Rupiah/TOE)

Types of Energy	2015	2016	2017	2018	2019	2020
Gasoline (RON 88)	173	158	155	155	155	155
Avtur	219	172	199	240	233	233
Kerosene	59	59	59	59	59	59
Gasoil CN 48	147	114	111	111	111	111
LPG 3 KG	70	70	70	70	70	70
LPG 12 KG	202	191	197	204	204	204
LPG 50 KG	200	175	205	226	226	226
Coal	22	20	26	25	25	30
Electricity for Household	191	193	241	252	251	227
Electricity for Industry	261	240	249	248	251	249
Electricity for Commercial Sector	293	274	284	284	287	283

Source: HEESI, 2020

2.6.1 Electricity Price

The price of electricity per kWh in 2020 is IDR 1,071.4 or lower than the previous year of IDR 1,130.3. The electricity price for household is still subsidized but the price for other sectors uses economical price. Electricity price is stipulated based on Minister of Energy and Mineral Resources Regulation No. 3 of 2020 on Electricity Tariff Supplied by PT. PLN (Persero). Electricity tariff by sector in the last five years is presented in Table 2.4.

Table 2.4 Average Electricity Price by Costumer Group

(Rupiah/kWh)

Year	Household	Industry	Business	Social	Govt. offices	Public Street Lighting	Average
2015	837.0	1,142.7	1,284.2	812.4	1,324.5	1,459.1	1,034.5
2016	843.7	1,051.8	1,201.2	816.0	1,234.7	1,415.3	991.4
2017	1,056.0	1,088.8	1,245.6	821.3	1,278.5	1,461.5	1,105.1

Table 2.4 Average Electricity Price by Customer Group (Continued)

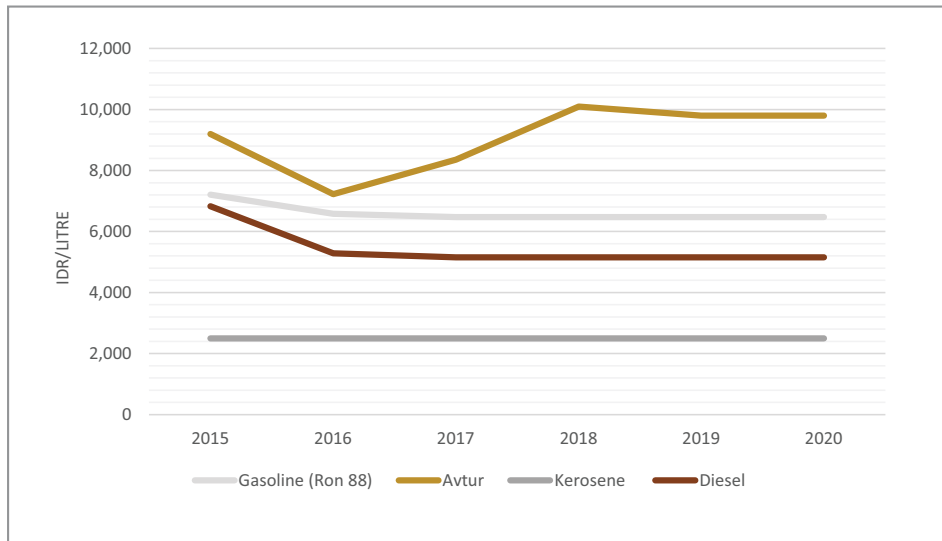
(Rupiah/kWh)

Year	Household	Industry	Business	Social	Govt. offices	Public Street Lighting	Average
2018	1,102.4	1,085.3	1,244.0	823.2	1,280.1	1,461.6	1,123,0
2019	1,098.8	1,100.7	1,258.3	830.7	1,291.5	1,465.1	1,129,6
2020	991.9	1,090.9	1,239.3	804.1	1,299.2	1,459.9	1,071,4

Source: MEMR Ministerial Reg, 2020

2.6.2 Fuel Price

Price of fuel that is still subsidized is diesel for goods transportation and kerosene for household. Meanwhile, since 2014, subsidy for oil fuel has been removed and the price follows the economical price. The trend of oil fuel price in the last five years is shown in Figure 2.11.



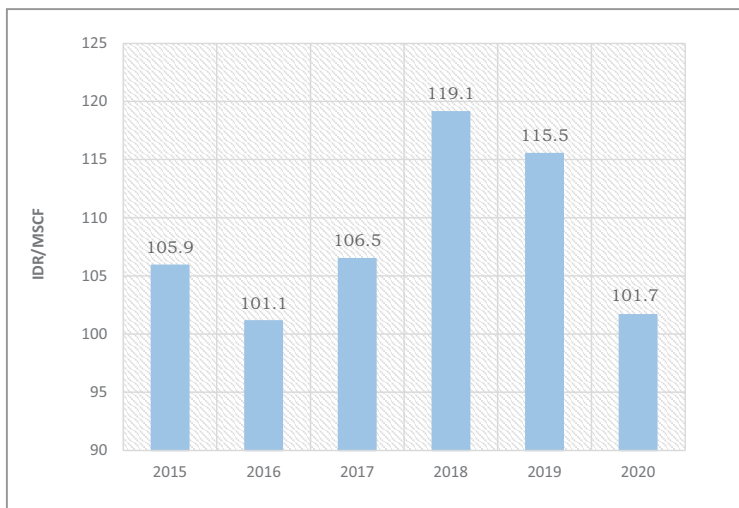
Source: HEESI, 2020

Figure 2.11 Trend of Fuel Price

2.6.3 Natural Gas Price

Gas price is stipulated based on the agreement between seller or supplier and buyer (consumer). However, since 2015, the government has issued regulation on gas price for industry due to the high gas price in industry. In 2016, the government issued policy related to gas price at 6 US dollar per MMBTU as stipulated in Presidential Regulation No. 40 of 2016 on Gas Price Stipulation. The government then issued the derivative of this Presidential Regulation through Minister of Energy and Mineral Resources Decision No. 89K/10/MEM/2020 on Certain Gas User and Price in Industry. In this Ministerial Decision, seven industries will receive gas at the price of 6 US dollar per MMBTU namely

the industry of fertilizer, petrochemical, oleochemical, steel, ceramics, glass, and rubber gloves. Based on this regulation, the price scheme is valid from 2020 to 2024. Gas price in the last five years is shown in Figure 2.12.

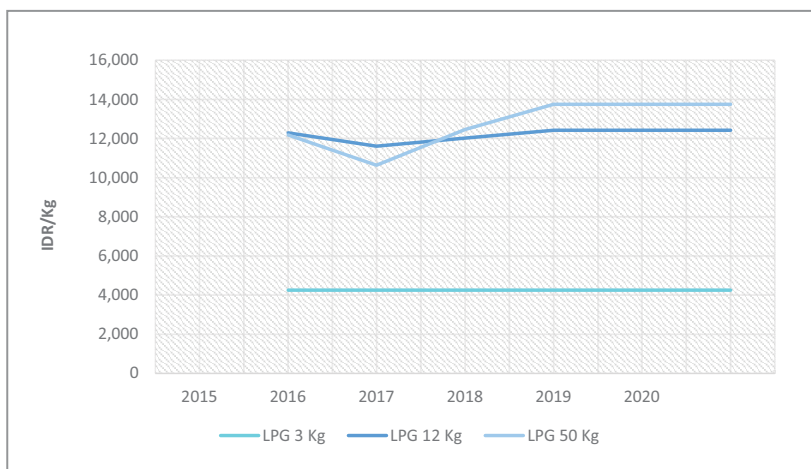


Source: Statistic PLN, 2020

Figure 2.12 Trend of Gas Price

2.6.4 LPG Price

LPG is one of the energy types which receive most subsidy that burdens the State Budget. The subsidized LPG is the 3 kg LPG for the poor or low-income household. In reality, the 3 kg LPG is used by high income household and commercial sector. In result, the subsidies do not meet the right target. The price of subsidized LPG is far below normal LPG. However, 85% of household uses subsidized LPG. The comparison between subsidized and non subsidized LPG is seen in Figure 2.13.

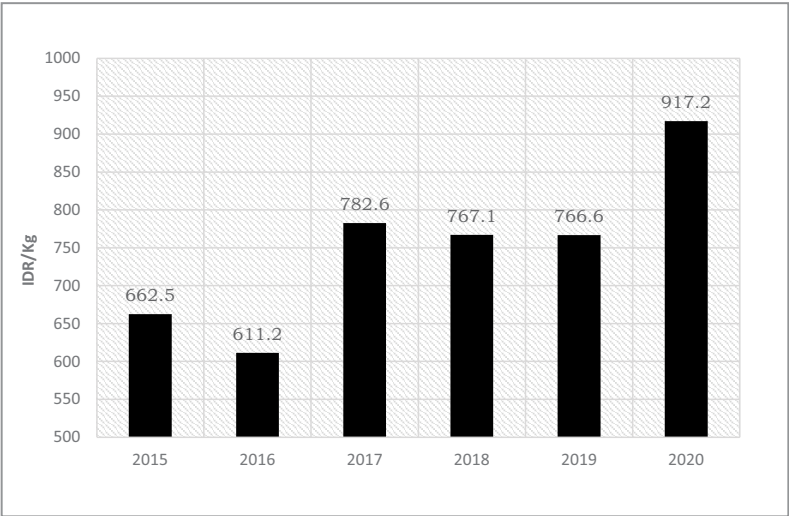


Source: HEESI, 2020

Figure 2.13 Trend of LPG Price

2.6.5 Coal Price

Coal price is regulated monthly based on Coal Reference Price issued by Directorate General of Mineral and Coal (Figure 2.14).

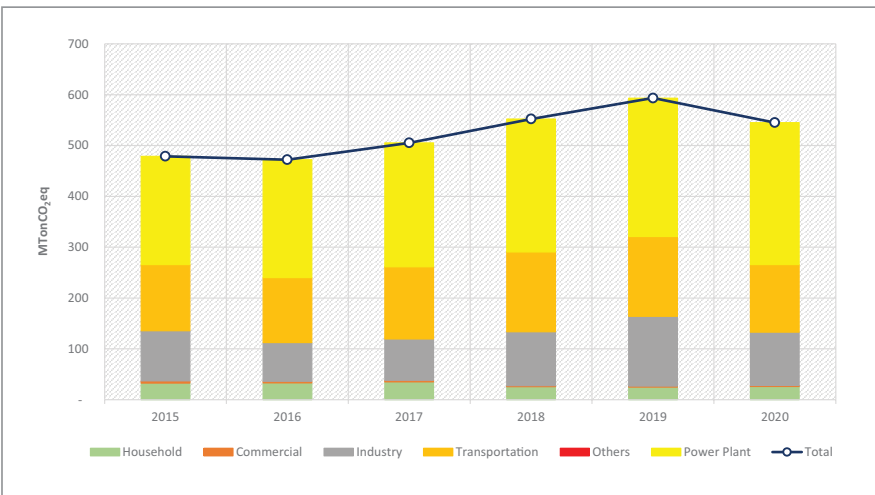


Source: Statistik PLN, 2020

Figure 2.14 Trend of Coal Price

2.7 EMISSION

Emission is the result of fuel combustion from energy consuming sectors and power plant activity. Total emission in 2020 is 545 Million Ton CO₂eq. The biggest contributor is power plant sector with 51%, followed by transportation with 24% and industry with 19%. The development of CO₂ emission in 2015-2020 is shown in Figure 2.15.



Source: Pusdatin MEMR, 2020

Figure 2.15 CO₂ Emission from Fuel Combustion

2.8 ENERGY INFRASTRUCTURE

2.8.1 Oil Refinery

At present, Indonesia's oil refinery capacity is 1,151 MBSD. The details of oil refineries and their capacity is resumed in Table 2.5.

Table 2.5 Indonesia Oil Refinery Capacity 2020

Refinery	Capacity (MBSD)
Tri Wahana Universal (TWU)*	18
Dumai	127
Sungai Pakning	50
Musi	127.3
Cilacap	348
Balikpapan	260
Balongan	125
Cepu	3.8
Kasim	10
Tuban (TPPI)	100
Total Capacity	1,169.1
Total Operating Capacity	1,151.1

*) Stop operating

Source: Oil and Gas Statistic Semester I 2020

In the last ten years, oil refinery capacity in Indonesia has not changed significantly. The latest additional capacity occurred in 2015 with the additional 12 MBSD to Tri Wahana Universal (TWU) refinery Unit II. TWU refinery is the oil refinery located in Bojonegoro, East Java which produces main products such as Straight Run Gasoline, Marine Diesel Oil, and Diesel. TWU refinery uses crude oil from Banyu Urip Field and Cepu Block. However, this refinery only operated for two years and it stopped operating in early 2018 due to lack of refinery raw material supply.

In energy balance data 2020, crude oil for refinery input is 302.3 Million Barrels from domestic production and import. Besides crude oil, oil refinery input also needs gas and intermedia. After being processed, oil refinery will produce gasoline, diesel oil, fuel oil, kerosene, avtur, avgas and other non-fuel refinery products such as LPG, lubricant, naptha, and others. Oil refinery production in 2020 is 251.5 Million Barrels and non fuel refinery production is 83.4 Million Barrels.

2.8.2 LPG Refinery

Currently, Indonesia's LPG refinery capacity is 4,740.3 MTPA (Million Ton Per Annum). The details of LPG refinery and its capacity is resumed in Table 2.6.

Table 2.6 Indonesia LPG Refinery Capacity 2020

LPG Refinery	Capacity (MTPA)
PT Badak NGL	1,000
PT Chevron*	90
PT Petrogas	14
PT Petrochina	600.0
PT Conoco Philips*	525
PT Saka Indonesia	113
PT Pertamina (Dumai)	68
PT Pertamina (Plaju)	131.0
PT Pertamina (Cilacap)	318
PT Pertamina (Balikpapan)	91
PT Pertamina (Balongan)	548
PT TPPI	175
PT Pertamina (P.Brandan)*	44
PT Maruta Bumi Prima*	17.0
PT Medco LPG Kaji*	73
PT Pertamina (Mundu)	37
PT Titis Sampurna	73
PT Sumber Daya Kelola (Tugu Barat)*	7.0
PT Bina Bangun Wibawa Mukti	55
PT Surya Esa Perkasa	82
PT Yudhistira Haka Perkasa*	44
PT Wahana Insannugraha	37
PT Media Karya Sentosa phase I*	58
PT Media Karya Sentosa phase II	84.0
PT Yudistira Energi	58
PT Gasuma Federal Indonesia	26
PT Pertasamtan Gas	259
PT Sumber Daya Kelola (Losarang)*	3.8
PT Arsynergy Resources	110
Total Capacity	4,740.3
Total Operating Capacity	3,878.5

Source: Oil and Gas Statistic Semester I 2020

From the total capacity of LPG refinery, 1,331 MTPA (28%) is oil refinery but LPG production from the oil refinery contributes 45% to the total national LPG production in 2020.

2.8.3 LNG Refinery

At present, Indonesia's LNG refinery capacity is 31 MMTPA. The details of LNG refineries and their capacity is summarized in Table 2.7.

Table 2.7 Indonesia LNG Refinery Capacity 2020

LNG Refinery	Capacity (MMTPA)
PT Arun LNG*	12.9
PT Badak	21.6
PT BP	7.6
PT Donggi Senoro	2.0
Total Capacity	44.1
Total Operating Capacity	31.2

*) Stop operating

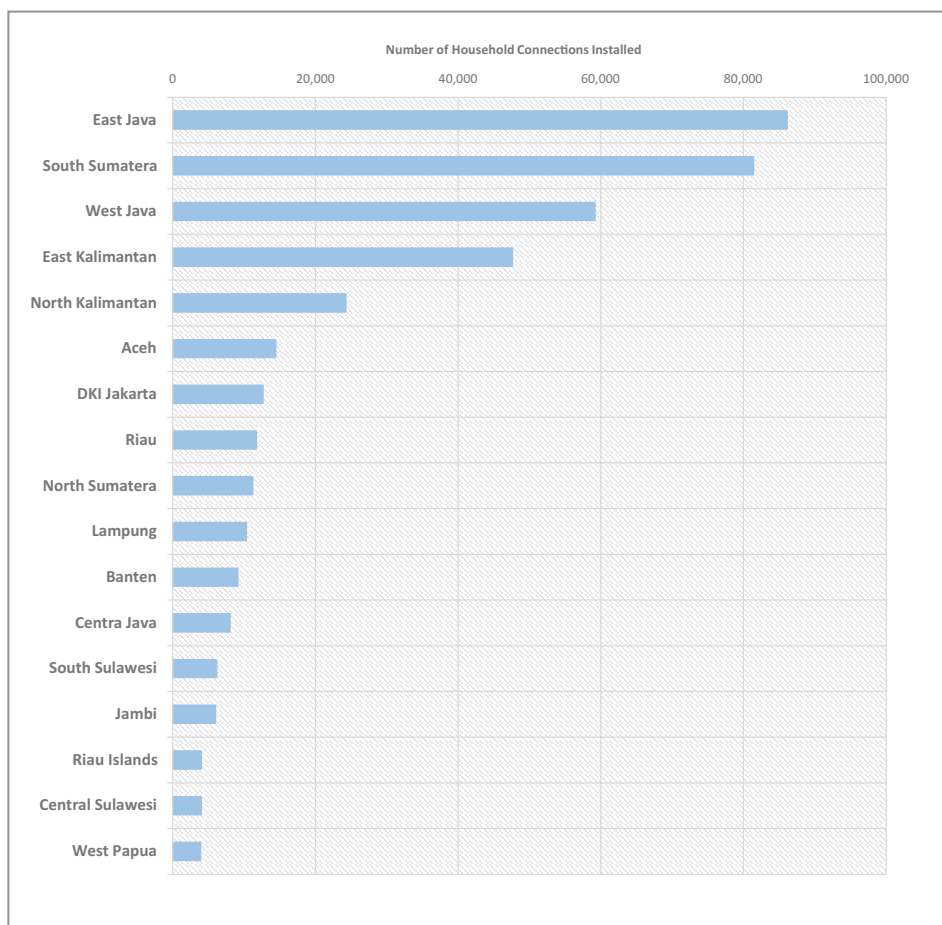
Source: Oil and Gas Statistic Semester I 2020

In October 2014, PT Arun's LNG refinery stopped operating due to the unavailable gas supply from Arun gas field. This refinery, located in Lhokseumawe, Aceh has been then transformed into regasification terminal by Pertamina through PT Perta Arun Gas to meet the gas supply for industry and electricity.

Total LNG production in 2020 from these three fields is 15,435.3 Thousand Metric Ton. Most LNG is exported especially to China, Korea, Japan and Taiwan based on long-term contract. Only 17% of LNG is used for domestic need especially to meet the demand from power sector.

2.8.4 City Gas Network

Until 2020, the government has connected 673 thousand households through city gas network infrastructure. As shown in Figure 2.16, the gas network infrastructure in the last ten years has covered 17 provinces in all areas in Indonesia. Most network is built in East Java (22%) and South Sumatera (20%) based on the gas sources.

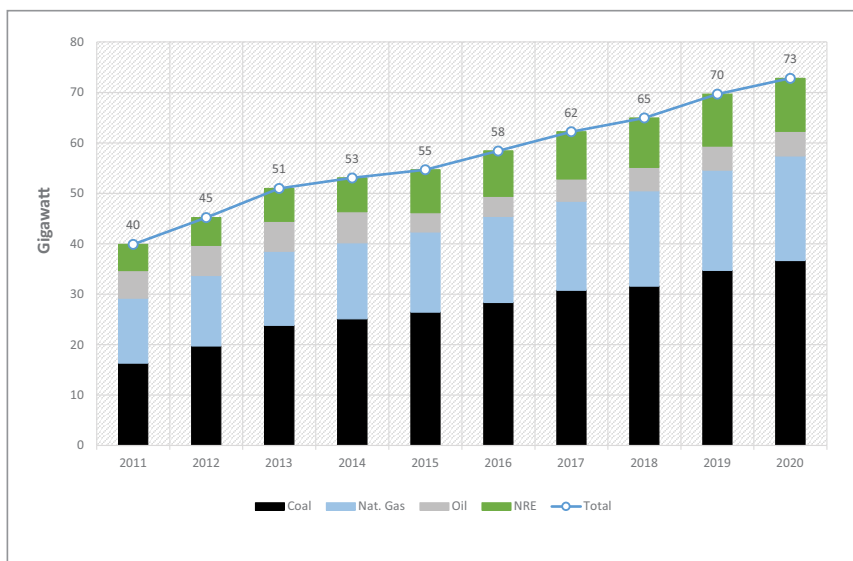


Source: Oil and Gas Statistic Semester I 2020 processed by Sec Gen NEC

Figure 2.16 Gas Network Infrastructure by Province

2.8.5 Power Plant

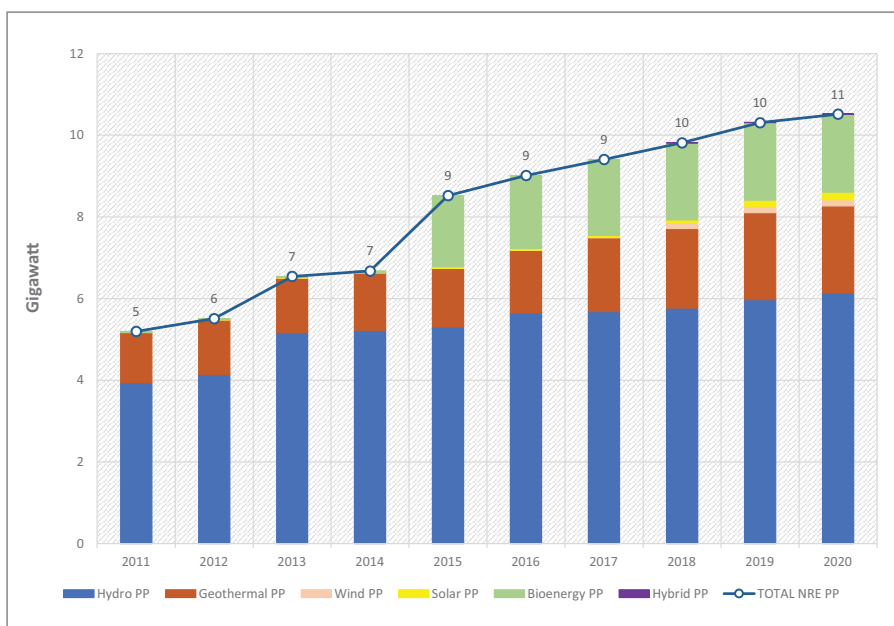
Until the end of 2020, Indonesia's total power plant capacity is 73 GW, consisting of 70 GW on grid power plant and 3 GW off grid power plant. This capacity shows that additional power plant capacity has been almost doubled in the last ten years as shown in Figure 2.17. Power plant is still dominated by coal which contributes almost half of the total national capacity. Gas follows with the contribution of 24%. Meanwhile, NRE based-power plant is only 15%. It has given additional 6 GW in the last ten years. From the total national capacity, most power plant is operated by PLN with the capacity of 43 GW (59.5%) followed by IPP with the capacity of 20 GW (28%). Meanwhile, power plant generated by Private Power Utility (PPU) and Non-Fuel Operation License contribute 4 GW (4.9%) and 6 GW (7.9%) each.



Source: HEESI, 2020 processed by Sec Gen NEC

Figure 2.17 Power Plant Installed Capacity by Type of Energy 2011-2020

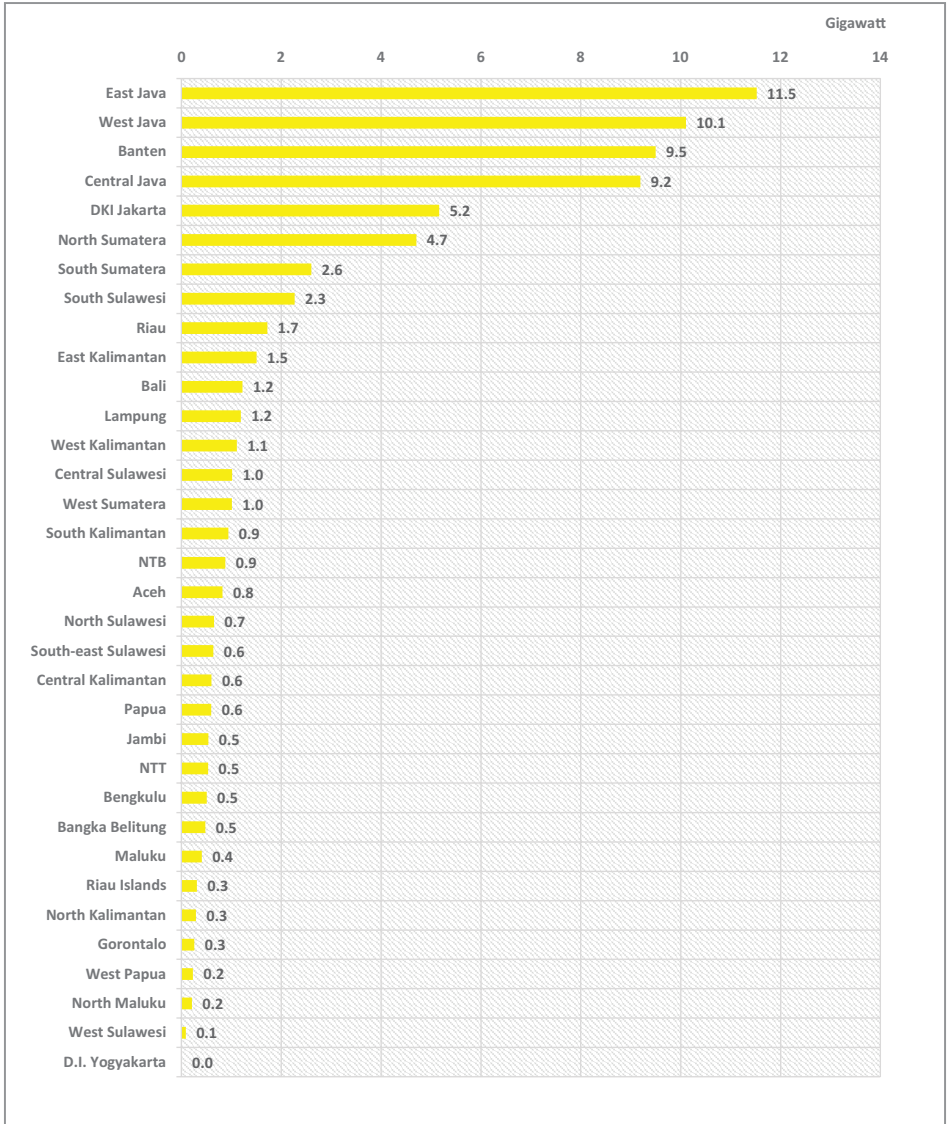
NRE for power plant is dominated by hydro (54%), geothermal (20%), and biomass (17%) as shown in Figure 2.18. Meanwhile, the use of solar, both on grid and off grid is only 186 MW.



Source: HEESI, 2020 processed by Sec Gen NEC

Figure 2.18 NRE Based-Power Plant Installed Capacity 2011-2020

Power plant infrastructure spread in all areas in Indonesia, but only 64% of power plant capacity is built in Java-Bali (Figure 2.19). Five provinces with the biggest capacity are in Java Island, except D.I. Yogyakarta which has fewer power plant. Meanwhile, power plant capacity is only 19% in Sumatera, 6% in Kalimantan, 7% in Sulawesi, and not more than 4% in eastern region of Indonesia (Papua-Maluku-Nusa Tenggara) from the total national capacity. This statistic shows that power plant infrastructure has not been developed equally.



Source: DG Electricity processed by Sec Gen NEC

Figure 2.19 Power Plant Capacity by Province 2020



3

2022 OUTLOOK

3.1 FINAL ENERGY CONSUMPTION

Total final energy consumption until 2022 is predicted to increase from 132 MTOE in 2019 into 129 MTOE or almost at the same level with final energy consumption in 2019 with the growing economy due to pandemic. Figure 3.1 shows that total final energy consumption in 2020 declines 11% from the previous year. However, it will increase 9% from 2020 to 2022.

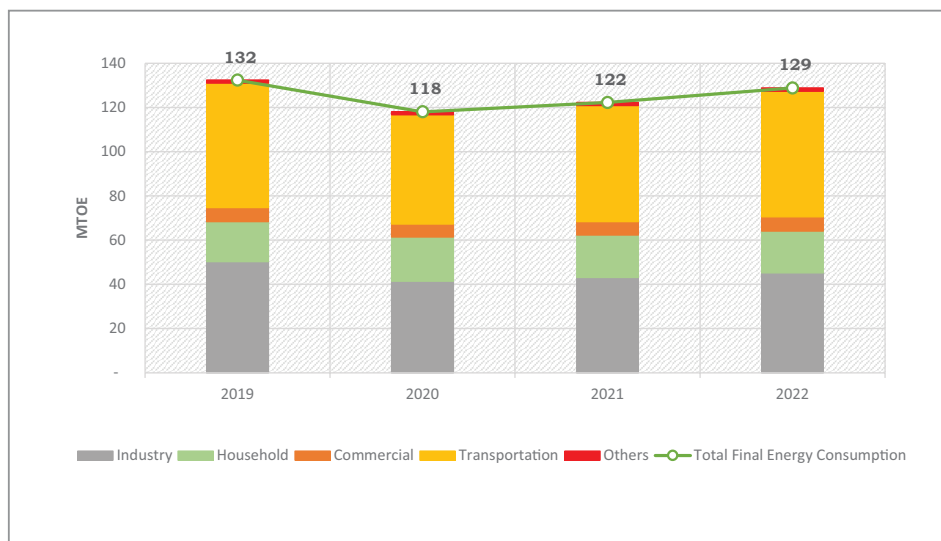


Figure 3.1 Final Energy Consumption by Sector

From its share, the biggest final energy consumers are transportation and industry, followed by household, commercial and other sectors. In 2022, energy consumption by transportation sector is predicted to increase 15% into 57 MTOE compared to in 2020. The recovering activities in several non essential sectors such as offices, mall, industry and tourism lead to the increasing fuel consumption in transportation. Meanwhile, final energy consumption in the industrial sector is projected to increase by 9% from 2020 to reach 45 MTOE in line with rising economic growth. Otherwise, the final energy consumption in household in 2022 will decline around 5% into 19 MTOE due to the declining activity in the house. In commercial and other sectors, the consumption increases 10% and 7% respectively. The comparison of final energy consumption shares per sector in 2020 and 2022 can be seen in Figure 3.2.

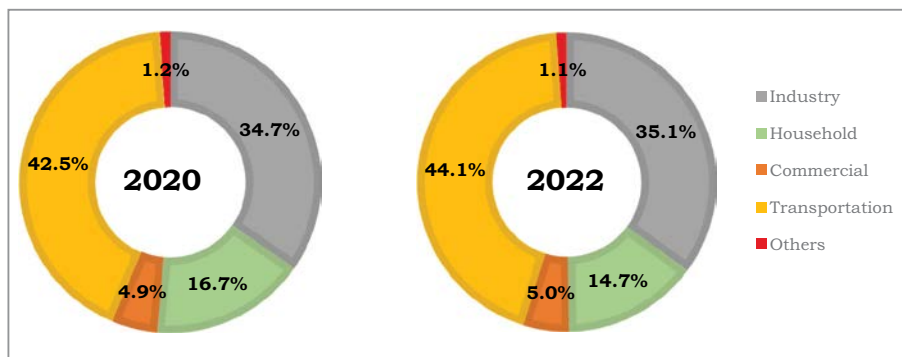


Figure 3.2 Comparison of Final Energy Consumption Share by Sector

Based on the energy types, the biggest final energy consumption until 2022 will be still dominated by fuel (41%), followed by electricity (19%), coal (17%), gas (17%) and NRE (6%). The trend of final energy consumption increase based on types of energy is shown in Figure 3.3.

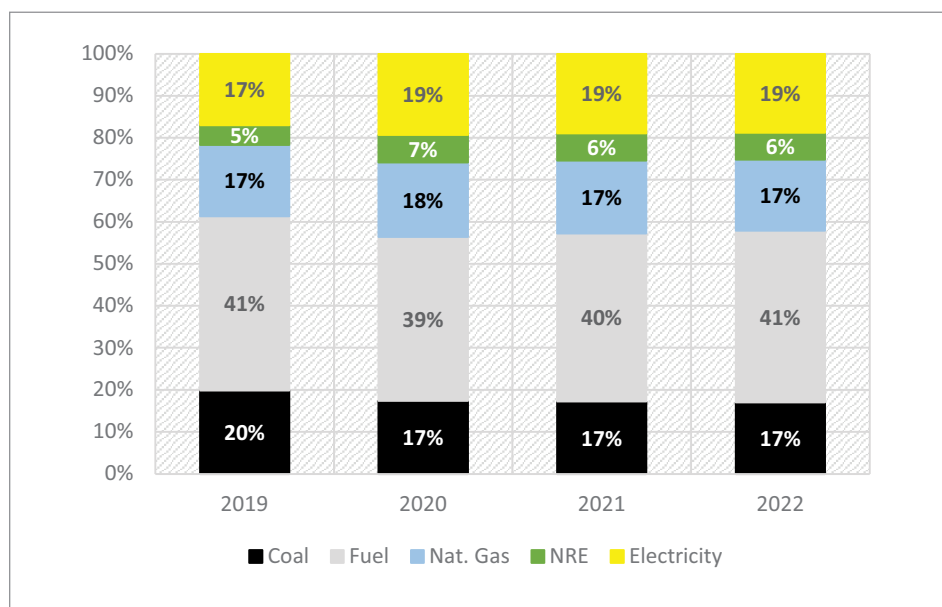


Figure 3.3 Final Energy Consumption by Type of Energy

The biggest increase in final energy consumption occurs in fuel due to the increasing avtur consumption from 2.39 MTOE in 2020 into 4.19 MTOE in 2022. Meanwhile the increase also occurs in coal and NRE by 7%, electricity by 6% and gas by 4%.

3.1.1 Household Sector

Energy consumed by household includes LPG, electricity, kerosene, and biomass (charcoal and firewood). Generally, energy consumption by household is used for cooking, lighting (lamp), cooling (AC) and others (water pump, iron, fridge). In the modelling, energy consumption in household is divided into Stove and Non-stove (fridge, lighting, AC and others). Stove uses LPG, kerosene, gas, electricity and biomass, while non-stove uses only electricity and kerosene.

The use of energy for stove is 95% derived from LPG while the rest is derived from kerosene, electricity, gas and biomass. In 2020, LPG consumption increases sharply due to Covid19 pandemic that forces everyone to stay at home. LPG consumption for cooking increases from 8.8 MTOE in 2019 into 9.5 MTOE in 2020. Along with the recovery of the pandemic, LPG consumption is predicted to slightly decline in 2022 into 8.7 MTOE. Until 2022, there has been no significant change in energy use for stove despite that since 2021 PLN has carried out dissemination to promote the use of induction stove by 1 million households by 2021 (based on GSEN target). To increase the use of induction stove, PLN makes a campaign and dissemination on the use of induction stove in PLN office, distribution channel and other marketplace. In addition, PLN also gives extra power services for housing developer that uses induction stove and develops induction stove-user friendly program with special price of extra power for institution, government institution, marketplace or other induction stove stakeholders who are committed to use induction stove. The energy demand for cooking in household sector can be seen in Figure 3.4.

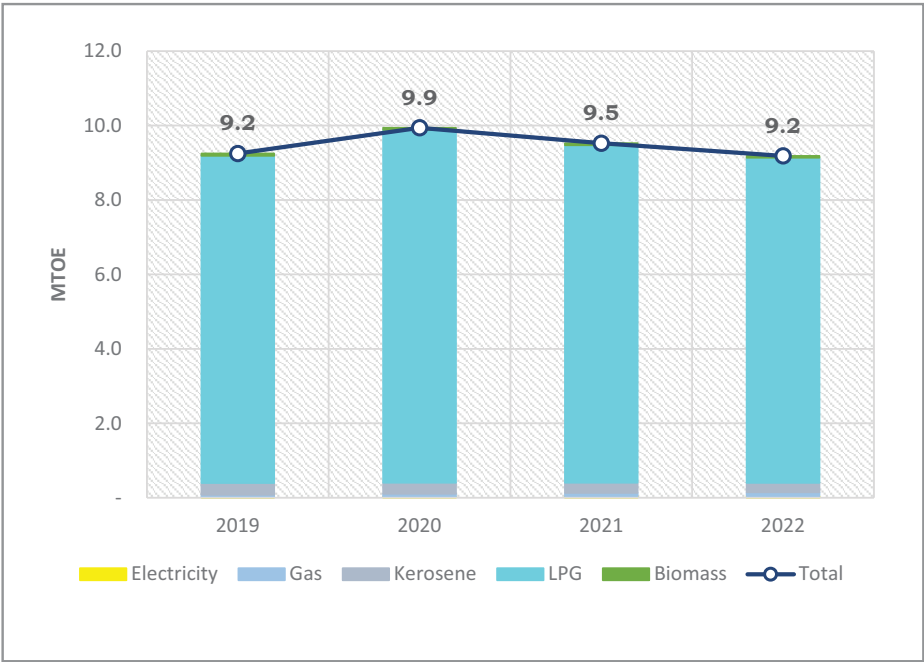


Figure 3.4 Energy Consumption for Cooking

Energy consumption in household for non stove category is only derived from electricity. It is used in fridgege, AC, electric pump, and other electronic devices. Kerosene is only used for lighting especially in rural and remote houses.

During the pandemic, electricity consumption has shown an increasing trend for non cooking activity from 8.8 MTOE in 2019 into 10 MTOE in 2020. With more activities at home, electricity consumption for AC and other electronic devices increases in 2020 compared to the previous year. However, in 2022 non stove electricity consumption will decrease after the pandemic is recovered as seen in Figure 3.5.

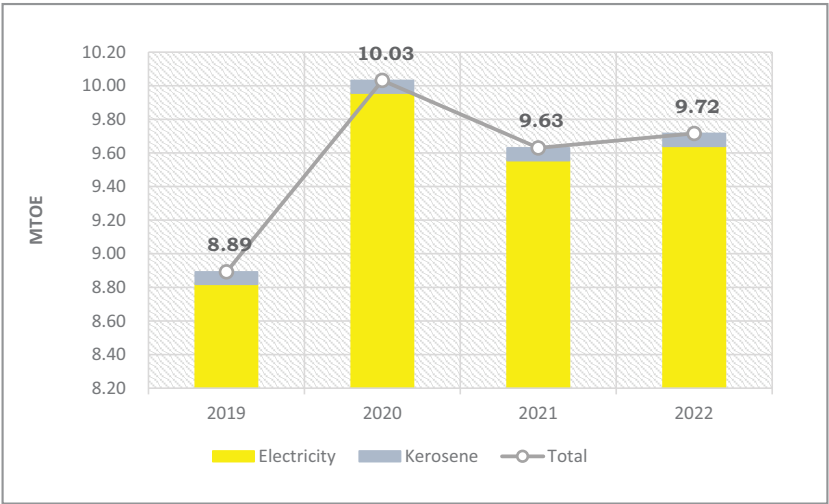


Figure 3.5 Energy Consumption for Non-cooking

Therefore, total energy consumption in households in 2022 will show a declining trend compared to the consumption in 2020 as shown in Figure 3.6.

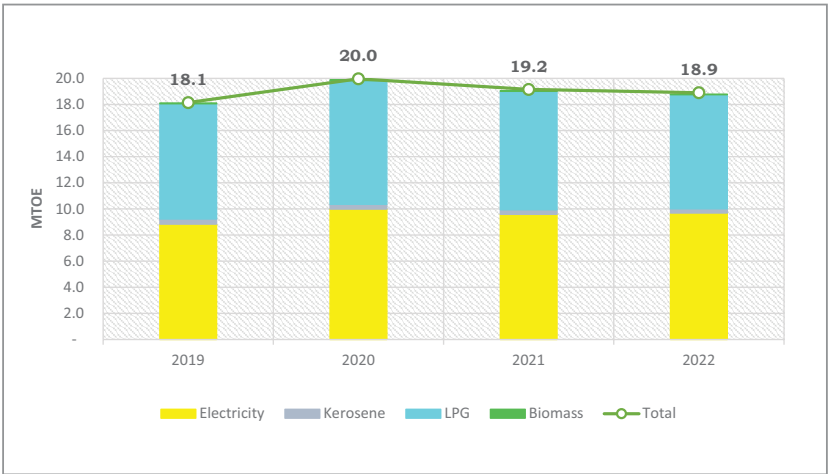


Figure 3.6 Final Energy Consumption by Household

3.1.2 Transportation Sector

Energy consumption in transportation is divided into eight categories namely car, motorcycle, bus, truck, commuter train, intercity train, sea transportation and air transportation. Meanwhile, energy used in transportation includes gasoline, diesel, biodiesel, gas, electricity, avtur, avgas and fuel oil.

In land transportation, types of energy used are gasoline and diesel. Gas is only limited for bus and taxi, while electricity is mainly used by commuter train and several electric vehicles and motorcycles operating in big city such as in Jakarta. Sea transportation uses fuel oil and diesel, while air transportation uses mainly avtur. Avgas is only used for propeller plane.

The biggest consumption in transportation is from motorcycle as its number has exceeded 120 million units in 2020. Energy consumption by motorcycle is 35% from the total consumption by transportation. The rest 30% of consumption is from truck and 16% from passenger cars. The overview of energy use in transportation sector by transportation mode is presented in Figure 3.7.

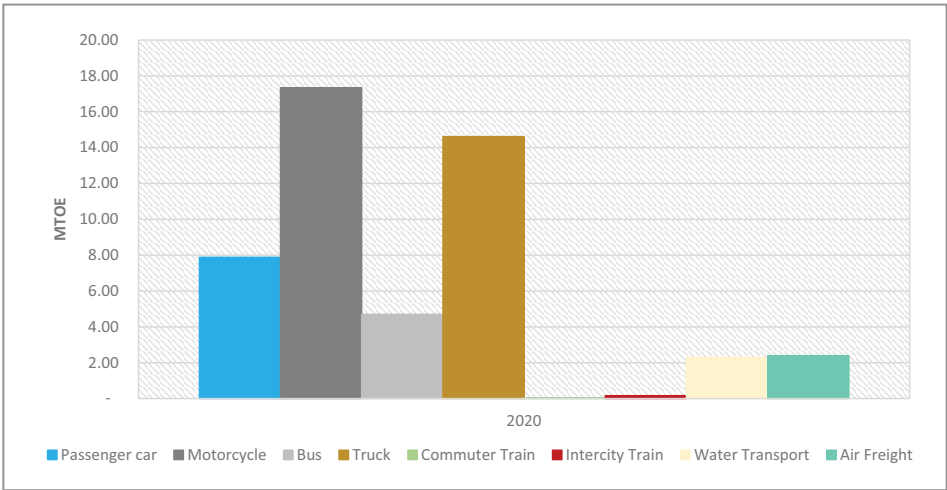


Figure 3.7 Energy Consumption by Transportation Mode

From the types of energy, transportation sector currently consumes gasoline by 51%, diesel by 29%, biodiesel by 14%, fuel oil by 1% and others (avgas, electricity and gas) by less than 1%. Gasoline in 2020 is mostly consumed by motorcycle (68%) and cars (29%). Meanwhile, diesel is mostly consumed by truck (66%) and bus (22%) as seen in Figure 3.8.

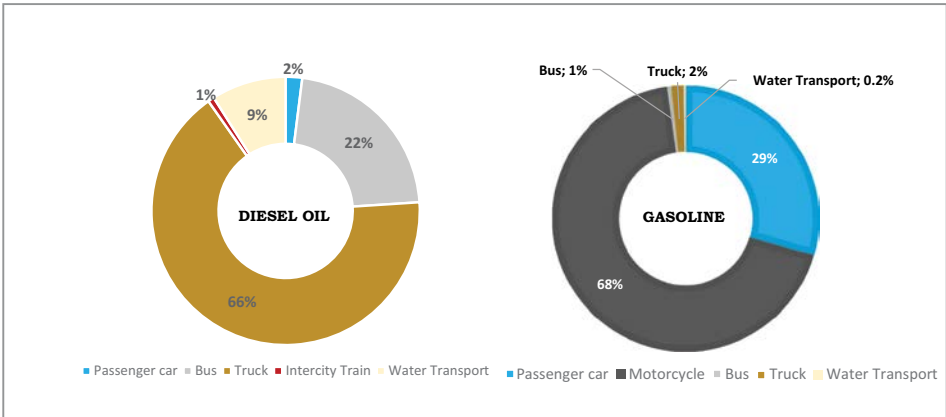


Figure 3.8 Portion of Gasoline and Diesel Oil Consumption by Transportation Mode

Total final energy consumption in transportation in 2020 declines from 56.2 MTOE in 2019 into 49.4 MTOE with the biggest decline in avtur due to the decreasing domestic and international flight. On the other hand, energy consumption in land transportation is still high because of the increasing frequencies of expedition services during the pandemic along with the growing e-commerce.

In 2022, energy consumption in transportation is predicted to increase to 56.9 MTOE when social activity will be gradually back to normal despite that the government will still implement the social activity restriction. Tourism sector is stagnant in the last two years. It starts to show an increasing activity that leads to increasing consumption in transportation. The trend of final energy consumption increase in transportation can be seen in Figure 3.9.

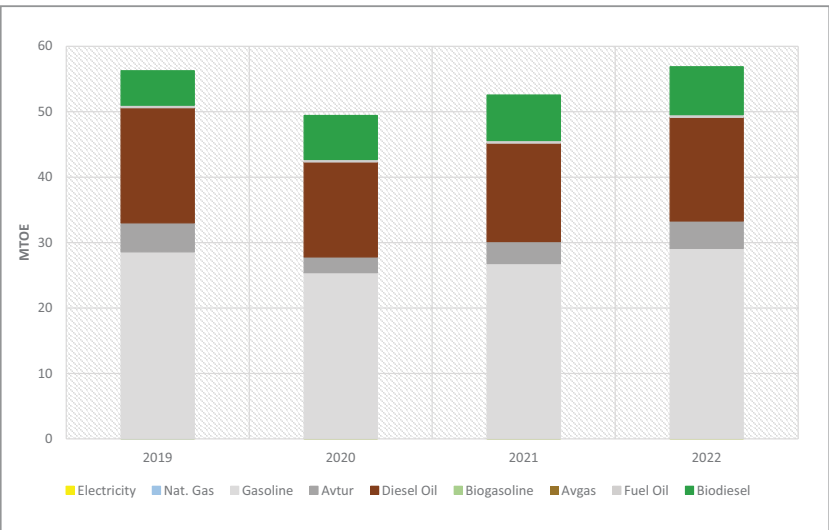
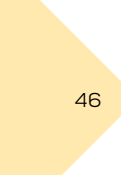


Figure 3.9 Final Energy Consumption in Transportation Sector



3.1.3 Industrial Sector

Energy consumption in industry is dominated by coal and gas. Coal is used in industry of cement, pulp and paper, textile and metal. Meanwhile, gas is used in industry of petrochemical, ceramics and metal. Other energy sources for industry are LPG, fuel, and electricity. In general, LPG is used in food industry, while fuel and biodiesel are used for several production processes or genset.

In 2022, industrial sector consumes 48% of coal, 27% of gas, 19% of electricity and the rest of fuel and biodiesel. Final energy consumption in industry in 2020 declines 21% into 41.4 MTOE compared to in 2019. However, in 2022, it is projected to increase around 8% into 45.2 MTOE compared to the consumption in 2020. The trend of energy consumption in industry is presented in Figure 3.10.

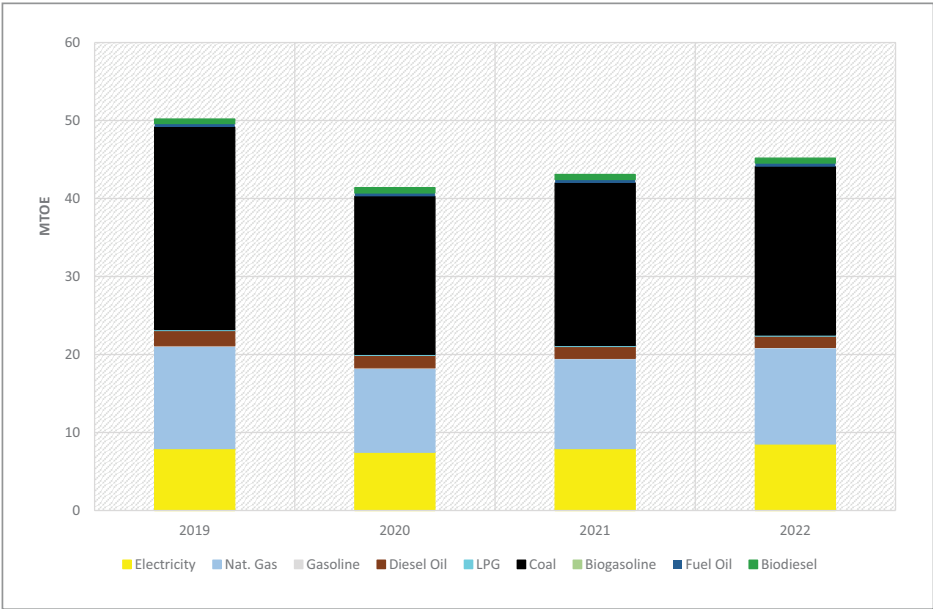


Figure 3.10 Final Energy Consumption in Industrial Sector

3.1.4 Commercial Sector

Energy consumption in commercial sector includes activities in offices, mall, hotel, restaurant, hospital, school and other services. Energy in commercial sector especially uses electricity for cooling (AC), water pump, and lighting. LPG is used for cooking especially in hotels and restaurants. Meanwhile, diesel and biodiesel are used only for genset as the back up of power supply.

In 2022, energy consumption in commercial sector will be dominated by electricity with 87% and LPG with 4%. In 2020, energy consumption in commercial sector declines 8% into 5.9 MTOE. However, in 2022 it is projected to increase into 6.4 MTOE with the gradual opening of offices, hotels, malls, restaurants, and schools. The trend of energy consumption in commercial sector is shown in Figure 3.11.

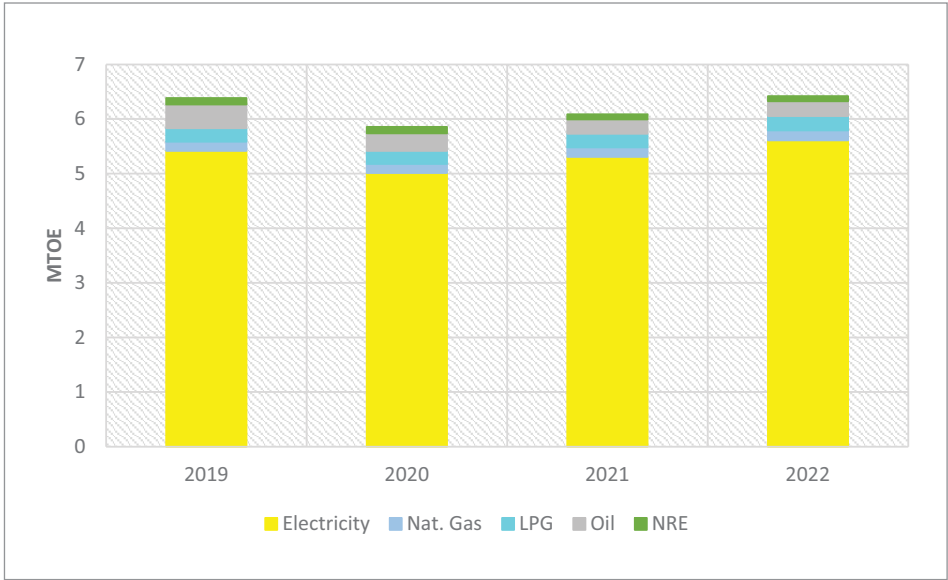


Figure 3.11 Final Energy Consumption in Commercial Sector

3.1.5 Other Sectors

Other sectors consist of three sectors namely agriculture, mining and construction. Energy demand in other sectors includes coal, diesel, biodiesel and electricity. Coal is used in mining, while diesel and biodiesel are used for genset as power supply back-up. Electricity is mainly used for lighting and other electronic devices.

Final energy consumption in other sectors, like in industrial sector, also experiences a decline of around 2% into 1.4 MTOE during the pandemic. The decline is influenced by the declining activity in mining and construction. Meanwhile, the activity in agriculture (including plantation) is increasing during the pandemic. In 2022, final energy consumption in other sectors will increase 7% into 1.5 MTOE when the economy is growing. The overview of final energy consumption in other sectors can be seen in Figure 3.12.

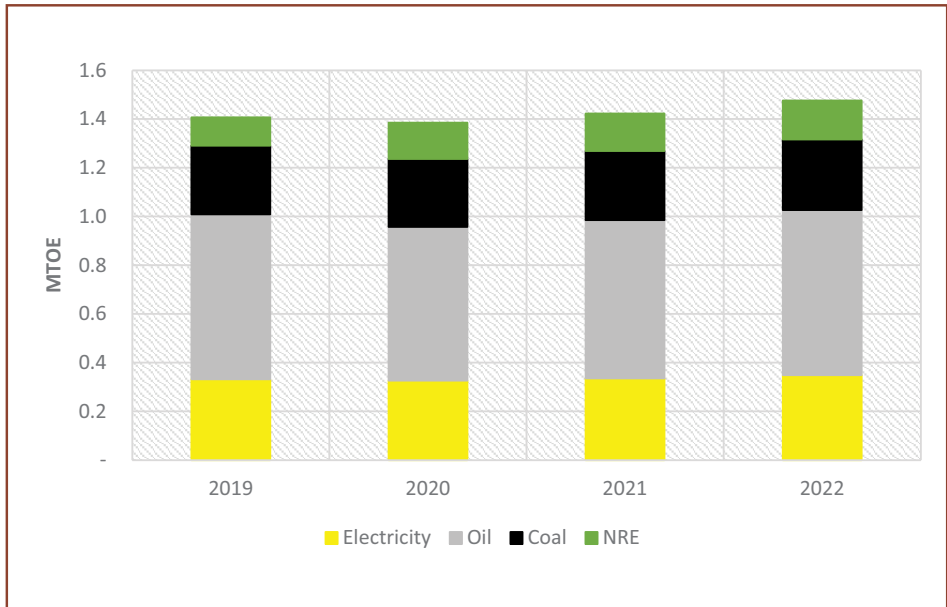


Figure 3.12 Final Energy Consumption in Other Sectors

3.2 ELECTRICITY CONSUMPTION PER SECTOR

Electricity demand will be the dominant energy in final energy in all sectors and it always grows higher than other types of energy since electricity is more environmentally friendly, cheaper and energy saving compared to other types of energy.

In 2022, electricity consumption is projected to increase 7% until 282 TWh. The increase of electricity consumption in 2022 is mostly influenced by the increasing consumption in industry and commercial sector around 12% and 11% respectively. Transportation sector shows 19% increase along with the promotion of electric vehicles especially bus, taxi, and online transportation operated by taxi companies in Jakarta, as well as Trans Jakarta bus serving the area of Jakarta. On the other hand, consumption in household is declining by 3%. The new normal activity post pandemic will lead to the increasing activity outside the house. This will reduce the electricity consumption in the house in 2022. The overview of electricity consumption per sector can be seen in Figure 3.13.

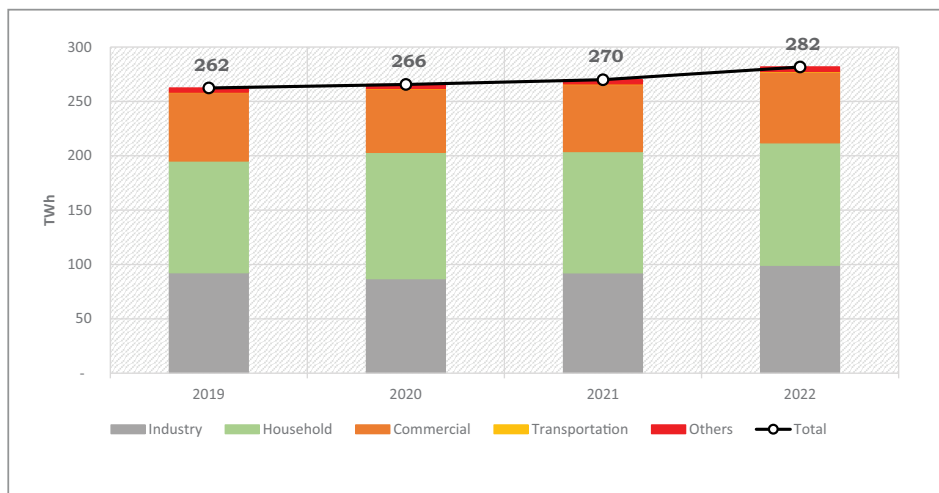


Figure 3.13 Electricity Consumption by Sector

3.3 ELECTRICITY CONSUMPTION PER PROVINCE

Based on data of electricity consumption by province, the biggest sales occur in West Java Province (20%) followed by East Java (15%) and DKI Jakarta (13%). This is influenced by the population in each province and the development of industrial and commercial sector. It is supported by the statistic which shows that West Java Province contributes the biggest household electricity customer (19%). Meanwhile, East Java has the biggest business and industrial units (17%). Jakarta Province's strategic position as the central government and the center of Indonesia economy also contributes to the high electricity demand in this province.

In total, electricity consumption in Java Island is 70% from the total national consumption, while consumption in Sumatera is 16% and consumption in Kalimantan is 5%. It shows that the growing infrastructure development outside Java is still far behind the condition in Java Island. The eastern part of Indonesia (Papua and Maluku) contributes not more than 2%. In addition, the population density is also one of the main factors in disparity since BPS data shows that 56% of population in Indonesia lives in Java Island. Total electricity customer in 2020 is 79% with 61% of it lives in Java Island. This fact further strengthens the need for accelerated development and equity outside Java.

Based on PLN sales per province data from PLN Statistics 2020 (Figure 3.14), electricity consumption per province in 2021 and 2022 is projected as presented in Table 3.1.

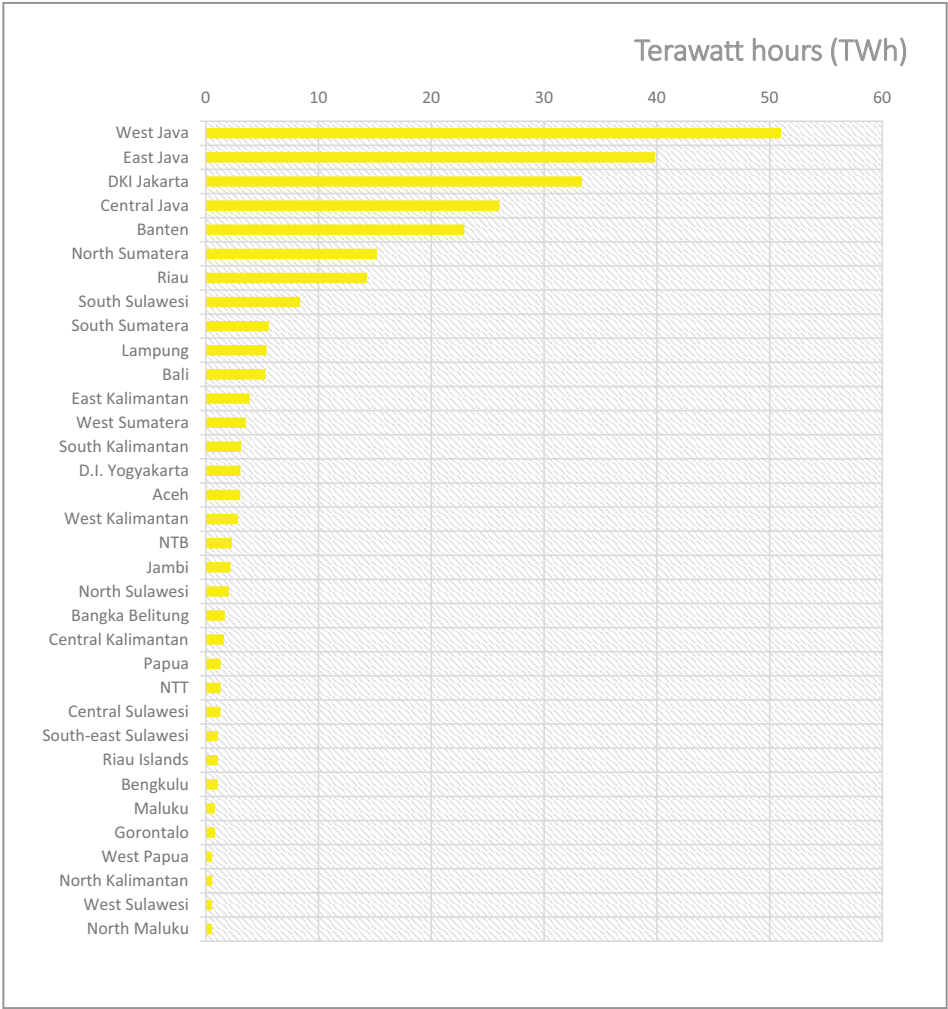


Figure 3.14 Electricity Consumption by Province in 2020

Table 3.1 Electricity Consumption per Province in 2021 - 2022

Province	Electricity Consumption (TWh)	
	2021	2022
Aceh	3.0	3.2
North Sumatera	15.1	16.0
West Sumatera	3.5	3.7
Riau	14.2	15.0
Riau Islands	1.0	1.1
Jambi	2.1	2.3
Bengkulu	1.0	1.1
South Sumatera	5.5	5.8
Bangka Belitung	1.6	1.7
Lampung	5.3	5.6
Banten	22.9	24.2
DKI Jakarta	33.3	35.2
West Java	51.0	53.9
Central Java	26.0	27.5
D.I. Yogyakarta	3.0	3.2
East Java	39.8	42.1
Bali	5.2	5.5
North Kalimantan	0.5	0.5
East Kalimantan	3.8	4.1
South Kalimantan	3.1	3.2
Central Kalimantan	1.6	1.6
West Kalimantan	2.8	2.9
NTB	2.2	2.4
NTT	1.2	1.3
Gorontalo	0.7	0.8
North Sulawesi	2.0	2.1
West Sulawesi	0.5	0.5
Central Sulawesi	1.2	1.3
South East Sulawesi	1.0	1.1
South Sulawesi	8.3	8.8
Maluku	0.7	0.8
North Maluku	0.5	0.5
Papua	1.3	1.3
West Papua	0.5	0.5
TOTAL	269.9	281.6

3.3.1 Power Plant Capacity by Type 2020-2022

The additional Indonesia power plant from 2020 to 2022 is planned to reach 12 GW based on RUPTL 2021-2030 with construction status. The additional power plant is used to meet electricity supply along with the projection of 6% electricity consumption increase until 2022. Coal fired power plant will still dominate the installed total capacity in 2022 of about 52% while gas fired power plant in 2022 shares about 29%. The domination of coal and gas power plant in Indonesia is predicted to survive until 2022 as there has been no change in energy price policy. Meanwhile, the share of NRE based-power plant will be only around 14% from the total capacity in Indonesia in 2022 (11.5 GW). The capacity of diesel-fired power plant will stay stagnant until 2022 shares 6% from the total national capacity. The details of power plant capacity until 2022 is shown in Table 3.2.

Table 3.2 Power Plant Capacity by Type 2020 - 2022

(GW)

Type of Power Plant	2020	2021	2022
PLTU/Steam Power Plant (PP)	36.7	41.3	43.8
PLTG/Gas-fired PP	5.3	5.5	5.5
PLTGU/Gas turbine PP	12.2	14.6	15.9
PLTMG/Combined cycle PP	3.2	3.3	3.3
PLTD/Diesed-fired PP	4.8	4.8	4.8
Total Fossil	62.2	69.48	73.2
PLTA/Hydro PP	5.6	6.0	6.0
PLTM/Minihydro PP	0.4	0.5	0.6
PLTMH/Microhydro PP	0.1	0.1	0.1
PLTB/Wind PP	0.2	0.2	0.2
PLTSa/Waste PP	0.0	0.0	0.0
PLTBioenergi/Bioenergy PP	1.9	1.9	1.9
PLTP/Geothermal PP	2.1	2.3	2.4
PLTS/Solar PP	0.1	0.1	0.3
PLTHibrid/Hybrid PP	0.0	0.0	0.0
Total NRE	10.4	11.1	11.5
Total	72.6	80.6	84.7

3.3.2 Additional Power Plant Capacity in 2021 and 2022

The additional power plant in 2021-2022 refers to data of power plant development plan with COD target only until 2022 and has been in construction status as included in RUPTL 2021-2030. The details of additional power plant capacity in 2021 and 2022 is shown in Table 3.3 below.

Table 3.3 Additional Power Plant Capacity 2021 – 2022

(MW)

Power Plant	2021	2022
PLTA/Hydro PP	400	-
PLT Bioenergi/Bioenergy PP	10	18
PLTG/Gas-fired PP	140	-
PLTGU/Gas Turbine PP	2,385	1,279
PLTM/Minihydro PP	125	105
PLTMG/Combined cycle PP	120	-
PLTP/Geothermal PP	136	108
PLTS/Solar PP	2	145
PLTU*/Steam PP	4,638	2,444
Total	7,956	4,099

It is targeted that in the next two years, there will be at least additional 12 GW power plant capacity in construction status consisting of 8 GW in 2021 and 4 GW in 2022. The additional power plant capacity is dominated by coal power plant (59%) and gas (30%). In total, NRE contributes 1 GW additional capacity or 9% from the total additional capacity.

With 12 GW additional power plant capacity until 2022, it is projected that the total electricity production will be 323 TWh in 2022.

a) Additional Power Plant Capacity in 2021

Coal utilization in additional power plant capacity will still dominate in 2021 of about 58%. The additional capacity of 4.7 GW Steam Power Plant (PLTU) in 2021 will be done in Java Island consisting of 4.2 GW (2,000 MW in PLTU Jawa-4, 1,900 MW in PLTU Jawa Tengah (PPP), and 315 MW in PLTU Lontar Exp). Therefore, the additional regional capacity in JMB (Java, Madura, Bali) will be equivalent to 89% of the total additional power plant in 2021. The additional PLTU by region is shown in Figure 3.15.

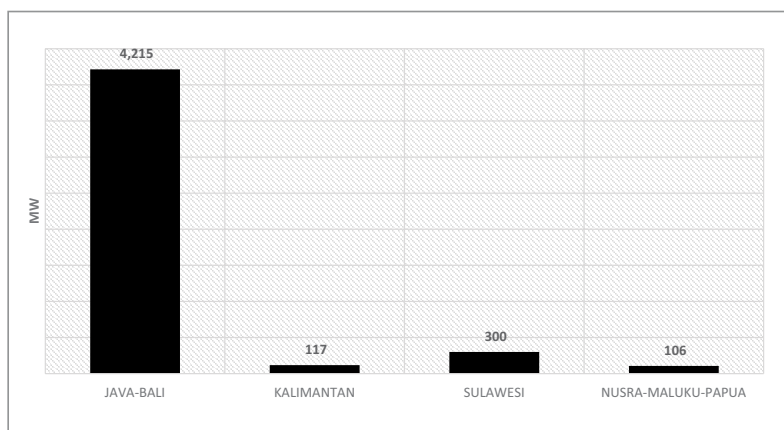


Figure 3.15 Additional Coal Power Plant

Gas will play a quite major role in additional power plant capacity of approximately 33% of the total additional capacity in 2021. The major gas utilization is supported by the development of Gas Turbine Combined Cycle (PLTGU) especially in JMB namely PLTGU Jawa-1 with the capacity of 1,760 MW, PLTGU Muara Tawar (add on block 2) with the capacity of 150 MW in East Java, and PLTGU Muara Karang with the capacity of 200 MW. The only PLTGU outside JMB region is PLTGU Riau with the capacity of 275 MW. PLTGU Riau is developed by PT Medco Power Indonesia. It is expected that in 2021, it will give additional capacity of 3.2 GW. Besides PLTGU, gas utilization in additional electricity capacity is also supported by the development of PLTG Bangkanai with the capacity of 140 MW in North Barito Regency-Central Kalimantan which uses Dual Fuel Technology (it may use gas or diesel oil). In addition, there is also PLTMGU Lombok Peaker in West Nusa Tenggara as the first combined cycle in Indonesia with gas engine in its combustion process. Meanwhile, diesel power plant will not change in 2021.

NRE utilization in 2021 is projected to contribute additional power plant capacity of 673 MW. The use of NRE in 2021 in this year's program will be still dominated by hydro power plant, geothermal power plant, and minihydro power plant which are continuously contributing to additional capacity of 400 MW, 136 MW, and 125 MW. Meanwhile, hydro power plant development in 2021 will include Poso Peaker Hydro Power Plant in Central Sulawesi with the capacity of 200 MW, Malea Hydro Power Plant in South Sulawesi with the capacity of 90 MW and Jatigede Hydro Power Plant in West Java with the capacity of 110 MW. Jatigede Hydro Power Plant has been developed since 2015 and later this power plant will use the water from Jatigede Water Reservoir in Cimanuk River with intake structure to flow the water from the tunnel to power house to produce electricity. The details of additional power plant capacity in 2021 is shown in Figure 3.16.

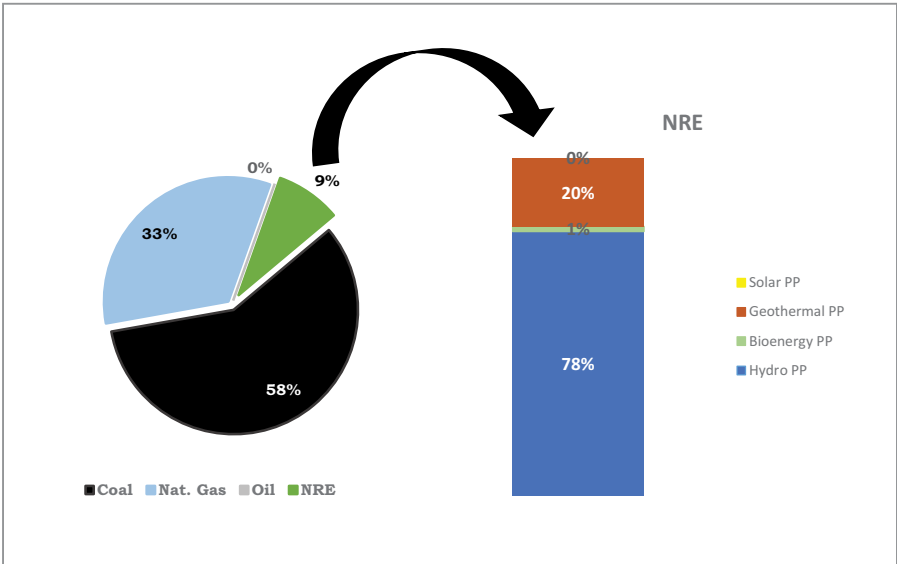


Figure 3.16 Additional Power Plant Capacity

Geothermal power plant development in 2021 will include Rantau Dedap Geothermal PP in South Sumatera with the capacity of 86 MW, Sorik Marapi Geothermal PP in North Sumatera with the capacity of 45 MW and Sokoria Geothermal PP in East Nusa Tenggara with the capacity of 5 MW.

Minihydro PP development in 2021 will include Tomata Minihydro PP, Parmonangan-2 Minihydro PP and Cisako-3 Minihydro PP with the capacity of 10 MW each, as well as Sei Wampu Minihydro PP, Lintau-1 Minihydro PP, and Anggoci Minihydro PP with the capacity of 9 MW each. There are also several other minihydro PP projects with smaller capacity. Besides these three types of energy, the additional capacity from NRE-based power plant is also supported by the development of small-scale power plants from other renewable energy such as: 2.4 MW of Rural Electric Program (*Lisdes*) Solar PP in East Nusa Tenggara for solar, 1 MW of Sumba Biomass PP for bioenergy, and 9 MW of Benowo Waste PP for waste utilization. Benowo Waste PP has been officially inaugurated by President Joko Widodo in May 2021. This PP is expected to utilize 1,600 ton of waste from Benowo Final Waste Dump each day.

b) Additional Power Plant Capacity in 2022

In 2022, coal steam power plant is targeted to contribute 60% from the total additional capacity in 2022. This project will include PLN's steam power plant in all regions such as Jawa-1 with the capacity of 924 MW, Kaselteng-2 with the capacity of 200 MW, Lombok with the capacity of 100 MW, Nabire-Kalibodo with the capacity of 14 MW, Cofiring in Talaud with the capacity of 6 MW and PLTU-MT Sumsel-8 with the capacity of 1,200 MW. PLTU-MT Sumsel 8 will be developed to utilize coal from PT Bukit Asam mining site in Muara Enim by using subcritical technology in its boiler for a more environmentally friendly process. Besides this data, Directorate General of Mineral and Coal data also shows that plan of additional capacity from non-PLN power plant namely PLTU owned by PT Halmahera Persada Lygend with the capacity of 60 MW, PLTU owned by PT Alchemist Metal Industry with the capacity of 120 MW, and PLTU owned by PT Well Harvest Winning Alumina Refinery with the capacity of 80 MW. These three PLTU are expected to operate in 2022.

The use of gas is expected to contribute 31% from the planned additional power plant capacity in 2022. This number is derived from the development of PLTGU Jawa-Bali 1 with the capacity of 779 MW and PLTGU Muara Tawar (add-on block 3 and 4) with additional capacity of 500 MW. PLTGU Muara Tawar uses gas from LNG regasification in FSRU Lampung. With this project, PLTGU Muara Tawar contributes 24% to the total gas utilization in electricity with special operation pattern namely as peaker that uses gas at the peak load. There is no additional capacity of Diesel power plant in 2022.

The additional capacity of 376 MW from NRE-based power plant in 2022 will rely on solar PP, geothermal PP, and Minihydro PP that contribute the additional capacity of 145 MW, 108 MW and 105 MW each. The capacity of 145 MW from solar energy is

expected to be derived from the development of Cirata Solar PP in West Java. Cirata Solar PP is also one of the national strategic projects. Cirata Solar PP has become the biggest floating solar PP in South East Asia with the investment of 129 million USD.

Meanwhile, geothermal PP development in 2022 will cover Lumut Balai geothermal PP in South Sumatera with the capacity of 55 MW, Sorik Marapu geothermal PP in North Sumatera with the capacity of 50 MW and Sokoria geothermal PP in East Nusa Tenggara with the capacity of 3 MW. Minihydro PP in 2022 includes Parluasan minihydro PP, Tunggang minihydro PP, Bone Bolango minihydro PP, Harjosari minihydro PP, and Pelangai Hulu minihydro PP with the capacity of 10 MW each. It will be also supported by several other minihydro PP with smaller capacity. Besides the three types of energy above, additional capacity in 2022 will be also supported by other NRE-based mini PP such as biomass PP, biogas PP and waste PP. The details of additional power plant capacity in 2022 can be seen in Figure 3.17.

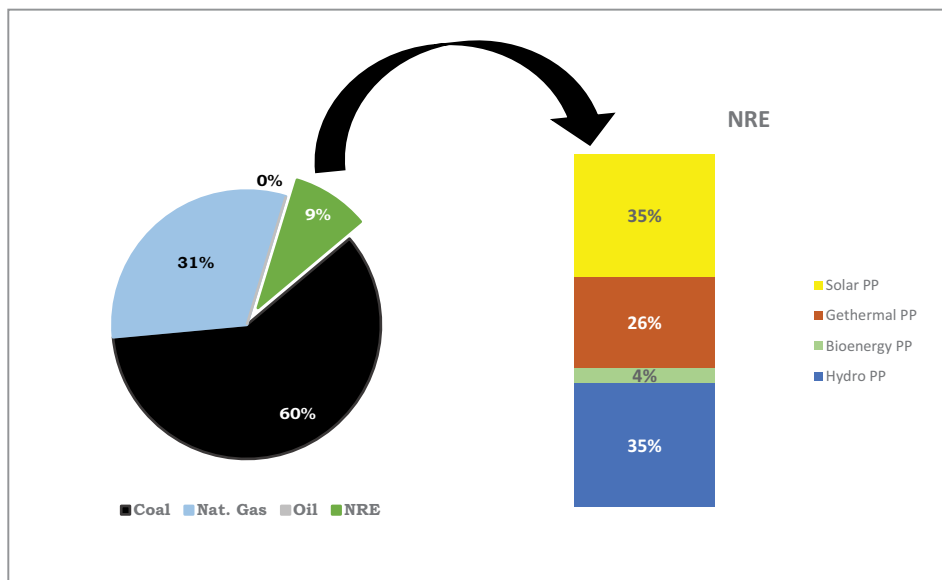


Figure 3.17 Additional Power Plant Capacity in 2022

3.3.3 Electricity Production by Type in 2020-2022

Electricity production in Indonesia until 2022 will increase 5.5% into 323 TWh from the production in 2020. Along with the power plant project development, electricity production from coal steam PP will reach 54% from the total electricity production in 2022. The comparison of electricity production by type (TWh) in 2020-2022 can be seen in Table 3.4.

Table 3.4 Electricity Production by Type 2020-2022

(TWh)

Power Plant	2020	2021	2022
PLTU/Steam PP	169	169	177
PLTU MT/Mine mouth PP	3	3	3
PLTG/Gas fired PP	8	7	7
PLTGU/Gas turbine PP	52	55	59
PLTMG/combine cycle PP	5	4	4
PLTD/Diesel fired PP	15	13	13
PLTA/Hydro PP	25	26	26
PLTM/Minihydro PP	2	2	3
PLTMH/Microhydro PP	0	0	0
PLTB/Wind PP	1	1	1
PLTSa/Waste PP	0	0	0
PLTBioenergi/Bioenergy PP	10	10	10
PLTP/Geothermal PP	17	18	19
PLTS/Solar PP	0	0	1
PLTHibrid/Hybrid PP	0	0	0
Total	306	310	323

3.4 PRIMARY ENERGY SUPPLY IN 2020-2022

The primary energy supply projection in 2020-2022 is made based on data from related units such as oil and gas production from SKK Migas or Directorate General of Oil and Gas and coal production projection from Directorate General of Mineral and Coal. Based on projection until 2022, Indonesia primary energy supply is 232 MTOE which will be dominated by coal with 87 MTOE. The high coal supply will be influenced by coal consumption in steam PP and industry. Meanwhile, oil supply in 2022 will be 64 MTOE. NRE supply in 2022 will be 27 MTOE and gas supply in 2022 will be 54 MTOE with electricity as the biggest user. NRE supply in 2022 will be partly used for power plant and biodiesel mixture. The primary energy supply in 2020-2022 is shown in Figure 3.18.

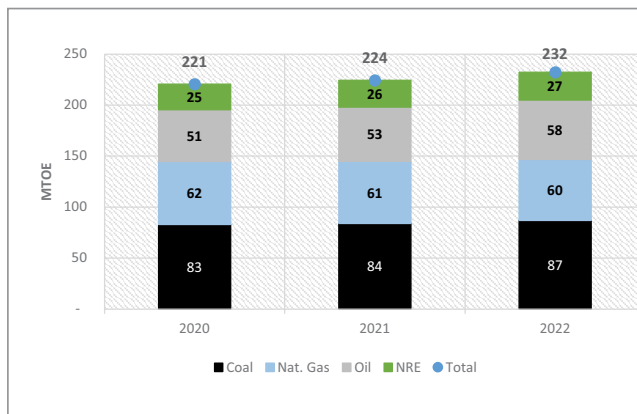


Figure 3.18 Primary Energy Supply in 2020-2022

3.4.1 Primary Energy Mix

Energy mix projection in 2020-2022 is fluctuating. Based on projection, primary energy mix in 2022 will be 37% for coal, 27% for oil and 23% for gas, and only 12% for NRE. The primary energy mix can be seen in Figure 3.19.

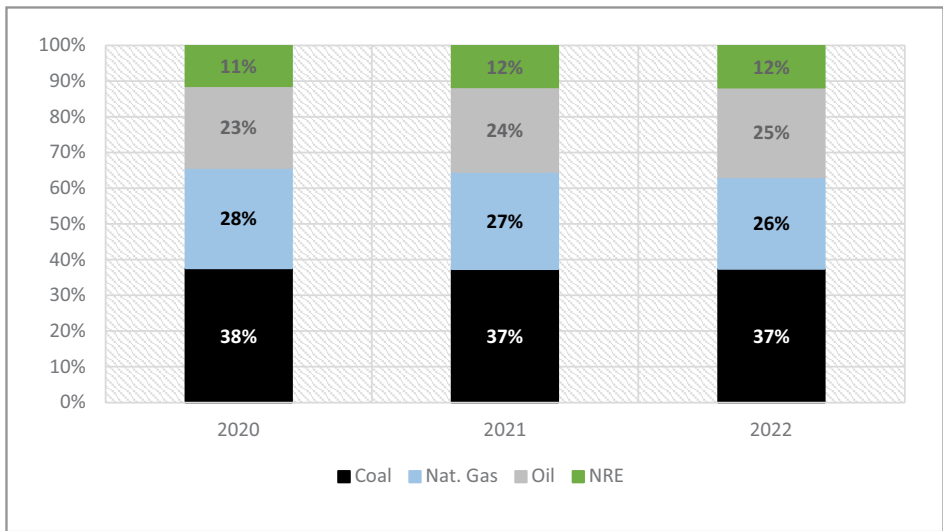


Figure 3.19 Primary Energy Mix in 2022

3.5 ELECTRICITY CONSUMPTION PER CAPITA

Electricity consumption per capita in 2022 will be 1,173 kWh/capita from 1,133 kWh/capita in 2020. This number will be far below the target in RUEN. In 2025, the target of electricity consumption per capita in RUEN is 2,500 kWh/capita. Electricity consumption per capita can be seen in Figure 3.20 below.

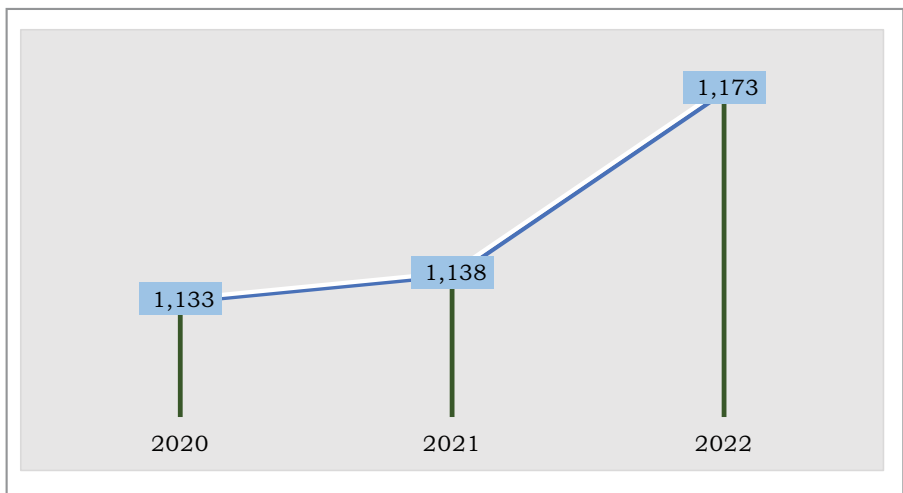


Figure 3.20 Electricity Consumption per Capita

3.5.1 Primary Energy Per Capita

Similar to electricity consumption per capita, the use of primary energy per capita is equivalent to economic growth. In 2022, primary energy per capita will be 0.84 TOE/Capita. It will increase from 0.82 TOE/Capita in 2020. This achievement is far from the RUEN target of 1.4 TOE/Capita in 2025 as presented in Figure 3.21.

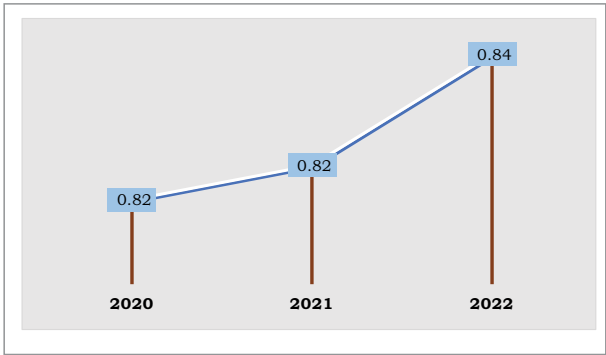


Figure 3.21 Primary Energy per Capita

3.6 EMISSION

Power plant sector is projected as the biggest emission contributor (48%), followed by transportation (26%) and industry (21%) in 2022. CO₂ emission in 2022 will increase into 582 million tonnes CO₂eq as presented in Figure 3.23. In the last three years, the biggest CO₂ emission contributors will be transportation with the average increase of 17%, followed by industry and power plant with 7% and 5% each. On the other hand, household shows CO₂ emission reduction of 8% due to the declining activities at home when the covid19 pandemic is recovering (Figure 3.22).

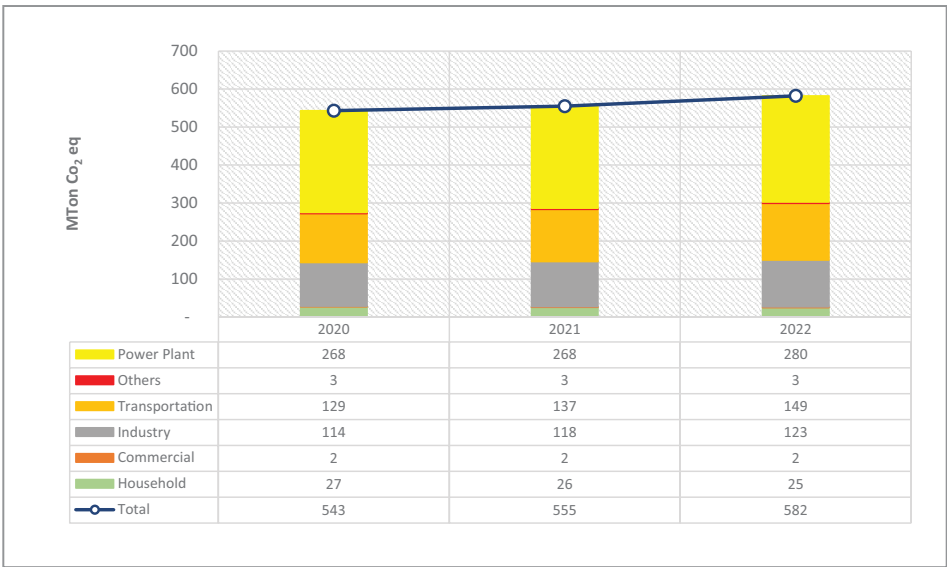


Figure 3.22 Emission per Sector in 2020 - 2022

Meanwhile, emission per capita in 2020-2022 is presented in Figure 3.23.

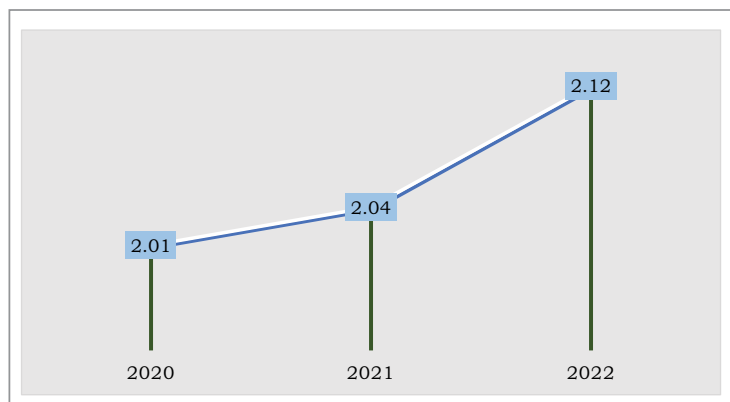


Figure 3.23 Emission per Capita in 2020 - 2022

The summary of several energy indicators based on KEN GR is shown in Table 3.5.

Table 3.5 Energy Indicator

Energy Indicator	Unit	2020	2021	2022
Primary Energy Supply	MTOE	221	224	232
Primary Energy Mix				
Coal	%	37.6	37.3	37.4
Oil		23.0	23.7	25.1
Gas		28.0	27.2	25.7
NRE		11.4	11.8	11.8
Electricity Consumption	kWh/Capita	1,133	1,138	1,173
Primary Energy Consumption	TOE/Capita	0.82	0.82	0.84
Energy Elasticity		0.17	1.17	1.08
Primary Energy Intensity	TOE/Billion Rupiah	20.0	20.2	19.8

3.7 ENERGY BALANCE 2022

Energy balance is the balance between supply and demand that shows the flow of energy supply and demand from production, transformation to end user. This flow shows the capability of national production for each type of energy, volume of export and import, supply for refinery and power plant, own use and losses, as well as national energy consumption per consumer sector in certain period (annually). Figure 3.24 shows energy flow in 2020 with energy supply consisting of 483.6 MTOE from energy production and 53.3 MTOE from import. Meanwhile, consumption consists of 304.5 MTOE from export and 128.9 MTOE from domestic use. Thus, total primary energy in 2022 will be 232.4 MTOE.

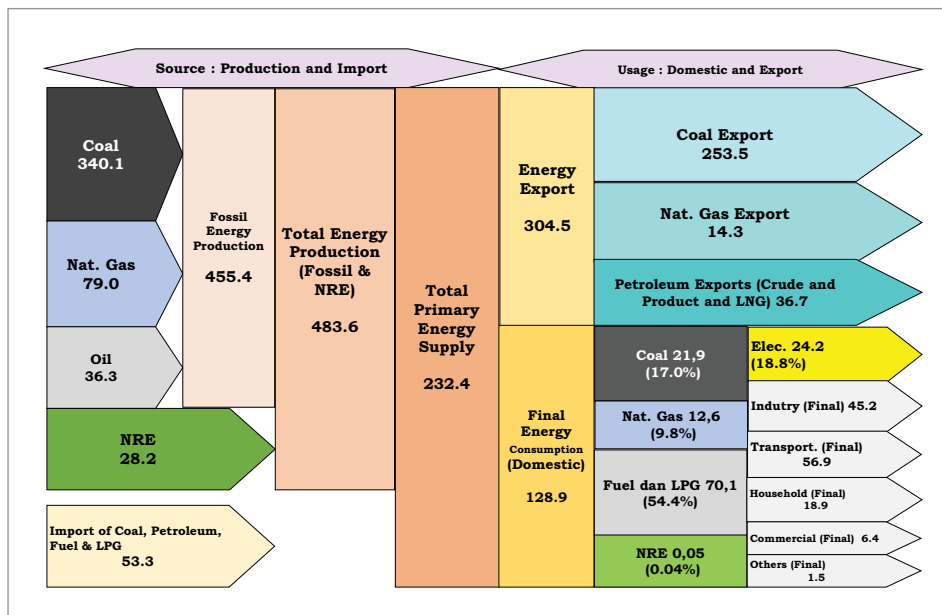


Figure 3.24 Energy Flow in 2022

3.8 ENERGY POLICY

The national energy policy as determined in Government Regulation No. 79 of 2014 is set as a guidance in national energy management to achieve energy independence and energy security to support the sustainable national development. RUEN, as the derivate of KEN, is stipulated in Presidential Regulation No. 22 of 2017 of which one of the targets is to achieve NRE target in primary energy mix by 23% in 2025 and 31% in 2050. In addition, Indonesia has set target and strong commitment to implement clean energy in enhancing energy supply through renewable energy utilization and investment. In NDC document, Indonesia is committed to reduce GHG emission by 29% through self effort and 41% through international cooperation in 2030. In line with commitment, the government has carried out improvement in formulating rules and regulations to achieve the target and commitment.

Several policies to achieve the targets are as follows:

3.8.1 Oil and Gas

The amendment to Oil and Gas Law is still in the process now. Improvements are continuously made to Law No. 22 of 2001 on Oil and Gas. A new regulation on oil is Minister of Energy and Mineral Resources No. 42 of 2018 on Priority to Use Oil for Domestic Need. The regulation says that Pertamina and Business Entity with Oil Processing License are required to prioritize oil supply from the domestic. Contractors

and their affiliates are required to offer oil from contractor split to Pertamina and/or Business Entity with Oil Processing License. It is aimed to maximally use oil for domestic refinery.

In addition, Ministry of Energy and Mineral Resources has also issued Ministerial Regulation No.08 of 2017 on Gross Split PSC scheme. Gross Split is a scheme that calculates the state and contractor's split in advance. Through Gross Split, the State will receive oil split and tax from exploration and exploitation activity that contributes to a firmer state revenue. The gross split calculation differs for each work area but the basic calculation lies in the Base Split percentage. The base split of oil is 57% for the state and 43% for the contractor. For gas, 52% is for the state and 48% for the contractor. Besides base split, the state and contractor may receive bigger share from additional ten variable components and two progressive components. It makes gross split more attractive to investors to operate oil and gas work area including unconventional work area that has more challenges. In gross split, operation cost is fully borne by contractors. It is different with cost recovery in which operation cost is borne by the government. Thus, contractors are encouraged to be more efficient as the operation cost is on their own. The more efficient than the more the contractors gain profit. Until June 2019, 42 contracts use gross split consisting of 21 ex termination, 16 tendered work areas, and five amendment work areas from cost recovery to gross split.

Offshore North West Java (ONWJ) operated by Pertamina Hulu Energi (PHE) is the first work area with gross split scheme. Gross split is focused on work areas with extended contract and new work area contract to respect the ongoing contracts until they expire.

In 2020, the Ministry of Energy and Mineral Resources stipulates Ministerial Regulation No. 12 of 2020 on the Third Amendment to Ministerial Regulation No. 08 of 2017 on Gross Split PSC. The amendment says that PSC contractors are not required to use gross split. They are free to choose between cost recovery or other scheme with some adjustment and improvement in their previous contracts.

Concerning gas price policy, the Government of Indonesia has also set gas price for certain industries at 6 USD/MMBTU based on Presidential Regulation No. 40 of 2016 on Gas Price Stipulation. The policy on gas price for industry is expected to promote utilization and competitiveness in domestic manufacture industry that will lead to significant contribution to national economic growth. As we know, the structure of gas component production cost in industry is quite high. For example, 26-30% production cost in ceramic industry is to buy gas. The declining gas price is expected to promote domestic industry competitiveness that leads to more competitive product price. The derivative rules of Presidential Regulation No. 40 of 2016 is Minister of EMR Regulation No. 8 of 2020 on Stipulation of Certain Consumer and Gas Price in Industry and its technical regulation in Minister of EMR Decision 89 K/10/MEM/2020 on Certain Consumer and Gas Price for Industry. This Ministerial Decision sets seven industries that receive gas price at 6 USD/MMBTU namely the industry of fertilizer, petrochemical, oleochemical, steel, ceramic, glass and rubber gloves. This price scheme will be valid from 2020 to 2024.

3.8.2 Coal

Indonesia makes Domestic Market Obligation (DMO) policy as an obligation for coal company to sell its part of the production to domestic consumption. The volume of DMO is regulated each year through Minister of EMR Decision. Based on Minister of EMR Decision No. 139.K/HK.02/MEM.B/2021 of 2021 on Coal Supply for Domestic, DMO is set at 25% of the coal company production in 2021. The regulation says company that does not comply with DMO will be given a sanction in the form of export restriction as well as fines and compensation funds. Coal DMO is expected to secure coal supply for domestic need. The Ministerial Regulation also sets coal selling price for power supply for public at USD 70 per ton. The price stipulation is done to secure PLN to buy domestic coal based on its financial capability without additional subsidy and increase in its basic tariff.

3.8.3 Renewable Energy

a. Geothermal

One of the factors that leads to the less developed geothermal energy in Indonesia is that geothermal tariff is not competitive due to the high cost and risk in geothermal exploration. In optimizing geothermal utilization, achieving NRE target, and carrying out Indonesia's commitment in reducing emission, the government has made some breakthroughs in geothermal sector. One of them is through deep slim hole exploration drilling funded by State Budget. This first government drilling program is done in Cikakak Subdistrict, Sukabumi Regency, West Java. This government's geothermal exploration is purposed to increase data accuracy that will reduce the risk in upstream side and reduce electricity tariff to be able to compete with other energy. This program is expected to be continued to create a more interesting investment and a more competitive geothermal tariff.

b. Net Metering

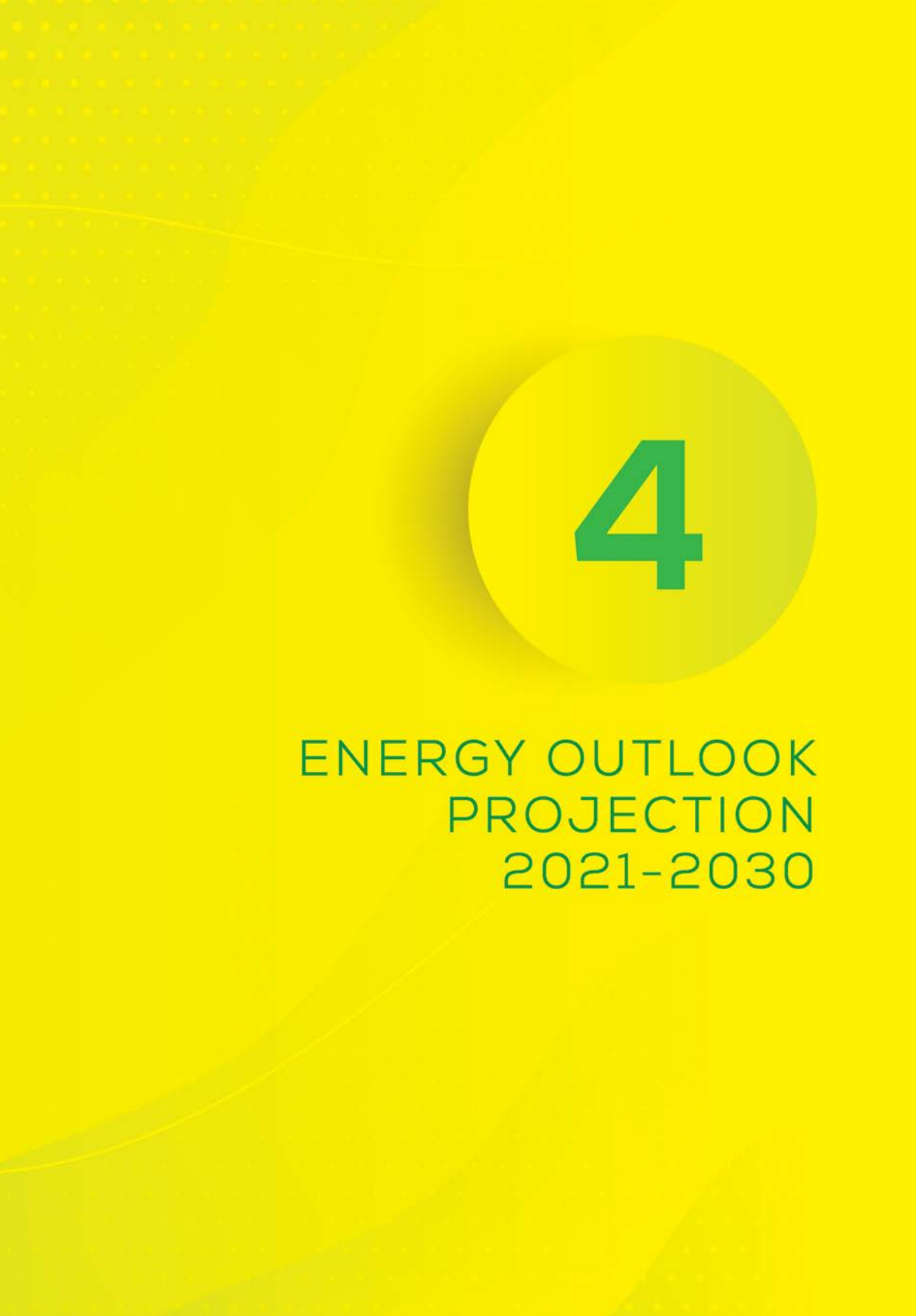
In 2018, the Government of Indonesia through the Ministry of Energy and Mineral Resources has issued Ministerial Regulation No. 49 of 2018 on Solar Panel Power Plant System by PT PLN (Persero)'s Consumer. One of the Article sets PT PLN's obligation to buy electricity from PLN's consumer Solar Panel. This regulation has been amended twice through Ministerial Regulation No. 13 of 2019 on the Amendment to Ministerial Regulation No. 49 of 2018 and through Ministerial Regulation No. 16 of 2019 on the second amendment to Ministerial Regulation No. 49 of 2018.

In 2021, MEMR will make improvements to the regulation, where one of the points that have been improved is related to increasing the provisions for the export of electricity by PLTS Rooftop customers to PT PLN (Persero) from previously only 65% of PLTS Roof production to 100% of connected power PLTS Roof. This is done based on input from various groups, both industry,

associations, and the community while still upholding the fairness regulated in the MEMR Number 26 of 2021 concerning Rooftop Solar Power Plants Connected to the Electric Power Network for the Public Interest.

c. Renewable Energy for Power Supply

The government through Ministry of Energy and Mineral Resources has also made improvements in renewable energy regulations in order to enhance and accelerate the use of renewable energy for domestic power supply. The Ministry of EMR has issued Ministerial Regulation No. 4 of 2020 on the Second Amendment to Ministerial Regulation No. 50 of 2017 on Renewable Energy Utilization for Power Supply. Several points in the amendment are the provisions in electricity purchase through direct appointment mechanism by PLN which is accommodated by a number of requirements and time arrangement of NRE-based electricity purchase through direct appointment by PLN. The amendment also revokes electricity purchase with Build, Own, Operate and Transfer/BOOT cooperation scheme. Previously, this scheme enables the transfer of all NRE based power plant asset to PLN that is assumed to be less profitable for NRE developer which ends the NRE projects to face financial issue or not bankable. It is hoped that the amendment to the regulation will promote investment in renewable energy sector.



4

ENERGY OUTLOOK PROJECTION 2021-2030

4

ENERGY OUTLOOK PROJECTION 2021-2030

4.1 COMPARISON OF FINAL ENERGY CONSUMPTION PER SECTOR

Final energy demand in each scenario is influenced by economic growth, population growth and policies made during projection period. If it is compared, total final energy consumption in both scenarios is not quite different since it uses the same economic and population growth assumption. The difference between the two scenarios is influenced by the policy of fuel to gas substitution, electric vehicle, and biofuel for transportation. In household sector, the difference is the substitution of LPG stove to gas, electricity, and DME stove as well as power plant capacity in Realistic Scenario with construction status in RUPTL. Energy demand in each scenario in 2030 is shown in Figure 4.1.

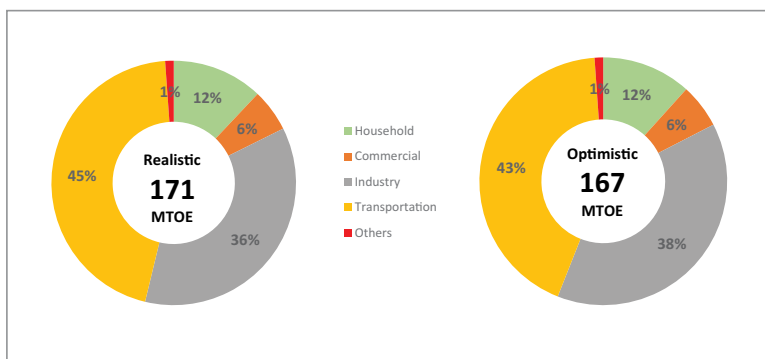


Figure 4.1 Comparison of Final Energy Demand in Both Scenarios in 2030

Based on type of energy, oil fuel becomes the biggest energy source in 2020 with 39% followed by electricity with 19%, coal with 17% and gas with 18%. Meanwhile, NRE share in final energy consumption is only 7% as seen in Figure 4.2.

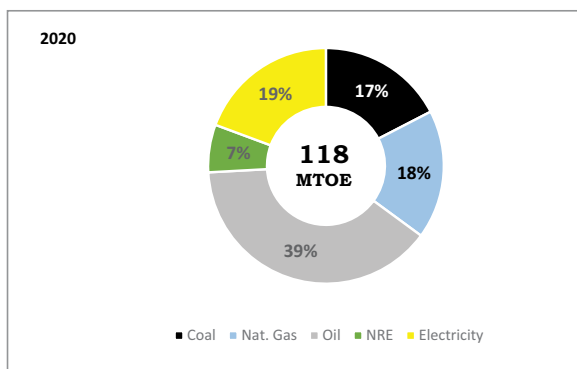


Figure 4.2 Share of Final Energy Consumption by Type of Energy in 2020

There is no significant change in energy use substitution in Realistic scenario. Thus, in 2030 the biggest energy demand is from fuel (41%). In Optimistic scenario, the role of fuel declines into 36% due to substitution policy from fuel to gas, electricity and biofuel. Meanwhile, NRE increases into 10% with the use of biogasoline. The complete overview of final energy demand share per type of energy in 2030 is presented in Figure 4.3.

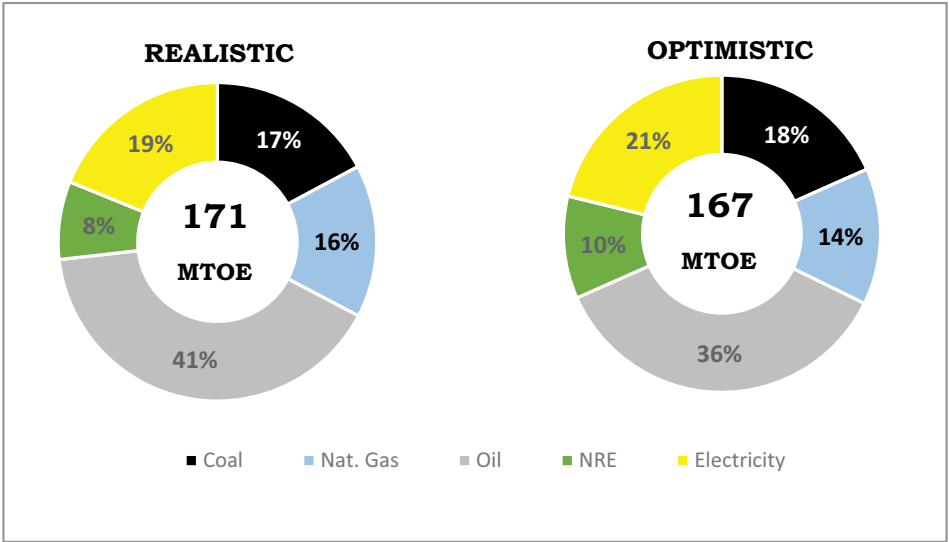


Figure 4.3 Comparison of Final Energy Demand in 2030

4.1.1 Industrial Sector

Coal, gas and electricity become the biggest energy sources consumed in industrial sector. Meanwhile, NRE (biomass) is mostly used for food as well as pulp and paper industry. Several food industries still use biomass (charcoal and wood) as fuel, while pulp and paper industry use renewable energy such as oil palm shells, rice straw, biogas and black liquor to substitute coal. Fuel is only used for genset (power supply back up).

Industrial sector has not developed innovation plan on energy substitution, except biodiesel program that has been implemented in all sectors and biomass substitution for pulp and paper industry. Energy demand projection in industry based on two scenarios is shown in Figure 4.4.

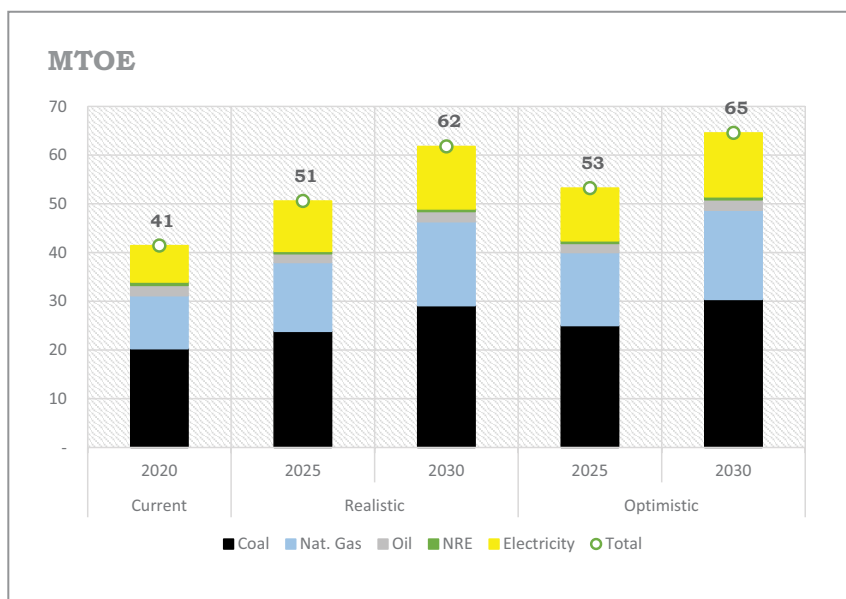


Figure 4.4 Final Energy Demand in Industrial Sector by Type of Energy

In total, types of industry with the most energy consuming category are industry of metal, cement, food and beverage. The biggest energy consumption in industrial sector in 2030 is 26% from metal industry, 21% from cement industry, and 14% from food industry as shown in Figure 4.5. The high energy consumption in these industries is influenced by the use of old machine. These industries have not used new technology that leads to wasteful energy consumption. In the future, these industries should be guided to use a more efficient new technology to support energy conservation and emission reduction program.

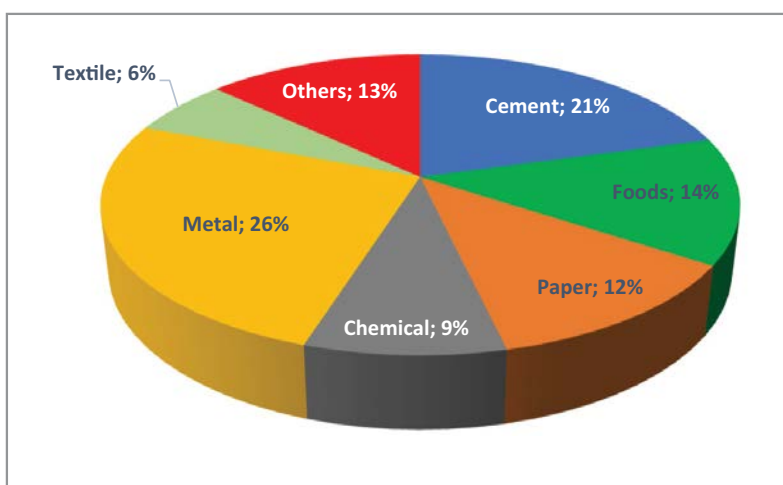


Figure 4.5 Share of Energy Demand sub sector Industries in Realistic Scenario in 2030

4.1.2 Transportation Sector

In 2020, 86% of energy consumption in transportation is derived from fuel. While the rest 14% is from biofuel (biodiesel). Meanwhile, consumption of gas and electricity is still relatively low. Gasoline consumption in 2020 is 31 million KL of which 31% of it is from import. If this condition is not anticipated, it will burden the state and influence trade balance. Therefore, the government is reducing fuel import especially gasoline through substitution to biofuel (biodiesel and biogasoline), gas and battery based-electric vehicle. The optimistic scenario in transportation sector refers to GSEN that fuel consumption is far lower than in realistic scenario since it is replaced by electricity, gas and biodiesel. Energy consumption projection in transportation sector per type of energy is presented in Figure 4.6.

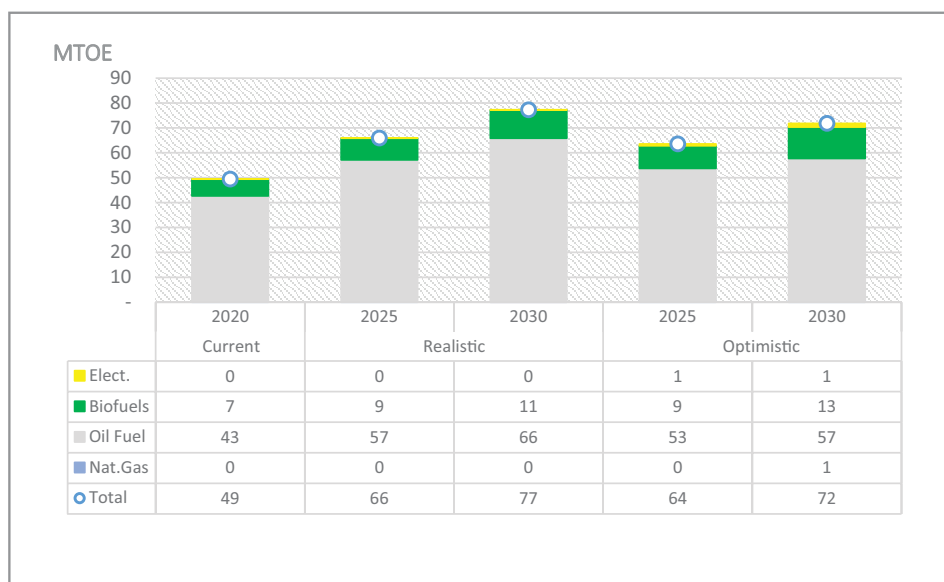


Figure 4.6 Energy Demand in Transportation Sector in Both Scenarios

The use of gasoline in cars and motorcycles is still dominating until 2030. As the result, energy demand for private vehicles is increasing and is still dominated by fuel. In order to reduce dependency on fuel especially gasoline, the government is promoting electric, gas and biogasoline fueled vehicles. In 2030, electricity consumption by cars is projected of about 287 thousand TOE (2.3%) in Optimistic scenario and 175 TOE in Realistic scenario. Meanwhile, biofuel consumption by cars in the same year is around 460 thousand TOE (4%) in Realistic scenario and 1.4 million TOE (11%) in Optimistic scenario. Gas consumption is increasing into 142 thousand TOE in Optimistic scenario and only 190 TOE in Realistic scenario. The share of final energy consumption by cars in 2030 is shown in Figure 4.7.

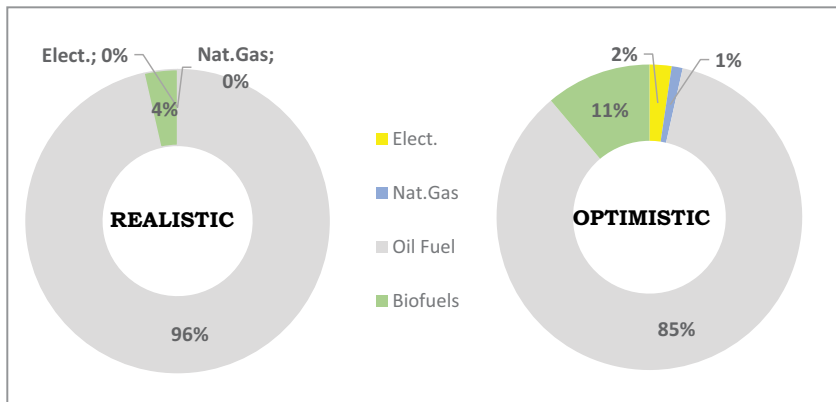


Figure 4.7 Final Energy Consumption by Car in 2030

For motorcycle, electricity consumption in 2030 is 800 thousand TOE (4%) in Optimistic scenario and 716 TOE (0.003%) in Realistic scenario. Meanwhile, biogasoline consumption is 2.3 MTOE (10%) in Optimistic scenario and 0.5 MTOE (2%) in Realistic scenario. In total, gasoline consumption declines into 86% in Optimistic scenario and stays at 98% in Realistic scenario from the total energy consumption by motorcycle. The share of final energy consumption by motorcycle in 2030 is presented in Figure 4.8.

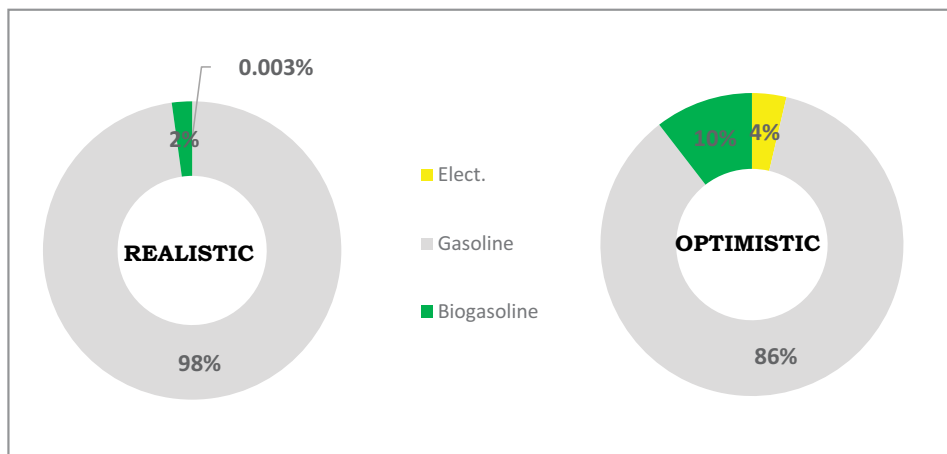


Figure 4.8 Final Energy Consumption by Motorcycle in 2030

4.1.3 Household Sector

Energy demand in household sector is not only influenced by the growing number of households but also by the pandemic.

LPG consumption in household in the next 10 years is projected to decline from 9.5 MTOE in 2020 into 8.4 MTOE (Realistic) and 2.2 MTOE (Optimistic). The declining LPG

consumption is influenced by the substitution of LPG stove to induction and DME stove. In Optimistic scenario, LPG consumption is projected to decline 7.3 MTOE in 10 years due to the use of induction stove and city gas for 1 million households per year as well as the use of DME starting in 2025. The final energy demand in household sector based on Realistic and Optimistic scenario is presented in Figure 4.9.

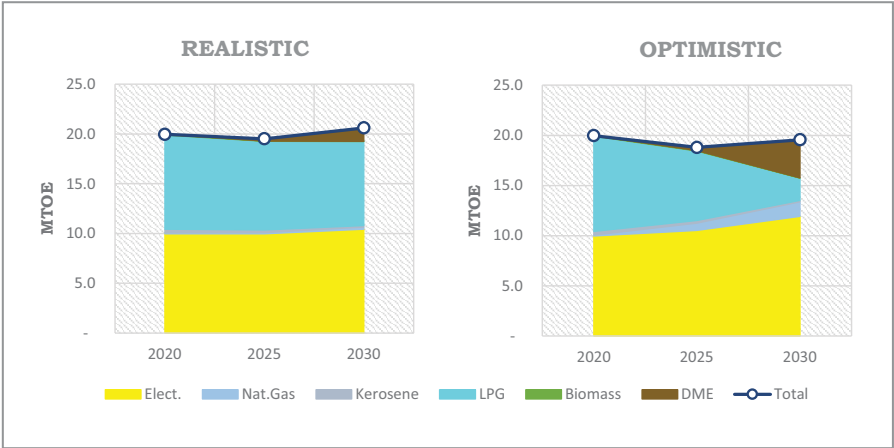
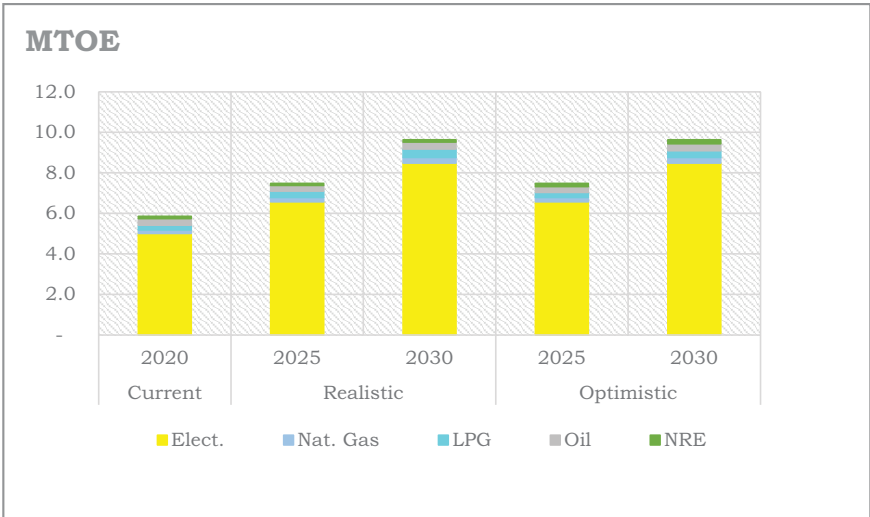


Figure 4.9 Energy Demand in Household Sector per Type of Energy

4.1.4 Commercial Sector

In 2020, energy demand in commercial sector is dominated by electricity (86%) and LPG (4%), while the rest is from gas, diesel and biodiesel. Until 2030, energy demand in commercial sector is projected to increase from 6 MTOE in 2020 into 10 MTOE for both scenarios (Realistic and Optimistic), but the energy demand composition for both scenarios is not quite different (Figure 4.10).



Picture 4.10 Energy Demand in Commercial Sector by Type of Energy

4.1.5 Other Sectors

Other sectors consist of three subsectors namely agriculture, mining and construction. Total final energy demand in other sectors in 2020 is 1.4 MTOE and it is projected to reach 1.9 MTOE (for both scenarios) in 2030 with the projection of per type of energy demand as shown in Figure 4.11.

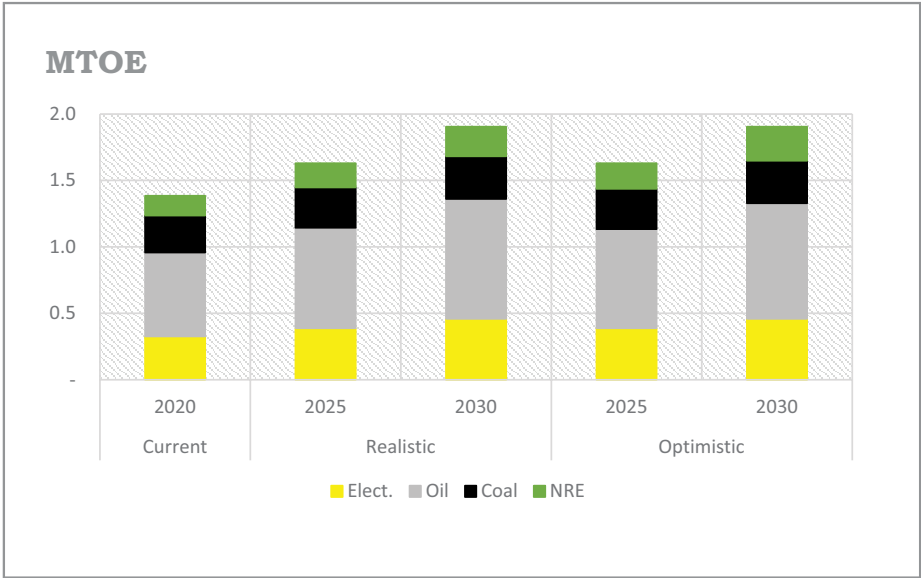


Figure 4.11 Energy Demand in Other Sectors by Type of Energy

4.2 ELECTRICITY CONSUMPTION PROJECTION PER SECTOR

Total electricity demand in 2020 is 266 TWh, while total electricity demand projection in 2030 will increase into 376 TWh in Realistic scenario and 398 TWh in Optimistic scenario. The share of electricity demand in 2020 is derived 44% from household sector, 33% from industry, 22% from commercial sector, 1% from other sectors, and 0.2% from transportation. Electricity consumption in 2030 in Optimistic scenario is projected to change and the biggest consumption will shift to industry with 37%, followed by household with 34%, commercial sector with 24%, transportation with 4% and other sectors with 1%. The increase of electricity consumption occurs in industry and transportation. The projection of electricity demand per sector in 2030 for each scenario is shown in Figure 4.12.

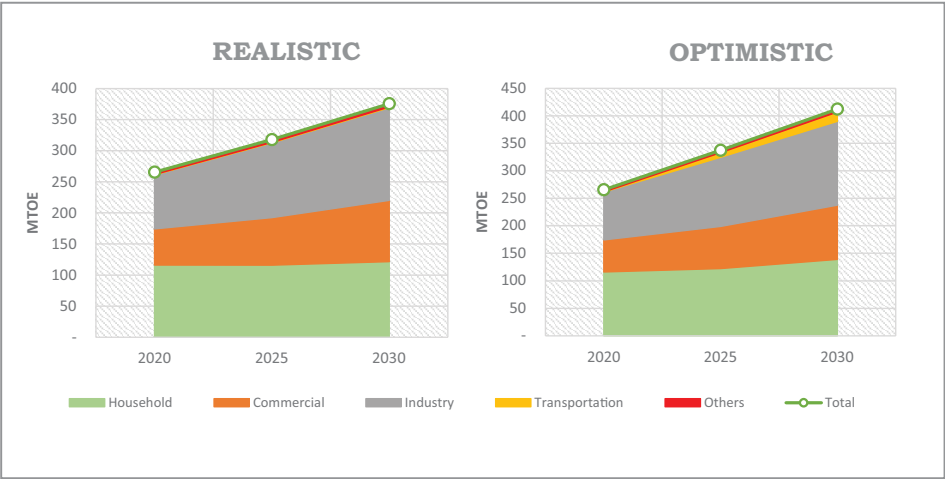


Figure 4.12 Electricity Demand by Sector in 2030

4.3 ELECTRICITY CONSUMPTION PER CAPITA

Overall, electricity consumption increase and number of population give the impact on electricity consumption per capita. Electricity per capita in both scenarios (Realistic & Optimistic) is still below the target of electricity per capita as stated in KEN at 2,500 kWh/capita in 2025 and 7,500 kWh/capita in 2050. Electricity projection per capita for each scenario is presented in Figure 4.13.

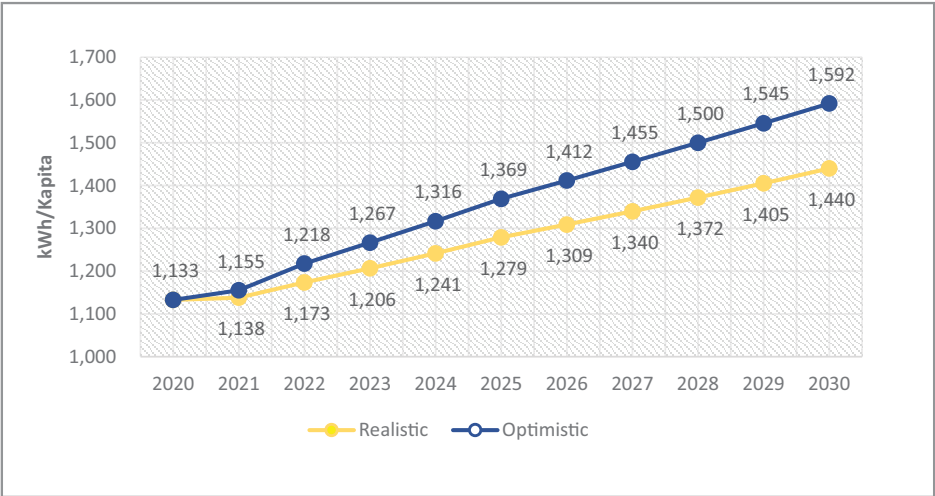


Figure 4.13 Electricity Consumption per Capita

4.4 ELECTRICITY PROJECTION

4.4.1 Electricity Production

Electricity production in 2030 increases into 425 TWh (Realistic) and 466 TWh (Optimistic) by taking into account 10% loss in transmission and distribution.

Based on type of power plant, the production of coal fired power plant in 2020 is around 60% or remains the same in Realistic scenario. However, in Optimistic scenario, electricity production share from coal will decline into 50%.

In Optimistic scenario, power plant is developed to optimize NRE based power plant which refers to RUPTL 2021-2030. In addition, the future power plant development is also directed to substitute fuel to NRE or gas so that the fuel share in 2030 will decline into around 3%. It means that Diesel power plant still exists but it is prioritized for remote and frontier islands with no access to energy. Electricity production per type of energy for the three scenarios can be seen in Figure 4.14.

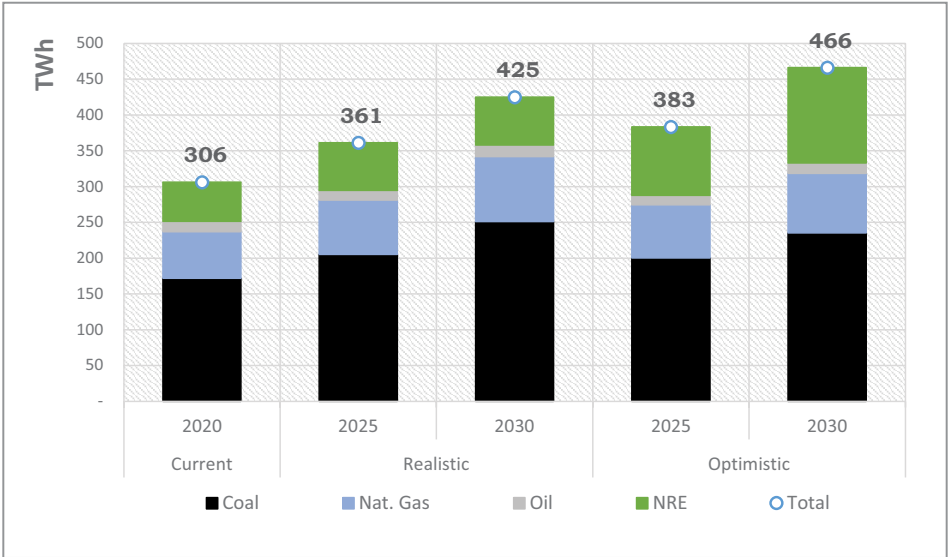


Figure 4.14 Electricity Production per Type of Energy

Electricity production from NRE power plant in 2030 will reach 67 TWh (Realistic) and 133 TWh (Optimistic) with the largest share from Hydro PP, Geothermal PP, and Bioenergy PP. Meanwhile, in the same year, electricity production share from Solar Power Plant compared to other NRE power plant in Optimistic scenario will increase around 10%. The low price of electricity component from Solar PP as well as the solar rooftop program in commercial building and luxurious houses as well as LTHSE will enable the acceleration of Solar PP penetration. Meanwhile, the biggest share is still from Hydro PP (40%, including Minihydro PP and Microhydro PP) and geothermal PP (32%). Bioenergy

PP (Biomass PP, Biogas PP, Waste PP and Hybrid PP) is in the third place (17%) while wind PP is only 1%. Meanwhile, the composition of NRE based power plant development in Realistic scenario is similar to current condition which is dominated by Hydro PP and Geothermal PP. The projection of electricity production from NRE PP is presented in Figure 4.15.

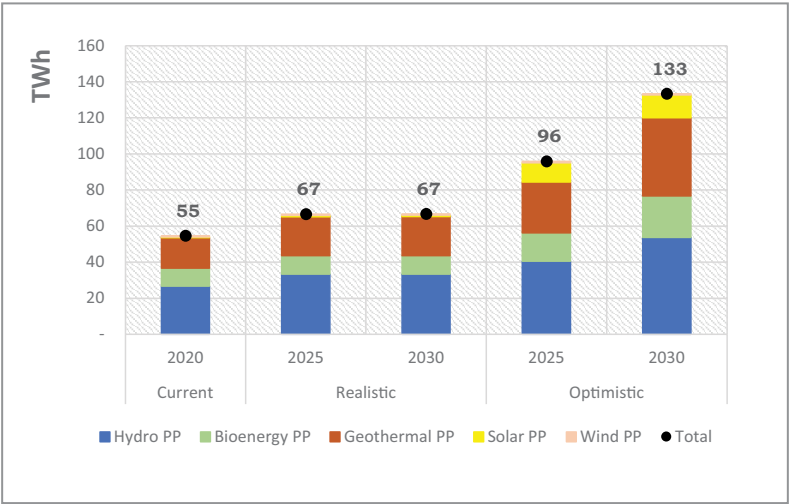


Figure 4.15 Electricity Production from NRE Power Plant

4.4.2 Power Plant Capacity

Power plant capacity in 2030 reaches 90 GW in Realistic scenario and 109 GW in Optimistic scenario. In Realistic scenario in the same year, the biggest power plant is from coal (53%) followed by gas (28%), NRE (14%), and the rest is from fuel. Meanwhile in Optimistic scenario in 2030, the use of NRE power plant is around 25% despite that the coal and gas share is still quite high with 46% and 24% each (Figure 4.16).

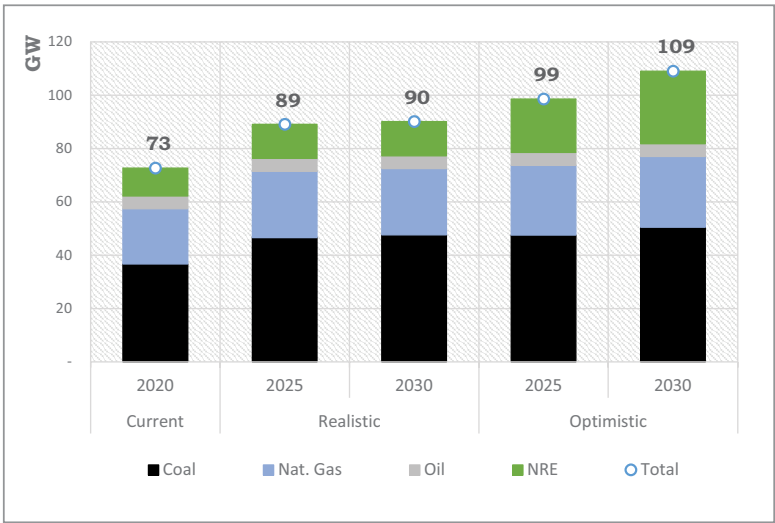


Figure 4.16 Power Plant Capacity per Scenario

NRE power plant capacity in 2030 will increase into 13 GW in Realistic scenario and 27 GW in Optimistic scenario. In Optimistic scenario, the capacity of Solar PP will grow faster compared to other power plants. Thus, in 2030 it will reach 5 GW or 18% from the total NRE power plant capacity. Meanwhile the biggest share is still from Hydro PP (45%, including Mini hydro PP and Micro hydro PP) and Geothermal (20%). The trend of power plant capacity in Realistic and Optimistic scenario can be seen in Figure 4.17.

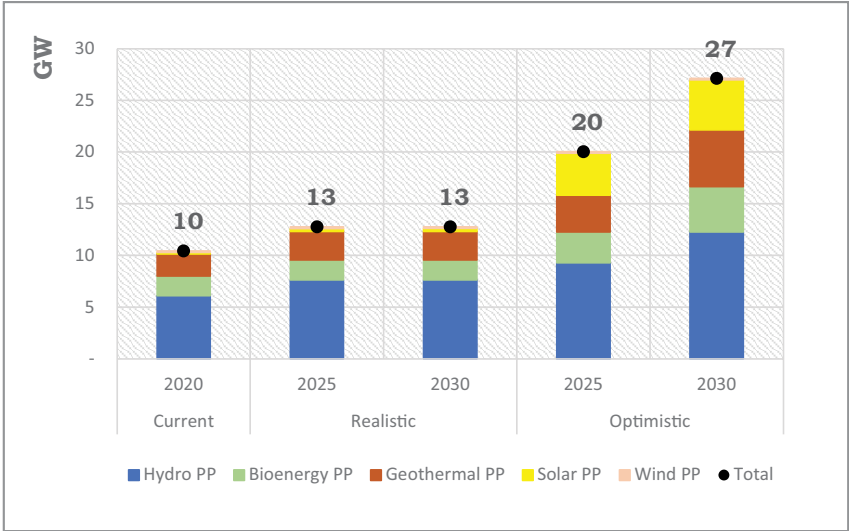


Figure 4.17 NRE Power Plant Capacity per Scenario

4.5 PRIMARY ENERGY SUPPLY PROJECTION PER TYPE

In Realistic scenario, the use of coal in power plant is projected to increase compared to the use of NRE. It influences primary energy mix in Realistic scenario so that the primary energy supply projection in 2025 will reach 256 MTOE. The share of coal dominates energy supply in Realistic scenario with 38% followed by oil with 26%, gas with 24% and NRE with 12%. Primary energy supply in Optimistic scenario is used to optimize NRE to increase NRE share in NRE energy mix in 2025 as shown in Figure 4.18.

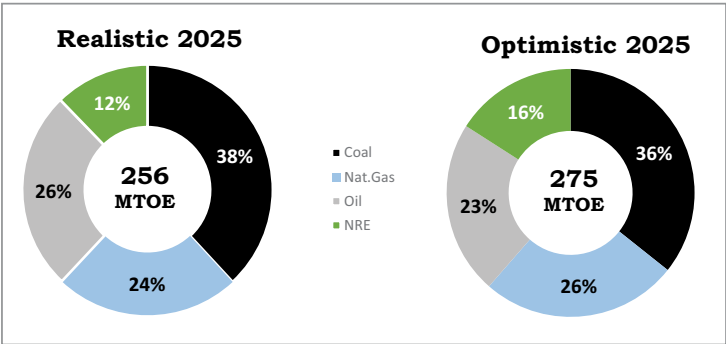


Figure 4.18 Primary Energy Mix in both Scenarios in 2025

In 2030, primary energy mix in Realistic scenario shows a higher share of coal, while NRE share stays the same (Figure 4.19). in Optimistic scenario, NRE share increases into 18% and coal share declines into 35%. The use of fossil fuel especially coal and NRE for power plant will highly influence the primary energy mix in the future.

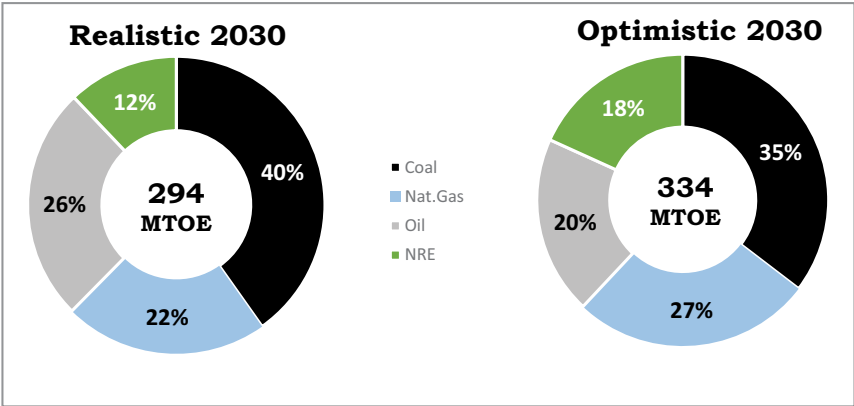


Figure 4.19 Primary Energy Mix in Both Scenarios in 2030

4.6 COMPARISON OF PRIMARY ENERGY MIX IN PROJECTION AND RUEN

Primary energy mix in RUEN in 2025 is 30% coal, 22% gas, 23% oil and 23% NRE. The comparison of primary energy mix projection in Realistic and Optimistic scenario and in RUEN is presented in Figure 4.20 below.

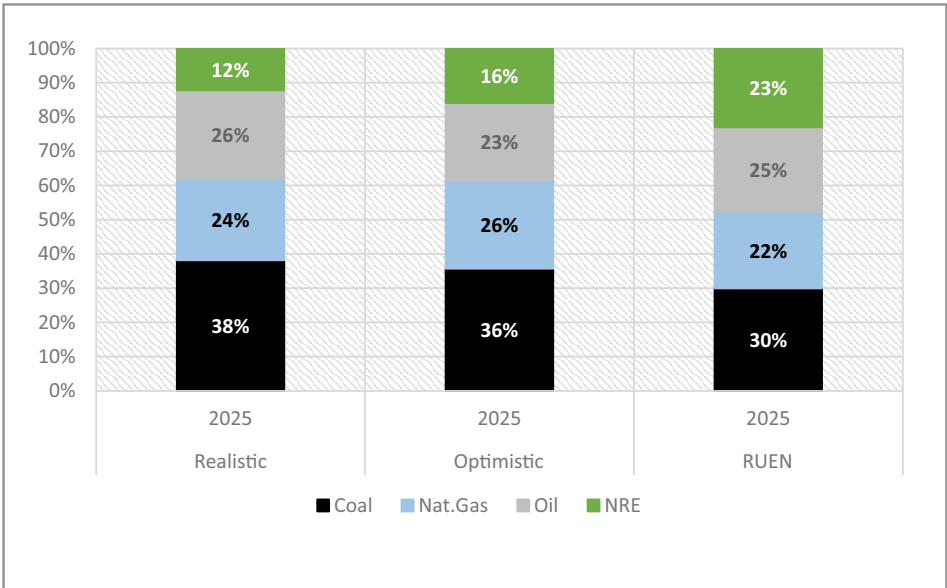


Figure 4.20 Comparison of Primary Energy Mix in Both Scenarios and in RUEN in 2025

If we take a look at the role of NRE in primary energy mix in both scenarios (Realistic and Optimistic) and in RUEN projection, the trend is shown in Figure 4.21 below.

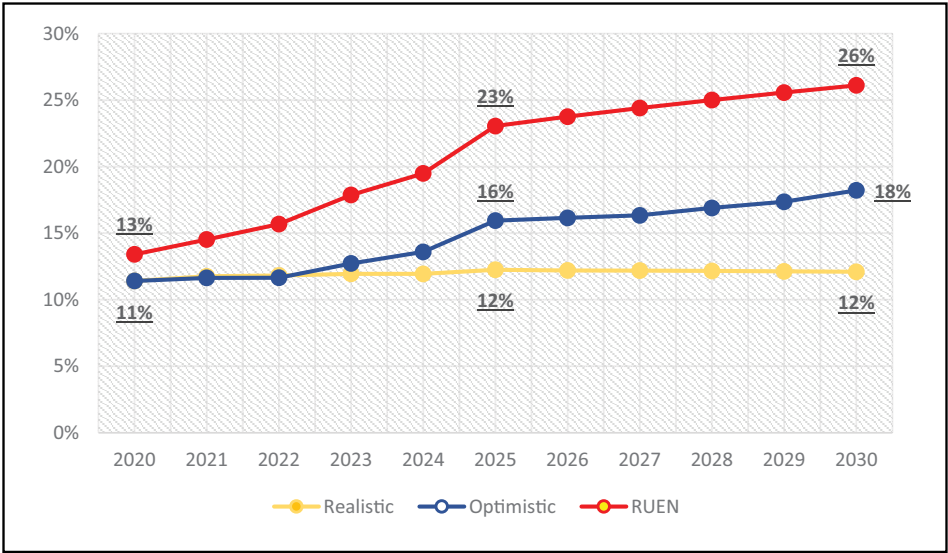


Figure 4.21 Comparison of NRE Mix in both Scenarios and in RUEN

4.7 PROJECTION OF CARBON EMISSION PER CONSUMER SECTOR

Total emission in 2030 is projected to increase into 782 Million Ton CO₂eq (Realistic) and 735 Million Ton CO₂eq (Optimistic). Both number is lower compared to emission target in energy NDC.

Emission in 2030 in Optimistic scenario spesifically in transportation, household and power plant sector is lower compared to emission in Realistic scenario. This condition is influenced by the substitution of LPG to electricity and gas in household massively. In transportation sector, emission reduction is influenced by the substitution of gas and biofuel as well electricity with higher scale compared to in Realistic scenario. Meanwhile, the low emission in power plant in Optimistic scenario is influenced by the high NRE power plant capacity compared to in Realistic scenario. The trend of GHG emission in both scenarios per sector is shown in Figure 4.22 below.

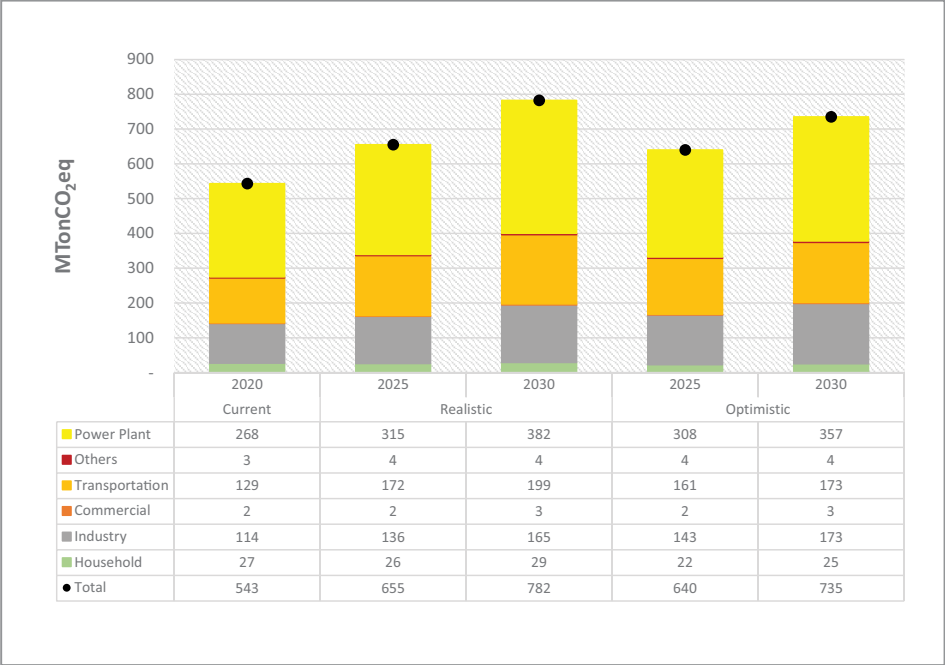


Figure 4.22 Carbon Emission in Both Scenarios

The overview of emission per capita is shown in Figure 4.23 below.

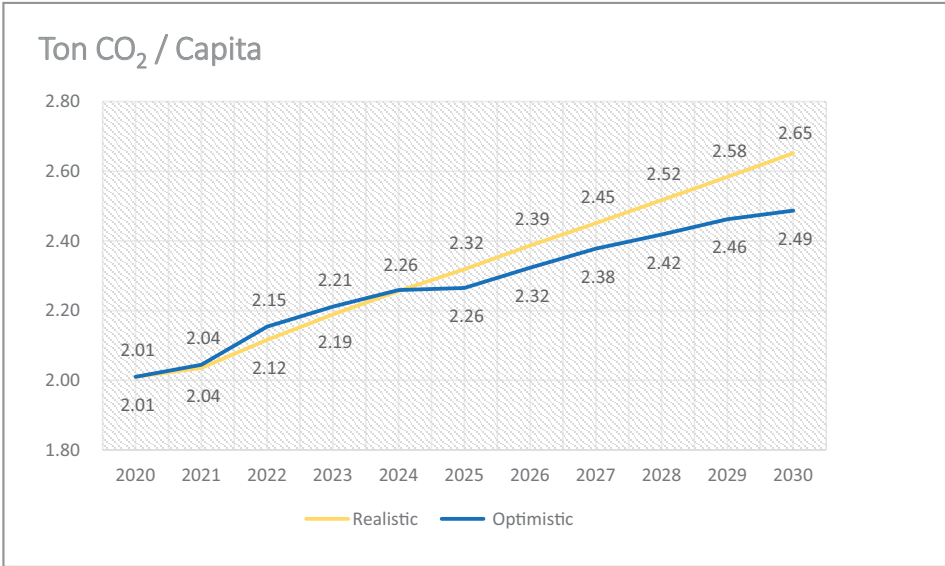


Figure 4.23 GHG Emission per Capita

The summary of several energy indicators based on Government Regulation on National Energy Policy is shown in Table 4.1.

Table 4.1 Energy Indicator per Scenario

Energy Indicators	Unit	Current	Realistic		Optimistic		RUEN	
		2020	2025	2030	2025	2030	2025	2030
Primary Energy Supply	MTOE	221	263	301	275	334	400	500
Primary Energy Mix								
Coal	%	38%	39%	40%	36%	35%	30%	29%
Oil		23%	26%	26%	23%	20%	25%	23%
Nat.Gas		28%	23%	22%	26%	27%	22%	22%
NRE		11%	12%	12%	16%	18%	23%	26%
Elect. Consumption	kWh/Capita	1,133	1,279	1,440	1,369	1,592	2,500	3,285
Primary Energy Consumption	TOE/Capita	0.816	0.905	0.995	0.972	1.131	1.400	1.754
Emission	Mton CO ₂ eq	543	655	782	640	735	893	1061
Emissions per Capita	Mton CO ₂ eq/Capita	2.01	2.32	2.65	2.26	2.49	3.13	3.58

5

NATIONAL ENERGY
RESILIENCE AND
PROGRAMS FOR
SUSTAINABLE ENERGY
DEVELOPMENT

NATIONAL ENERGY RESILIENCE AND PROGRAMS FOR SUSTAINABLE ENERGY DEVELOPMENT

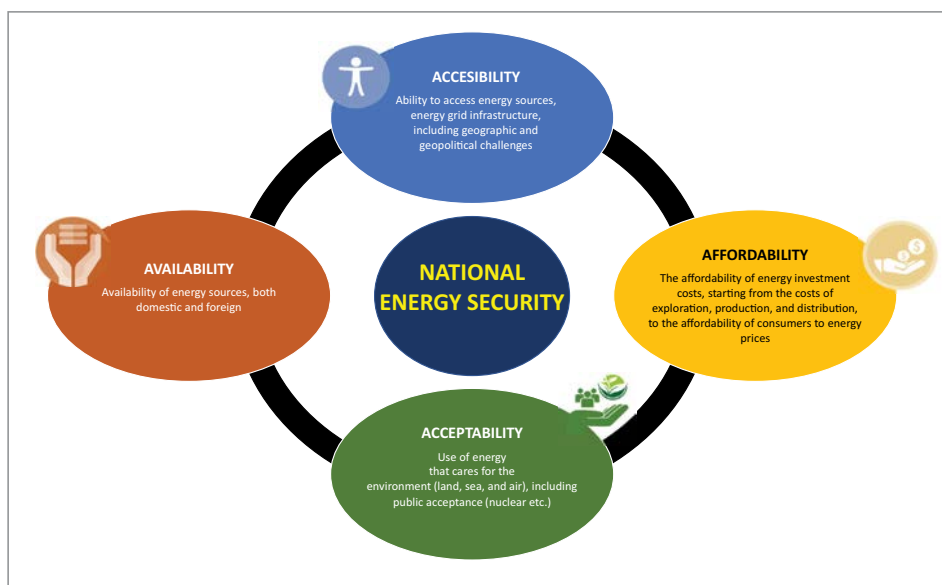
5.1 NATIONAL ENERGY RESILIENCE CONDITION

Energy resilience based on Article 1 point 10 Government Regulation No. 79 of 2014 on National Energy Policy (KEN) is defined as a condition where energy supply and people's access to energy with affordable price for a long-term is secured by considering the protection to the environment.

The condition of energy resilience is dynamic that is influenced by various strategic environmental factors both internal and external, such as the development of energy supply and demand, investment to energy infrastructure development, as well as impact of policy and regulation on energy sector and related cross sectors. The dynamic condition in energy resilience needs to be monitored periodically to identify the change of development in energy sector to secure sustainable energy supply and utilization, promote value added increase in energy use for national economy and people's welfare, as well as to meet the target in KEN.

Assessment to Indonesia energy resilience has been done with energy resilience model with 4A aspects namely Availability (the availability of energy sources both from domestic or abroad), Affordability (the affordability of energy investment cost, from exploration, production and distribution cost to consumer's affordability toward the price of energy), Accessibility (the ability to access energy source, energy network infrastructure, including geographical and geopolitical challenges), Acceptability (the use of environmentally friendly energy in land, sea and air environment including people's acceptance on nuclear and so on); weighting method with Hierarchy Process Analysys; and assessment by experts. Since the assessment in 2004 to 2019, Indonesia energy resilience level increases each year.

The energy resilience assessment is done to see Indonesia's ability in providing energy domestically through aspects and indicators selected during the previous year period.



Source: Indonesia Energy Security, 2019

Figure 5.1 Energy Resilience Model

Based on energy resilience value, capability is measured to secure primary energy supply from energy supply side to meet people's demand on energy. Energy resilience is measured from energy infrastructure to distribute and supply energy to the people, energy economical price from producer side and price affordability from consumer side as well as to increase the use of sustainable and environmentally friendly new and renewable energy.

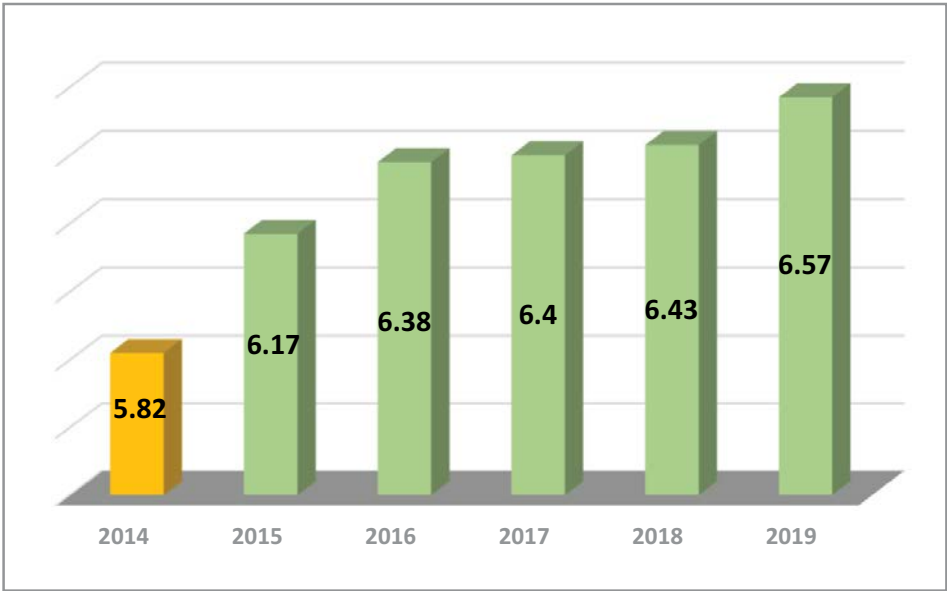
The assessment to energy resilience level to evaluate the implementation of policy and the achievement on government's program target in energy sector is done in order to improve the independent, fair and sustainable energy supply and utilization management. The approach in determining energy resilience level is classified into 4 levels as shown in the following table.

Table 5.1 Level in Energy Resilience Assessment

ASSESSMENT CLASSIFICATION	VALUE/COLOR SCALE
Highly Resilience	8 - 10
Resilience	6 - 7.99
Less Resilience	4 - 5.99
Vulnerable	2 - 3.99
Highly Vulnerable	0 - 1.99

Source: Indonesia Energy Security, 2019

Since the energy resilience assessment in 2014, the level of national energy resilience has been increasing until 2019 (Figure 5.2). In 2014, the energy resilience level is 5.82 or less resilience. It increases into 6.57 in 2019. In general, there has been some improvements in aspect and indicator such as: the increase in energy productivity, the reduction of fuel subsidy, the increase in fuel storage capacity, the reduction in electricity blackout, electrification ratio up to 100%, the reduction in electricity subsidy, the fulfilment of coal DMO, the increase in coal proven reserves, and energy use efficiency.



Source: Indonesia Energy Security, 2019

Figure 5.2 Energy Resilience Level

The increase of future energy resilience needs improvement in all aspects and indicators. One of them is to carry out programs in GSEN:

1. Accelerating NRE power plant with the capacity of 38 GW in 2035 (Solar PP and other NRE)
2. Increasing crude oil production up to 1 million bopd and acquisition of oil field abroad to meet refinery demand,
3. Increasing the capacity of existing and new grass root refineries,
4. Supplying gas based energy for industrial area and transportation sector (such as gas for fuel),
5. Increasing the use of battery electric vehicle,
6. Optimizing biofuel production (biodiesel or biohydrocarbon),
7. Increasing city gas network,
8. Increasing domestic LPG production,
9. Promoting the use of induction stove,
10. Developing DME production,

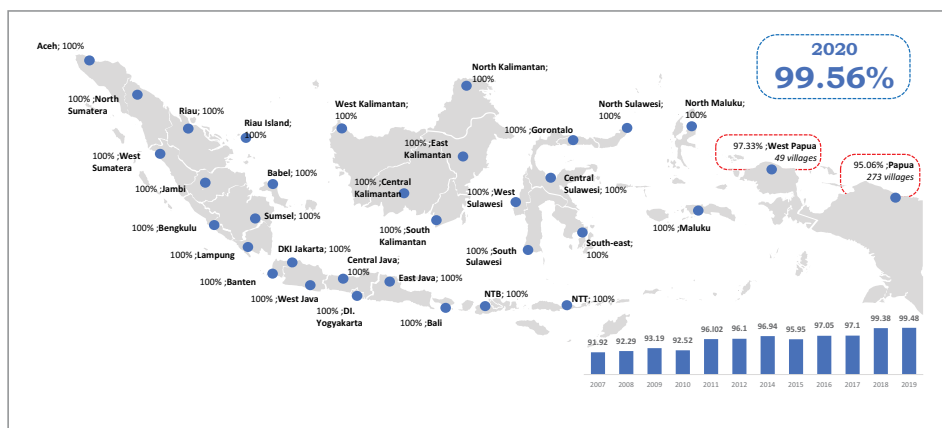
11. Developing gas transmission, LNG receiving terminal, and energy buffer reserve infrastructure,
12. Developing the production of methanol, fertilizer & syngas, as well as synergy between coal mining and smelter,
13. Developing electricity transmission and distribution, smart grid, off grid, and nuclear power plant based on demand as well as establishing Nuclear Energy Programme Implementing Organization (NEPIO)
14. Promoting efficiency, energy conservation and innovation in energy such as Hydrogen, NH₃, CCS and CCUS.

5.2 ELECTRICITY FOR RURAL AREAS AND ENERGY SUPPLY FOR OUTERMOST, REMOTE AND DISADVANTAGED AREAS

As a country with vast area and high population, access to energy for people especially in rural areas as well as in outermost, remote and disadvantaged areas is a challenge for the Government of Indonesia. Many ways and strategies have been taken to supply energy to those areas to give a just development.

Based on MEMR data in 2020, the national electrification ratio is 99.2%. It means that around 0.8% people are not yet electrified. Meanwhile, electrified ratio in rural areas in 2020 is 99.5%. It means that there still 403 unelectrified villages.

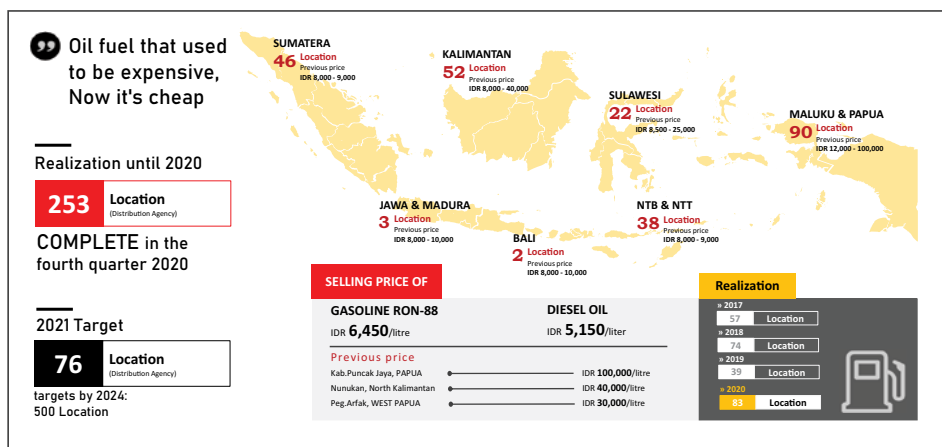
Electrified village is the number of villages which receive electricity from PLN, Non-PLN and LTSHE. The Non-PLN electrified village is village which receives electricity from regional government, private sector or self-subsistence that has been consolidated and verified by Office of Energy and Mineral Resources in local provincial government. To meet the target of 100% electrification ratio, the government has allocated 12.02 trillion Rupiah for infrastructure fund. Several strategies by MEMR to increase the electrification ratio is to carry out grid extension massively to connect villages or households located near PLN grid. The government will also build mini grid and NRE based power plant for people lining in remote areas. There is also plan to build Electricity Recharge Station and Electrical Power Supply (APDAL) for people living scatteredly. There are 20,711 APDAL or known as electricity box for 285 villages in 2021 from State Budget 2021 and PLN SPEL (Recharge Station). If these programs could not meet the needs, diesel power plant is the last option to electrify these areas provided that this diesel power plant must be replaced by NRE based power plant in the next five years. The distribution of electrified village in 2020 is presented in Figure 5.3.



Source: DG Electricity, 2021

Figure 5.3 Eletrified Village Distribution in 2020

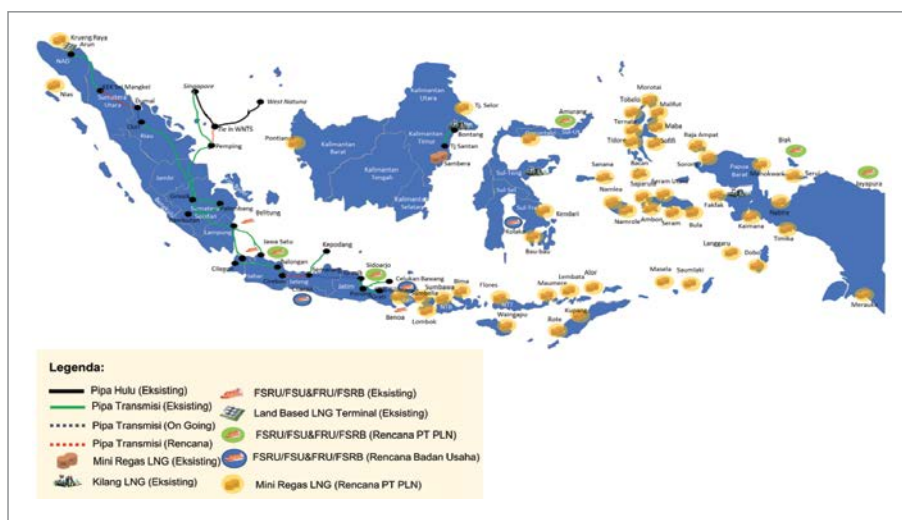
The government has also stipulated One Price Fuel program since 2017 to create energy justice in all areas in Indonesia. This policy is motivated by the high fuel price disparity in several areas especially in eastern Indonesia. The disadvantaged, outermost and frontier area become the government's focus in implementing One Price Fuel. It is expected that fuel with affordable price could be accessed by the people in rural areas and in disadvantaged, outermost and frontier area. The Ministry of Energy and Mineral Resources has stipulated Ministerial Regulation No. 36 of 2016 on the Acceleration of One Price for Certain Types of Fuel and Designated Fuel Nationally to support this policy. This regulation mandates Business Entity with fuel distribution license to establish distributor in certain location especially in areas where distributor of certain types of fuel and designated fuel has not existed yet to let people buy fuel with retail price as stipulated by the government. One Price Fuel program is designed for Gasoline RON 88 (Premium) at the price of IDR 6,450 per liter and diesel at the price of IDR 5,150 per litre with the total target of 253 locations in 2020 and total volume of 147,971 KL for premium and 70,432 KL for diesel. The Oil and Gas Downstream Regulatory Body has also stipulated target with the total additional 330 one fuel price locations in 2020-2024. The distribution of one fuel price locations until 2020 is presented in Figure 5.4 below.



Source: Capaian Kinerja 2020 dan Program 2021 KESDM

Figure 5.4 One price fuel distribution until 2020

In gas supply side, the government is promoting the development of Mini LNG Terminal for remote areas that are not covered by gas pipeline. Mini LNG Terminal in Sambera, East Kalimantan becomes the pilot project of LNG for power plant. This terminal has been used for Sambera Gas fired power plant, Kutai Kertanegara, East Kalimantan with the capacity of 2x20 MW. The development of this Mini LNG Terminal is based on the Ministry of Energy and Mineral Resources Decision No.13 of 2020 on Assignment on LNG Supply and Infrastructure Development and Fuel to LNG Conversion for Power Plant. This Ministerial Decision sets the target to develop 52 LNG power plant with total capacity of 1.7 GW and total gas supply of 167 BBTUD. This program is expected to save 2.6 million Kilo Liter fuel. The map of gas infrastructure development is presented in Figure 5.5 below.



Source: DG Oil and Gas, 2021

Figure 5.5 Map of Gas Infrastructure Development

The government continuous to distribute energy to meet people's demand. Mini LNG Terminal, One Price Fuel, electricity for rural areas, and electrification ratio increase are expected to ease the access to energy and affordable energy price to promote regional economic productivity and growth.

5.3 CITY GAS

In the last several years, the Ministry of EMR has developed city gas for households with no charge. The city gas for households is developed in cities or areas near gas sources.

The number of connected households becomes one of the indicators in national priority program namely gas distribution network for households that is aimed for energy diversification, subsidy reduction, clean energy supply, and kerosene to LPG conversion program to accelerate fuel reduction. Through this program, people are expected to obtain a cleaner and safer fuel. Furthermore, GSEN sets city gas, DME and induction stove program to stop LPG import by 2030.

The target of city gas network in RUEN in 2025 is to connect 4.7 million households. In line with RUEN, city gas network in RPJMN 2020-2024 is 4 million household connections in 2024, but the first development in 2009-2020 has connected only 528 thousand households. This number is far below the target due to budgeting issue. To give an overview, 1 household connection costs IDR 8-10 million.

City gas for household is still funded by the State Budget. The government takes over the project of city gas for households since it is considered as less interesting for business entity due to minimum profit from the project. Concerning the limited state budget, the Public Private Partnership (PPP) involving PT Pertamina and PT Perusahaan Gas Negara (PGN) is being formulated. To support this PPP scheme, the government gives various facilities for the project, such as project development facility (PDF), viability gap fund (VGF), and infrastructure guarantee. This scheme is expected to accelerate the development of gas network infrastructure.

To achieve the target, PGN is formulating roadmap of city gas additional consumer. Starting in 2022, PGN sets the target to connect 1 million households. The acceleration is projected to reduce LPG import up to 144,000 ton per year, employ 83,000 workers, increase 70% local content, save more than 100 billion rupiah per year, and support economic growth in order to increase people's competitiveness and power purchase.

5.4 PROMOTING INDUCTION STOVE AS FOSSIL FUEL SUBSTITUTION

Since 2015 to 2020, the cumulative of LPG subsidy from the government is 219 trillion rupiah. The burden from subsidy is quite high since 70% of LPG is imported.

On the other hand, electricity supply in Java-Bali based on RUPTL 2021 shows that the electricity installed capacity in Java-Bali is 41 GW, while peak load is 29 GW. It means that the reserve margin is above 30% in Java-Bali. One of the efforts to increase electricity consumption is promoting the use of induction stove in Java-Bali. It is predicted that energy consumption for induction stove with 2,000 Watt is around 82 kWh/month based on the study by Research and Development Agency MEMR. Meanwhile, based on data by Directorate General of Electricity, the number of household consumer in Java-Bali with 1,300 VA and above is around 10.3 million households with the details as shown in Table 5.2.

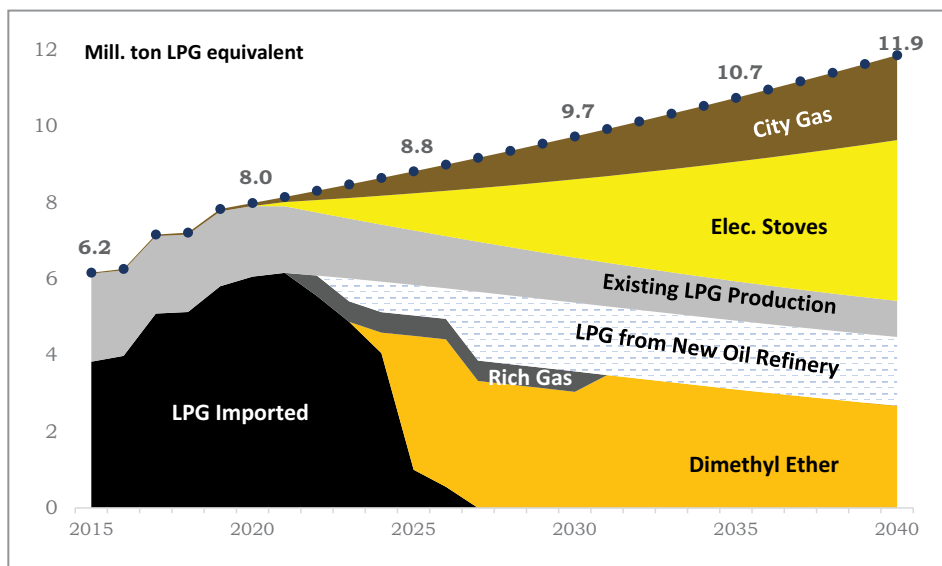
Table 5.2 Details of Electricity Consumer in Java-Bali

CUS- TOMER TYPE	BANTEN	JAKARTA	WEST JAVA	CENTRAL JAVA, DI. YOGYAKARTA	EAST JAVA	BALI	TOTAL
R1 / 1300 VA	703,393	1,776,808	2,113,980	913,834	1,076,458	356,599	6,941,072
R1 / 2200 VA	184,514	612,357	572,898	293,191	328,576	138,488	2,130,024
R2 / 3500- 5500 VA	76,101	360,279	193,741	112,246	148,194	78,769	969,330
R3 / 6600 VA	12,776	114,590	27,869	16,605	27,441	20,721	220,002

Source: DG Electricity, 2021

It is assumed that the additional use of induction stove for 1 million households per year will increase electricity consumption of 82 GWh or 414 MW. If 1,300 VA or above consumers use induction stove, then electricity consumption will increase 841 GWh or electricity load will increase 4.2 GW.

In GSEN, there are several programs to reduce LPG import such as City Gas, increase of production capacity in grass root refineries, development of DME, and the use of induction stove. The use of induction stove is projected to start in 2021 with additional 1 million households. Starting in 2022 to 2030, the program is targeted to reach additional 2 million households per year. Thus, total induction stove users by the end of 2030 are 19 million households. If programs in GSEN run smoothly, there will be no LPG import by 2027. GSEN program on LPG import reduction is presented in Figure 5.6. below.



Source: Grand Strategy of National Energy, 2021

Figure 5.6 GSEN Program on LPG Import Reduction

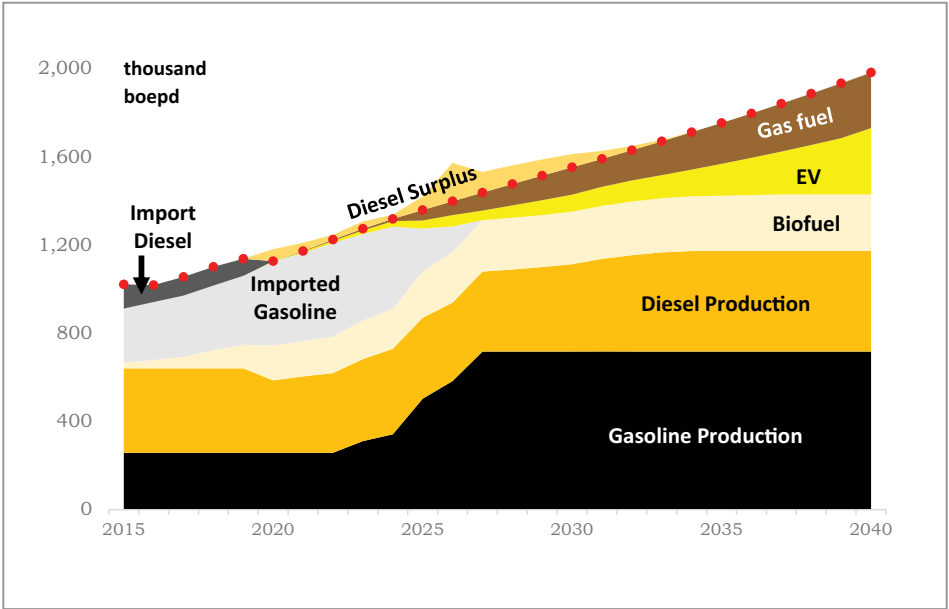
It is expected that induction stove will reduce LPG subsidy, reduce LPG import and increase electricity consumption especially in Java-Bali that experiences electricity surplus. Based on the calculation, induction stove is predicted to consume 4.2 GW of the 12 GW surplus.

5.5 THE DEVELOPMENT OF ELECTRIC VEHICLE

Transportation is the biggest energy consumer compared to other sectors. Fuel consumption in transportation is 83.1% in 2015 and continues to increase into 90.5% in 2020. On the other hand, 33% of fuel supply is derived from import. Therefore, there has been no fuel diversification that it will burden the trade balance.

The government has regulated the use of electric vehicle in Presidential Regulation No. 55 of 2019 on Acceleration of Battery Electric Vehicle for Land Transportation. This Presidential Regulation sets the duties of related ministries in the area of infrastructure development, research and development, regulator, and 35% local content component. To promote this Presidential Regulation, the government through Ministry of Industry has formulated roadmap of electric vehicle development as set in Ministerial Regulation No. 27 of 2020. In this Ministerial Regulation, the target of electric vehicle in 2030 is 627 thousand units, while electric motorcycle is 9 million units. Currently, the Ministry of Industry is coordinating with Coordinating Ministry of Maritime and Investment to amend Minister of Industry Regulation No. 27 of 2020 on roadmap of battery electric vehicle.

In GSEN, there are programs to reduce fuel import through gas fueled vehicle, the use of biofuel, and the use of battery electric vehicle. In 2030, the target is 2 million units of electric car and 12 million units of electric motorcycle. The battery electric vehicle program is expected to stop fuel import before 2030 as shown in Figure 5.7 below.



Source: Grand Strategy of National Energy, 2021

Figure 5.7 GSEN Program to Stop Fuel Import

5.6 CARBON TAX AND EMISSION INTENSITY REDUCTION

Based on Article 65 paragraph 1 Law No. 32 of 2009 on Environmental Protection and Management, everyone has the right to a good and healthy environment as part of human rights. Considering the importance of reducing green house gas emission, the Government of Indonesia has ratified Paris Agreement through Law No. 16 of 2016 as Indonesia’s commitment in reducing GHG to international world.

Carbon has important economical value that reflects sustainable resources management for people’s welfare. The Government of Indonesia needs substantial fund to carry out GHG emission reduction mitigation and meet the target of Nationally Determined Contribution (NDC). Therefore, carbon economical value needs to be regulated in Presidential Regulation to be immediately legalized and implemented.

The government of Indonesia is Drafting Presidential Regulation on Carbon Economical Value and Presidential Regulation on National GHG Inventory and Low Carbon

Development. In 2020, both drafts have been synchronized and are expected to be completed in 2021.

The draft of Presidential Regulation on Carbon Economical Value is aimed as the basis and guidance of GHG emission reduction as well as the arrangement to control carbon emission in national development. The target of emission reduction in 2030 is 29% (834 million ton CO₂) with own efforts and 41% (1,081 million ton CO₂) with international support.

The effort to meet NDC target is done through climate change mitigation and climate change adaptation. It will be formulated in a roadmap that will further regulated in Ministerial Regulation. The mitigation on climate change is carried out in energy sector, waste, industrial process and the use of product, agriculture, forestry and other sectors based on the development of science and technology. The mitigation plan is done through GHG inventory, GHG emission baseline formulation and stipulation, climate change mitigation target, and climate change mitigation action plan.

The GHG inventory is reported by regent/major to governor, governor to minister, business player to related minister, related minister to minister, and minister to president through coordinating minister of maritime and investment. It is done once a year.

GHG emission baseline is formulated based on annual GHG inventory report, GHG emission historical data, scientific data basis on GHG emission, and economic and socio-cultural aspect. The calculation of GHG emission reduction from climate change mitigation is done by comparing baseline GHG emission and GHG inventory.

To meet NDC target, the government, regional government, business player and society hold Carbon Economic Value through:

1. Carbon trade in domestic and foreign trade through emission trade and emission offset trade
2. Performance based-payment based on emission reduction done by government, regional government or business/activity
3. Carbon levy through incentive mechanism and/or disincentive based on performance/mitigation
4. Other efforts based on the development of science and technology

GHG emission trade system will use cap and trade scheme. Cap is defined as the upper margin of GHG emission set by the government or state administrator. Trade allows the trade of gap between GHG emission and Cap. Meanwhile, offset is the use of carbon credit from mitigation activities outside emission trade system to reduce GHG emission. GHG emission trade system allows emission producer to buy and sell emission contract to meet the mandatory of regulated emission Cap. This concept of GHG emission trade has been implemented in many other countries in the world.

In 2021, the Ministry of Energy and Mineral Resources has carried out carbon trade test with 80 units steam PP by using cap and trade scheme. The steam PP consists of 54 PLN units and 26 IPP units where 10 units are located in mine mouth and 70 units are non-mine mouth. The steam PP capacity is divided based on Minister of Energy and Mineral Resources Regulation No. 9 of 2020 on Electricity Supply Efficiency by PT PLN (Persero). Next, cap is calculated based on average GHG emission intensity value in 2019 on steam PP as the participants of carbon trade test by taking into account the surplus of emission quota allocation. The details of Cap in carbon tax test in 2021 is presented in Table 5.3.

Table 5.3 Cap in Carbon Trade Test in 2021

Power Plant Type	Installed capacity (MW)	Cap Value (ton CO ₂ /MWh)
Coal PP	X > 400	0.918
Coal PP	100 ≤ X ≤ 400	1.013
Mine Mouth Coal PP	100 ≤ X ≤ 400	1.094

Source: DG Electricity, 2021

There are 28 carbon trade transactions between power plant units and total carbon transfer of 42.45 thousand ton CO₂ resulting in more than IDR 1.2 billion incentive. PT PLN (Persero) has also conducted emission trade between steam PP voluntarily.

The government of Indonesia has formulated Roadmap of Carbon Market in Energy Sector. The Cap uses weighted average per cluster that will be implemented in 2021-2024. After 2024, the Cap uses ideal benchmark per cluster. In 2021, sectoral scope of carbon market in power plant sector. In 2022-2024, the market scope is expanded to oil and gas sector especially Carbon Capture Storage (CCS) and Carbon Capture, Utilization, and Storage (CCUS). After 2024, carbon market sectoral scope is projected to expand to industrial and aviation sector. In 2022-2024, independent verifactor with the capability of monitoring carbon market is expected to be established. After 2024, the Government of Indonesia will establish a national institution to run carbon market and carry out independent verifactor accreditation as carbon market verifactor.

In October 2021, the government and the House of Representatives has set carbon tax tariff at the lowest of IDR 30 per kilogram CO₂ especially in power plant sector in Draft of Law on taxation rules harmonization. Tax collection will be done if power plant produces CO₂ above Cap (emission upper margin) set by the government. Carbon tax collection is planned in April 2022. Tax carbon is expected to change people’s behavior and increase people and business player’s awareness to reduce GHG emission.

5.7 PREPARING ENERGY TRANSITION

Energy transition is done through transformation of fossil fuel using clean technology, acceleration of new and renewable energy use, and energy conservation. Energy transformation into Net Zero Emission is expected to create energy independence, climate resilience and low carbon.

The use of fossil fuel with clean technology is done through the technology of CCT (Clean Coal Technology) and CCUS (Carbon Capture, Utilization, and Storage). CCT is high efficiency technology in steam PP to reduce emission in several pollutants and waste as well as to increase energy from each ton of coal. The Government of Indonesia has developed Ultra Super Critical (USC) technology in steam PP from Jawa-7 steam PP located in Serang (Banten) in 2017. In 2021, there are several USC steam PP being developed in Jawa 9 & 10 steam PP, Jawa Tengah steam PP (Batang) and Jawa 4 steam PP (Tanjung Jati B).

CCUS technology is a technology that enable CO₂ capture before being released to the atmosphere. It then carries or flows CO₂ into the storage to be stored for a very long time. In 2020, the Government of Indonesia has carried out feasibility study of CCUS pilot project in Gundih field Central Java. The total CO₂ reduction potential in this pilot project is projected to reach 2.92 million ton in 10 years. Indonesia has major geological storage resources as potential CCUS location in Java, Sumatera and Kalimantan. The feasibility study is also being done in these areas to develop CCUS.

The acceleration on new renewable energy use is done by substituting primary energy and final energy through the use of biofuel, co-firing technology, fuel source from Refuse-Derived Fuel (RDF) and Solid Recovered Fuel (SFR), additional NRE power plant capacity with the focus on Rooftop solar PP, and non electricity NRE and non-biofuel utilization such as briquette, biogas and Compressed Biomethane Gas (CBG).

Through the use of battery electric vehicle, the battery industry and hydrogen technology are also expected to support the acceleration of new renewable energy use.

Beside saving money, energy conservation program also reduces emission. In renewable energy sector, Smart Energy program optimizes renewable energy production system by maximizing efficiency, reducing cost, and enhancing security. In electricity, Smart Grid development is expected to reduce number of power plant needed to supply electricity and to integrate renewable energy sources.

Innovation in technology is required to reach Net Zero Emission and to create sustainable clean energy. The technology used today is hydro, solar, wind, bioenergy, and geothermal technology. Meanwhile, the technology being developed is nuclear, coal gasification, liquified coal, tidal, hydrogen, and Carbon Capture Storage.

Innovation in renewable energy use in the domestic is expected to create a more competitive renewable energy price. It needs participation, synergy and cooperation from all related stakeholders to accelerate energy transition in Indonesia.



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